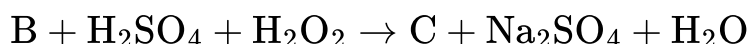


# The D and F Block Elements

## Question1

Consider the following reactions :



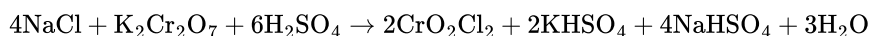
In the product ' C ', ' X ' is the number of  $\text{O}_2^{2-}$  units, ' Y ' is the total number oxygen atoms present and ' Z ' is the oxidation state of Cr. The value of  $X + Y + Z$  is \_\_\_\_ .

**JEE Main 2026 (Online) 21st January Morning Shift**

**Answer: 13**

**Solution:**

The first reaction is the well-known **Chromyl chloride test**. When a chloride salt like sodium chloride ( $\text{NaCl}$ ) is heated with potassium dichromate ( $\text{K}_2\text{Cr}_2\text{O}_7$ ) and concentrated sulfuric acid ( $\text{H}_2\text{SO}_4$ ), deep red vapors of chromyl chloride are evolved.



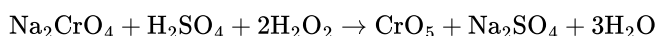
Thus, Product A is **Chromyl chloride** ( $\text{CrO}_2\text{Cl}_2$ ).

When chromyl chloride gas is passed through an aqueous solution of sodium hydroxide ( $\text{NaOH}$ ), a yellow solution of sodium chromate is formed.



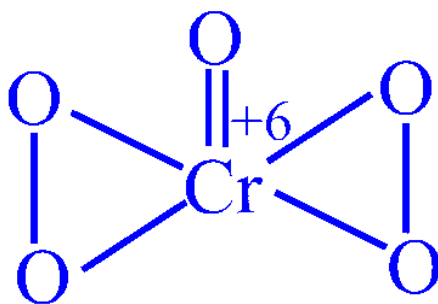
Thus, Product B is **Sodium chromate** ( $\text{Na}_2\text{CrO}_4$ ).

When a solution of a chromate acts with hydrogen peroxide ( $\text{H}_2\text{O}_2$ ) in an acidic medium (provided by  $\text{H}_2\text{SO}_4$ ), a deep blue-colored compound known as chromium pentoxide is formed.



Thus, Product C is **Chromium pentoxide** ( $\text{CrO}_5$ ).

Chromium pentoxide has a famous "butterfly structure".



- **Finding X:** In its structure, there are two peroxide (O – O) linkages. Each peroxide linkage represents one  $\text{O}_2^{2-}$  unit. Therefore, the number of  $\text{O}_2^{2-}$  units is 2.

So,  $X = 2$ .

- **Finding Y:** The chemical formula is  $\text{CrO}_5$ , meaning there are 5 oxygen atoms in total.

So,  $Y = 5$ .

- **Finding Z:** Out of the 5 oxygen atoms in the butterfly structure, one is bonded by a double bond (normal oxide with an oxidation state of  $-2$ ), and the remaining four form the peroxide linkages (with an oxidation state of  $-1$  each).

Let the oxidation state of Chromium (Cr) be  $Z$ . We can calculate it as follows:

$$Z + 1(-2) + 4(-1) = 0$$

$$Z - 2 - 4 = 0$$

$$Z = +6$$

So,  $Z = 6$ .

$$\therefore X + Y + Z = 2 + 5 + 6 = 13$$

## Question2

Among the following oxides of  $3d$  elements, the number of mixed oxides are \_\_\_\_ .

$\text{Ti}_2\text{O}_3$ ,  $\text{V}_2\text{O}_4$ ,  $\text{Cr}_2\text{O}_3$ ,  $\text{Mn}_3\text{O}_4$ ,  $\text{Fe}_3\text{O}_4$ ,  $\text{Fe}_2\text{O}_3$ ,  $\text{Co}_3\text{O}_4$

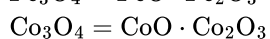
**JEE Main 2026 (Online) 22nd January Evening Shift**

**Answer: 3**

**Solution:**

**Mixed oxides** are those oxides which can be written as a combination of two simpler oxides of the same metal, like  $\text{MO} \cdot \text{M}_2\text{O}_3$ .

Now, check the given oxides and see which ones can be written in this form:



Only three  
mixed oxides

So, the number of mixed oxides in the given list is 3.

-----

## Question3

Number of paramagnetic ions among the following d- and f-block metal ions is \_\_\_\_\_.

$\text{Mn}^{2+}$ ,  $\text{Cu}^{2+}$ ,  $\text{Zn}^{2+}$ ,  $\text{Yb}^{2+}$ ,  $\text{Sc}^{3+}$ ,  $\text{La}^{3+}$ ,  $\text{Gd}^{3+}$ ,  $\text{Lu}^{3+}$ ,  $\text{Ti}^{4+}$ ,  $\text{Ce}^{4+}$

(Atomic number of Mn = 25, Cu = 29, Zn = 30, Yb = 70, Sc = 21, La = 57, Gd = 64, Lu = 71, Ti = 22, Ce = 58)

**JEE Main 2026 (Online) 2nd April Evening Shift**

**Answer: 3**

### Solution:

To find the number of **paramagnetic ions**, we check whether each ion has **one or more unpaired electrons**.

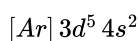
If all electrons are paired, the ion is **diamagnetic**.

We will write the electronic configuration of each ion.

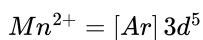
#### 1. $\text{Mn}^{2+}$

Atomic number of Mn = 25

Neutral Mn:



For  $\text{Mn}^{2+}$ , remove two electrons from 4s:

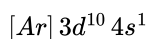


There are 5 unpaired electrons, so it is **paramagnetic**.

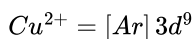
#### 2. $\text{Cu}^{2+}$

Atomic number of Cu = 29

Neutral Cu:



For  $\text{Cu}^{2+}$ , remove one electron from 4s and one from 3d:

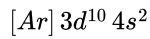


There is 1 unpaired electron, so it is **paramagnetic**.

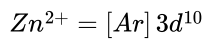
#### 3. $\text{Zn}^{2+}$

Atomic number of Zn = 30

Neutral Zn:



For  $Zn^{2+}$ :

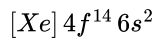


All electrons are paired, so it is **diamagnetic**.

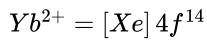
#### 4. $Yb^{2+}$

Atomic number of Yb = 70

Neutral Yb:



For  $Yb^{2+}$ :

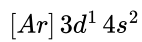


All electrons are paired, so it is **diamagnetic**.

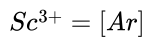
#### 5. $Sc^{3+}$

Atomic number of Sc = 21

Neutral Sc:



For  $Sc^{3+}$ :

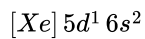


No unpaired electron, so it is **diamagnetic**.

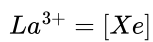
#### 6. $La^{3+}$

Atomic number of La = 57

Neutral La:



For  $La^{3+}$ :

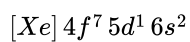


No unpaired electron, so it is **diamagnetic**.

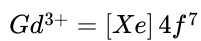
#### 7. $Gd^{3+}$

Atomic number of Gd = 64

Neutral Gd:



For  $Gd^{3+}$ , remove  $6s^2$  and  $5d^1$ :

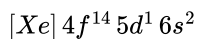


This has 7 unpaired electrons, so it is **paramagnetic**.

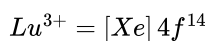
#### 8. $Lu^{3+}$

Atomic number of Lu = 71

Neutral Lu:



For  $Lu^{3+}$ :

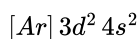


All electrons are paired, so it is **diamagnetic**.

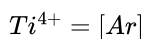
### 9. $Ti^{4+}$

Atomic number of Ti = 22

Neutral Ti:



For  $Ti^{4+}$ :

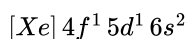


No unpaired electron, so it is **diamagnetic**.

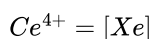
### 10. $Ce^{4+}$

Atomic number of Ce = 58

Neutral Ce:

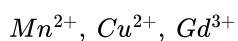


For  $Ce^{4+}$ , remove four electrons:



No unpaired electron, so it is **diamagnetic**.

**Paramagnetic ions are:**



So, the **number of paramagnetic ions** is

3

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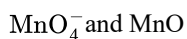
## Question4

$MnO_4^{2-}$ , in acidic medium, disproportionates to :

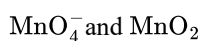
**JEE Main 2026 (Online) 21st January Morning Shift**

**Options:**

A.



B.



C.

Mn<sub>2</sub>O<sub>7</sub> and MnO<sub>2</sub>

D.

Mn<sub>2</sub>O<sub>7</sub> and MnO

**Answer: B**

**Solution:**

In MnO<sub>4</sub><sup>2-</sup> (manganate ion), oxidation state of Mn is:

$$x + 4(-2) = -2 \Rightarrow x - 8 = -2 \Rightarrow x = +6$$

In acidic medium, it **disproportionates**, i.e., Mn in +6 state is converted into a higher and a lower oxidation state:

- Oxidation: +6 → +7 gives MnO<sub>4</sub><sup>-</sup> (permanganate)
- Reduction: +6 → +4 gives MnO<sub>2</sub>

Balanced equation in acidic medium:



So, MnO<sub>4</sub><sup>2-</sup> disproportionates to MnO<sub>4</sub><sup>-</sup> **and** MnO<sub>2</sub>.

**Correct option: B**

-----

## Question5

**On heating a mixture of common salt and  $K_2Cr_2O_7$  in equal amount along with concentrated  $H_2SO_4$  in a test tube, a gas is evolved. Formula of the gas evolved and oxidation state of the central metal atom in the gas respectively are:**

**JEE Main 2026 (Online) 21st January Evening Shift**

**Options:**

A.

$Cr_2O_2Cl_2$  and +3

B.

$Cr_2O_2Cl_2$  and +6

C.

$CrO_2Cl_2$  and +6

D.

$CrO_2Cl_2$  and +5

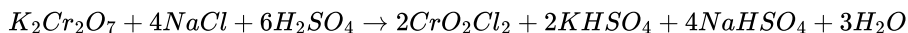
**Answer: C**

**Solution:**

Gas evolved is **chromyl chloride**, formed in the **chromyl chloride test**.

### Step 1: Reaction (NCERT)

On heating  $K_2Cr_2O_7$  with a chloride salt (common salt  $NaCl$ ) and conc.  $H_2SO_4$ , **red fumes of chromyl chloride** are obtained:



So, the gas is  $CrO_2Cl_2$ .

### Step 2: Oxidation state of Cr in $CrO_2Cl_2$

Let oxidation state of Cr be  $x$ .

- Oxygen:  $2 \times (-2) = -4$
- Chlorine:  $2 \times (-1) = -2$

$$\text{Sum} = x - 4 - 2 = 0$$

$$x - 6 = 0 \Rightarrow x = +6$$

**Correct option**

**Option C:  $CrO_2Cl_2$  and +6**

---

## Question6

**Given below are some of the statements about Mn and  $Mn_2O_7$ . Identify the correct statements.**

- A. Mn forms the oxide  $Mn_2O_7$ , in which Mn is in its highest oxidation state.**
- B. Oxygen stabilizes the Mn in higher oxidation states by forming multiple bonds with Mn.**
- C.  $Mn_2O_7$  is an ionic oxide.**
- D. The structure of  $Mn_2O_7$  consists of one bridged oxygen.**

**Choose the correct answer from the options given below :**

### JEE Main 2026 (Online) 21st January Evening Shift

**Options:**

A.

A, B, C and D

B.

A, B and D Only

C.

A, B and C Only

D.

A, C and D Only

**Answer: B**

## Solution:

**A. Mn forms the oxide  $\text{Mn}_2\text{O}_7$ , in which Mn is in its highest oxidation state.**

- To find the oxidation state of Mn in  $\text{Mn}_2\text{O}_7$ , we set the sum of oxidation states to zero. Let Mn's oxidation state be 'x'. Oxygen's oxidation state is typically -2.
- $2(x) + 7(-2) = 0$
- $2x - 14 = 0$
- $2x = 14$
- $x = +7$
- Manganese is a Group 7 element with an electronic configuration of  $[\text{Ar}]3d^54s^2$ . The highest possible oxidation state for manganese is +7, which involves the loss of all 7 valence electrons.
- Therefore, Mn is indeed in its highest oxidation state (+7) in  $\text{Mn}_2\text{O}_7$ .
- This statement is **correct**.

**B. Oxygen stabilizes the Mn in higher oxidation states by forming multiple bonds with Mn.**

- In high oxidation states, transition metals strongly attract electrons, increasing the covalent character of their bonds with oxygen. This is particularly true for elements like Mn in a +7 oxidation state.
- The formation of multiple bonds (e.g.,  $\text{Mn}=\text{O}$ ) allows for better delocalization of electron density, which helps to stabilize the high positive charge on the central metal atom. This is a common phenomenon for high oxidation state metal oxides (e.g.,  $\text{CrO}_3$ ,  $\text{V}_2\text{O}_5$ ).
- This statement is **correct**.

**C.  $\text{Mn}_2\text{O}_7$  is an ionic oxide.**

- Ionic oxides are typically formed when the metal has a low oxidation state and a significant electronegativity difference with oxygen leads to electron transfer (e.g.,  $\text{MnO}$ ).
- However, as the oxidation state of a metal increases, its ability to polarize electron clouds increases, and the covalent character of its bonds with oxygen becomes prominent.
- $\text{Mn}_2\text{O}_7$  is a dark green, viscous liquid with a melting point of  $-20^\circ\text{C}$ . These properties are characteristic of a molecular covalent compound, not an ionic solid which would have a high melting point. It is highly covalent due to the high electronegativity of oxygen and the high charge density of  $\text{Mn}^{7+}$ .
- Therefore,  $\text{Mn}_2\text{O}_7$  is a covalent oxide, not an ionic oxide.
- This statement is **incorrect**.

**D. The structure of  $\text{Mn}_2\text{O}_7$  consists of one bridged oxygen.**

- The structure of  $\text{Mn}_2\text{O}_7$  can be represented as  $\text{O}_3\text{Mn} - \text{O} - \text{MnO}_3$ . It consists of two  $\text{MnO}_4$  tetrahedra sharing a common oxygen atom.
- The oxygen atom that links the two manganese atoms is a bridging oxygen. Each Mn atom has three terminal oxygen atoms and one bridging oxygen atom.
- This statement is **correct**.

Based on the analysis, statements A, B, and D are correct, while statement C is incorrect.

The final answer is 

|   |
|---|
| B |
|---|

---

## Question 7

Given below are two statements :

**Statement I :** The first ionization enthalpy of Cr is lower than that of Mn .

**Statement II :** The second and third ionization enthalpies of Cr are higher than those of Mn .

**In the light of the above statements, choose the correct answer from the options given below :**

### JEE Main 2026 (Online) 22nd January Evening Shift

**Options:**

A.

Statement I is true but Statement II is false

B.

Both Statement I and Statement II are true

C.

Both Statement I and Statement II are false

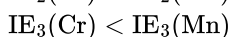
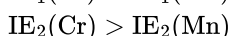
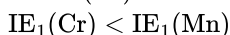
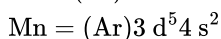
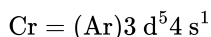
D.

Statement I is false but Statement II is true

**Answer: A**

**Solution:**

First, write the electronic configurations of Cr and Mn.



**For first ionization enthalpy ( $\text{IE}_1$ ):** In Cr, the electron is removed from  $4s^1$ . After losing this one  $4s$  electron, Cr becomes  $\text{Cr}^+$  with a stable half-filled  $3d^5$  configuration. So removing the first electron is comparatively easier, hence  $\text{IE}_1(\text{Cr}) < \text{IE}_1(\text{Mn})$ .

**For second ionization enthalpy ( $\text{IE}_2$ ):** After losing one electron,  $\text{Cr}^+$  already has the stable  $3d^5$  configuration. Removing the second electron would disturb this stable half-filled  $3d$  arrangement, so it needs more energy. But in Mn, after the first electron loss, removing the next electron can lead to  $\text{Mn}^{2+}$  with stable  $3d^5$ . Therefore,  $\text{IE}_2(\text{Cr}) > \text{IE}_2(\text{Mn})$ .

**For third ionization enthalpy ( $\text{IE}_3$ ):**  $\text{Mn}^{2+}$  is very stable due to  $3d^5$ . Removing one more electron from this stable state is difficult, so  $\text{IE}_3$  for Mn becomes high. In comparison,  $\text{Cr}^{2+}$  does not have this extra stability, so removing the third electron is relatively easier. Hence,  $\text{IE}_3(\text{Cr}) < \text{IE}_3(\text{Mn})$ .

---

## Question8

Given below are two statements :

**Statement I :** The number of pairs, from the following, in which both the ions are coloured in aqueous solution is 3.



**Statement II :**  $\text{Th}^{4+}$  is the strongest reducing agent among  $\text{Th}^{4+}$ ,  $\text{Ce}^{4+}$ ,  $\text{Gd}^{3+}$  and  $\text{Eu}^{2+}$ .

In the light of the above statements, choose the correct answer from the options given below

### JEE Main 2026 (Online) 28th January Morning Shift

Options:

A.

Statement I is false but Statement II is true

B.

Both Statement I and Statement II are false

C.

Both Statement I and Statement II are true

D.

Statement I is true but Statement II is false

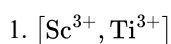
**Answer: B**

**Solution:**

**Statement I**

An ion is **coloured in aqueous solution** if it has **partially filled  $d$ -orbitals** (so that  $d$ - $d$  transitions are possible). Ions with  $d^0$  or  $d^{10}$  configurations are **colourless**.

Now check each pair:



- $\text{Sc}^{3+}$  : Sc ( $Z = 21$ ) is  $[\text{Ar}] 3d^1 4s^2$ ; so  $\text{Sc}^{3+} = [\text{Ar}] 3d^0$  (colourless)
- $\text{Ti}^{3+}$  : Ti ( $Z = 22$ ) is  $[\text{Ar}] 3d^2 4s^2$ ; so  $\text{Ti}^{3+} = [\text{Ar}] 3d^1$  (coloured)

$\Rightarrow$  **Both not coloured**

1.  $[\text{Mn}^{2+}, \text{Cr}^{2+}]$

- $\text{Mn}^{2+} = [\text{Ar}] 3d^5$  (coloured)
- $\text{Cr}^{2+} = [\text{Ar}] 3d^4$  (coloured)

$\Rightarrow$  **Both coloured** 

1.  $[\text{Cu}^{2+}, \text{Zn}^{2+}]$

- $\text{Cu}^{2+} = [\text{Ar}] 3d^9$  (coloured)
- $\text{Zn}^{2+} = [\text{Ar}] 3d^{10}$  (colourless)

$\Rightarrow$  **Both not coloured**

1.  $[\text{Ni}^{2+}, \text{Ti}^{4+}]$

- $\text{Ni}^{2+} = [\text{Ar}] 3d^8$  (coloured)
- $\text{Ti}^{4+} = [\text{Ar}] 3d^0$  (colourless)

$\Rightarrow$  **Both not coloured**

So, the number of pairs in which **both ions are coloured** is only 1, not 3.

Hence, **Statement I is false**.

### Statement II

A **reducing agent** is a species that **gets oxidised** (it donates electrons).

- $\text{Ce}^{4+}$  is a well-known **strong oxidising agent** (it gets reduced to  $\text{Ce}^{3+}$ ).
- $\text{Th}^{4+}$  is also in a high oxidation state and is **not** a strong reducing agent.
- $\text{Gd}^{3+}$  is a very stable common oxidation state for Gd; not strongly reducing.
- $\text{Eu}^{2+}$  readily gets oxidised to  $\text{Eu}^{3+}$ , so it acts as a **strong reducing agent** among the given species.

Therefore, the strongest reducing agent among  $\text{Th}^{4+}$ ,  $\text{Ce}^{4+}$ ,  $\text{Gd}^{3+}$  and  $\text{Eu}^{2+}$  is  $\text{Eu}^{2+}$ , not  $\text{Th}^{4+}$ .

Hence, **Statement II is false**.

### Correct Option

Since **both Statement I and Statement II are false**, the correct choice is:

**Option B: Both Statement I and Statement II are false.**

-----

## Question9

**Consider the following statements about manganate and permanganate ions. Identify the correct statements.**

**A. The geometry of both manganate and permanganate ions is tetrahedral.**

**B. The oxidation states of Mn in manganate and permanganate are +7 and +6, respectively.**

**C. Oxidation of Mn(II) salt by peroxodisulphate gives manganate ion as the final product.**

**D. Manganate ion is paramagnetic and permanganate ion is diamagnetic.**

**E. Acidified permanganate ion reduces oxalate, nitrite and iodide ions.**

**Choose the correct answer from the options given below :**

**JEE Main 2026 (Online) 28th January Evening Shift**

**Options:**

A.

A and D Only

B.

A, B and C Only

C.

A, C and D Only

D.

A, D and E Only

**Answer: A**

**Solution:**

Manganate ion  $\rightarrow \text{MnO}_4^{-2}$

Permanganate ion  $\rightarrow \text{MnO}_4^{-}$

(A) In both ions, Mn is surrounded by four oxygen atoms. So, the shape of both  $\text{MnO}_4^{-2}$  and  $\text{MnO}_4^{-}$  is tetrahedral (often written as  $d^3$ s hybridisation in this context).

(B) For  $\text{MnO}_4^{-}$ , let oxidation state of Mn be  $x$ . Then  $x + 4(-2) = -1 \Rightarrow x = +7$ .

For  $\text{MnO}_4^{-2}$ ,  $x + 4(-2) = -2 \Rightarrow x = +6$ .

So, Mn is +7 in permanganate and +6 in manganate (the statement given is reversed).

(C) When  $\text{Mn}^{2+}$  salt is oxidised by peroxodisulphate  $\text{S}_2\text{O}_8^{2-}$ , the product formed is permanganate ion, not manganate:

$\text{Mn}^{2+} + \text{S}_2\text{O}_8^{2-} \rightarrow \text{MnO}_4^{-}$  (Permanganate ion)

(D) In  $\text{MnO}_4^{-}$ , Mn is in +7 state, which corresponds to  $3d^0$  (no unpaired electrons), so it is diamagnetic.

In  $\text{MnO}_4^{-2}$ , Mn is in +6 state, which corresponds to  $3d^1$  (one unpaired electron), so it is paramagnetic.

(E) Acidified permanganate ion acts as a strong oxidising agent, so it oxidises (i.e., “reduces” itself while oxidising) oxalate, nitrite and iodide ions.

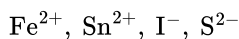
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**Question10**

**Which of the following sets includes all the species that will change the orange colour of  $\text{K}_2\text{Cr}_2\text{O}_7$  in acidic medium?**

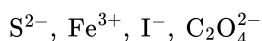
**JEE Main 2026 (Online) 2nd April Morning Shift**

**Options:**

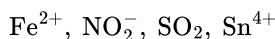
A.



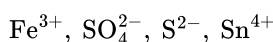
B.



C.



D.

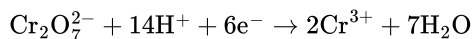


**Answer: A**

**Solution:**

In acidic medium,  $\text{K}_2\text{Cr}_2\text{O}_7$  acts as a strong oxidising agent.

Its orange colour is due to  $\text{Cr}_2\text{O}_7^{2-}$ , which gets reduced to green  $\text{Cr}^{3+}$ :



So, the species that will change the orange colour must be **reducing agents**, because they will reduce dichromate to  $\text{Cr}^{3+}$ .

Now check each option.

**Option A:**  $\text{Fe}^{2+}$ ,  $\text{Sn}^{2+}$ ,  $\text{I}^-$ ,  $\text{S}^{2-}$

- $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$  : reducing agent
- $\text{Sn}^{2+} \rightarrow \text{Sn}^{4+}$  : reducing agent
- $\text{I}^- \rightarrow \text{I}_2$  : reducing agent
- $\text{S}^{2-}$  : strong reducing agent

All these can reduce acidified dichromate, so all will change the orange colour.

**Option B:**  $\text{S}^{2-}$ ,  $\text{Fe}^{3+}$ ,  $\text{I}^-$ ,  $\text{C}_2\text{O}_4^{2-}$

- $\text{S}^{2-}$  : reducing agent
- $\text{Fe}^{3+}$  : not a reducing agent here
- $\text{I}^-$  : reducing agent
- $\text{C}_2\text{O}_4^{2-}$  : reducing agent

Since  $\text{Fe}^{3+}$  will not reduce dichromate, this option is incorrect.

**Option C:**  $\text{Fe}^{2+}$ ,  $\text{NO}_2^-$ ,  $\text{SO}_2$ ,  $\text{Sn}^{4+}$

- $\text{Fe}^{2+}$  : reducing agent
- $\text{NO}_2^-$  : can act as reducing agent
- $\text{SO}_2$  : reducing agent
- $\text{Sn}^{4+}$  : already in higher oxidation state, not a reducing agent

So this option is incorrect.

**Option D:**  $\text{Fe}^{3+}$ ,  $\text{SO}_4^{2-}$ ,  $\text{S}^{2-}$ ,  $\text{Sn}^{4+}$

- $\text{Fe}^{3+}$  : not reducing
- $\text{SO}_4^{2-}$  : not reducing
- $\text{S}^{2-}$  : reducing
- $\text{Sn}^{4+}$  : not reducing

So this is also incorrect.

Therefore, the correct answer is:

|          |
|----------|
| Option A |
|----------|

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## Question11

**Among  $\text{Fe}^{2+}$ ,  $\text{Fe}^{3+}$ ,  $\text{Cr}^{2+}$  and  $\text{Zn}^{2+}$ , the ion that shows positive borax bead test and with highest ionisation enthalpy is :**

### JEE Main 2026 (Online) 2nd April Evening Shift

**Options:**

A.



B.



C.



D.



**Answer: D**

**Solution:**

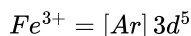
For a **positive borax bead test**, the ion must generally have **unpaired electrons** in its *d*-orbitals, because transition metal ions give characteristic bead colours due to electronic transitions.

Let us check each ion:

- $Fe^{2+} : [Ar] 3d^6 \rightarrow$  has unpaired electrons, so it gives borax bead test.
- $Fe^{3+} : [Ar] 3d^5 \rightarrow$  has unpaired electrons, so it gives borax bead test.
- $Cr^{2+} : [Ar] 3d^4 \rightarrow$  has unpaired electrons, so it gives borax bead test.
- $Zn^{2+} : [Ar] 3d^{10} \rightarrow$  all electrons are paired, so it does **not** give a positive borax bead test.

So, among the given ions, we compare  $Fe^{2+}$ ,  $Fe^{3+}$ ,  $Cr^{2+}$  for **highest ionisation enthalpy**.

Now, higher positive charge and greater effective nuclear charge generally mean higher ionisation enthalpy. Also,  $Fe^{3+}$  has the stable half-filled configuration:



This makes removal of another electron more difficult.

Therefore, the ion that:

- shows **positive borax bead test**, and
- has **highest ionisation enthalpy**

is



So, the correct option is **D.  $Fe^{3+}$** .

-----

## Question 12

Correct statements from the following are

- A. Potassium dichromate is an oxidising agent and it oxidises  $FeSO_4$  to  $Fe_2(SO_4)_3$  in acidic medium.**
- B. Sodium dichromate can be used as primary standard in volumetric estimation.**
- C.  $CrO_4^{2-}$  and  $Cr_2O_7^{2-}$  are interconvertible in aqueous solution by varying the pH of the solution.**
- D. Cr — O — Cr bond angle in  $Cr_2O_7^{2-}$  is  $126^\circ$ .**

Choose the correct answer from the options given below :

**JEE Main 2026 (Online) 4th April Morning Shift**

**Options:**

A.

A, B and C Only

B.

A, C and D Only

C.

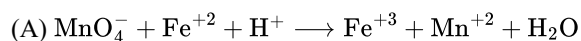
A and C Only

D.

B and D Only

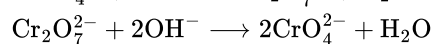
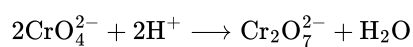
**Answer: B**

**Solution:**

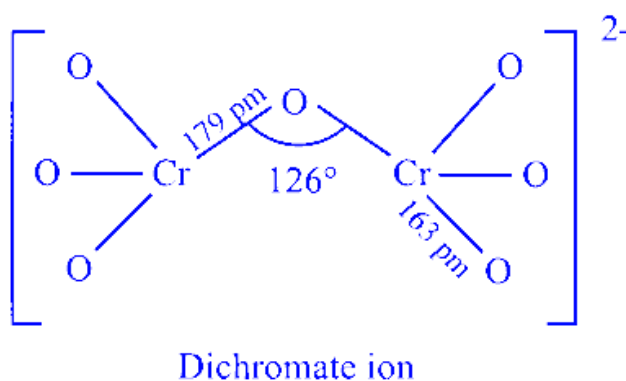


(B)  $\text{Na}_2\text{Cr}_2\text{O}_7$  generally not used as a primary standard in volumetric analysis because it is deliquescent.

(C) In acidic medium  $\text{Cr}_2\text{O}_7^{2-}$  exists and it get converted into  $\text{CrO}_4^{2-}$  in basic medium.



(D)  $\text{K}_2\text{Cr}_2\text{O}_7$  is a good oxidising agent and bond angle of  $\text{Cr} - \text{O} - \text{Cr}$  bond in  $\text{Cr}_2\text{O}_7^{2-}$  is  $126^\circ$



---

## Question13

The correct statements among the following are,

A. Mo(VI) and W(VI) are less stable than Cr(VI).

B.  $\text{Ce}^{4+}$  and  $\text{Tb}^{4+}$  are oxidant while  $\text{Eu}^{2+}$  and  $\text{Yb}^{2+}$  are reductant.

C. Cm and Am have seven unpaired electrons.

D. Actinoid contraction is greater from element to element than lanthanoid contraction.

Choose the correct answer from the options given below :

### Options:

A.

A and B Only

B.

C and D Only

C.

B and D Only

D.

A and C Only

**Answer: C**

### Solution:

#### Statement A

Mo(VI) and W(VI) are less stable than Cr(VI).

This statement is **incorrect**.

In group 6 elements:

- Cr, Mo and W commonly show the +6 oxidation state.
- The higher oxidation states become **more stable** for heavier members of the group.

So,

- Cr(VI) is strongly oxidising and less stable comparatively,
- while Mo(VI) and especially W(VI) are **more stable**.

Hence, **A is false**.

#### Statement B

$\text{Ce}^{4+}$  and  $\text{Tb}^{4+}$  are oxidant while  $\text{Eu}^{2+}$  and  $\text{Yb}^{2+}$  are reductant.

This statement is **correct**.

Reason:

- $\text{Ce}^{4+}$  tends to change to  $\text{Ce}^{3+}$ , so it acts as an **oxidising agent**.
- $\text{Tb}^{4+}$  also tends to reduce to  $\text{Tb}^{3+}$ , so it is an **oxidant**.
- $\text{Eu}^{2+}$  tends to change to  $\text{Eu}^{3+}$ , so it donates electrons and acts as a **reducing agent**.
- $\text{Yb}^{2+}$  similarly tends to go to  $\text{Yb}^{3+}$ , so it is also a **reductant**.

Hence, **B is true**.

#### Statement C

Cm and Am have seven unpaired electrons.

This statement is **incorrect**.

Let us see the electronic configurations:

- Am :  $[Rn] 5f^7 7s^2$
- Cm :  $[Rn] 5f^7 6d^1 7s^2$

Now count unpaired electrons:

- In Am,  $5f^7$  gives **7 unpaired electrons**
- In Cm,  $5f^7$  gives **7 unpaired**, and  $6d^1$  gives **1 more unpaired**

So Cm has **8 unpaired electrons**, not 7.

Hence, **C is false**.

### Statement D

Actinoid contraction is greater from element to element than lanthanoid contraction.

This statement is **correct**.

Reason:

- In actinoids, the  $5f$  electrons shield the nuclear charge even more poorly than the  $4f$  electrons do in lanthanoids.
- Therefore, the decrease in atomic and ionic size across actinoids is **greater**.

Hence, **D is true**.

### Final conclusion

Correct statements are:

$B$  and  $D$  only

So the correct option is:

Option C

## Question 14

Consider  $|x|$  is the difference in oxidation states of Mn in highest manganese fluoride and highest manganese oxide. The ions with  $|x|$  number of unpaired electrons from the following are :

- A.  $\text{Sc}^{3+}$
- B.  $\text{Zn}^{2+}$
- C.  $\text{V}^{2+}$
- D.  $\text{Fe}^{2+}$
- E.  $\text{Co}^{2+}$

Choose the correct answer from the options given below :

## JEE Main 2026 (Online) 4th April Evening Shift

### Options:

A.

A and B Only

B.

C, D and E Only

C.

C and E Only

D.

B and E Only

**Answer: C**

### Solution:

The highest oxidation state of manganese in its **fluoride** is in  $\text{MnF}_4$ , so oxidation state of Mn is +4.

The highest oxidation state of manganese in its **oxide** is in  $\text{Mn}_2\text{O}_7$ , so oxidation state of Mn is +7.

Therefore,

$$|x| = |7 - 4| = 3$$

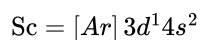
So, we need to find the ions having **3 unpaired electrons**.

Now check each ion one by one.

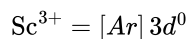
**A.  $\text{Sc}^{3+}$**

Sc has atomic number 21.

Electronic configuration of Sc:



For  $\text{Sc}^{3+}$ , remove 2 electrons from 4s and 1 from 3d:

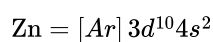


Unpaired electrons = 0

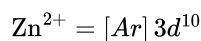
So, this is **not correct**.

**B.  $\text{Zn}^{2+}$**

Zn has atomic number 30.



For  $\text{Zn}^{2+}$ :



Unpaired electrons = 0

So, this is **not correct**.

**C.  $\text{V}^{2+}$**

V has atomic number 23.

$$V = [Ar] 3d^3 4s^2$$

For  $V^{2+}$ :

$$V^{2+} = [Ar] 3d^3$$

In  $3d^3$ , there are 3 unpaired electrons.

So, this is **correct**.

**D.  $Fe^{2+}$**

Fe has atomic number 26.

$$Fe = [Ar] 3d^6 4s^2$$

For  $Fe^{2+}$ :

$$Fe^{2+} = [Ar] 3d^6$$

In  $3d^6$ , number of unpaired electrons = 4.

So, this is **not correct**.

**E.  $Co^{2+}$**

Co has atomic number 27.

$$Co = [Ar] 3d^7 4s^2$$

For  $Co^{2+}$ :

$$Co^{2+} = [Ar] 3d^7$$

In  $3d^7$ , number of unpaired electrons = 3.

So, this is **correct**.

Thus, the correct ions are:

$V^{2+}$  and  $Co^{2+}$

So the correct answer is:

|                        |
|------------------------|
| Option C: C and E Only |
|------------------------|

---

## Question 15

**Pairs of elements with the same number of electrons in their respective 4f orbital are**

**[Atomic number. Eu-63, Gd-64, Dy-66, Ho-67, Tm-69, Yb-70, Lu-71, Hf-72]**

**A. (Eu and Gd)**

**B. (Dy and Ho)**

**C. ( Yb and Hf )**

**D. ( Lu and Tm)**

**Choose the correct answer from the options given below :**

**JEE Main 2026 (Online) 4th April Evening Shift**

**Options:**

A.

B and C Only

B.

A and B Only

C.

A and D Only

D.

A and C Only

**Answer: D**

**Solution:**

We need to compare the number of electrons present in the  $4f$  orbital for each given element.

For lanthanoids, we write the electronic configurations and count the electrons in  $4f$ .

Given elements:

- Eu (63) =  $[Xe] 4f^7 6s^2$
- Gd (64) =  $[Xe] 4f^7 5d^1 6s^2$
- Dy (66) =  $[Xe] 4f^{10} 6s^2$
- Ho (67) =  $[Xe] 4f^{11} 6s^2$
- Tm (69) =  $[Xe] 4f^{13} 6s^2$
- Yb (70) =  $[Xe] 4f^{14} 6s^2$
- Lu (71) =  $[Xe] 4f^{14} 5d^1 6s^2$
- Hf (72) =  $[Xe] 4f^{14} 5d^2 6s^2$

Now check each pair:

**A. Eu and Gd**

- Eu has  $4f^7$
- Gd has  $4f^7$

So, same number of  $4f$  electrons. This is correct.

**B. Dy and Ho**

- Dy has  $4f^{10}$
- Ho has  $4f^{11}$

Not same. This is incorrect.

### C. Yb and Hf

- Yb has  $4f^{14}$
- Hf has  $4f^{14}$

Same number of  $4f$  electrons. This is correct.

### D. Lu and Tm

- Lu has  $4f^{14}$
- Tm has  $4f^{13}$

Not same. This is incorrect.

So the correct pairs are:

A and C

Hence, the correct option is:

|          |
|----------|
| Option D |
|----------|

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## Question 16

**Which of the following is NOT a physical or chemical characteristics of interstitial compounds?**

### JEE Main 2026 (Online) 5th April Morning Shift

#### Options:

A.

They have high melting points, higher than those of pure metals.

B.

They are very soft and ionic in nature.

C.

They retain metallic conductivity.

D.

They are chemically inert and usually non-stoichiometric.

**Answer: B**

#### Solution:

The main physical and chemical properties of interstitial compounds are :

(1) They have very high melting points, even higher than the pure metals from which they are formed.

- (2) They are extremely hard. For example, some borides can be almost as hard as diamond.
- (3) They continue to show metallic conductivity, meaning they can conduct electricity like metals.
- (4) They are chemically inactive (inert) and often do not have a fixed composition, which is called being non-stoichiometric.
- 

## Question 17

A paper dipped in a dil.  $\text{H}_2\text{SO}_4$  solution of 'X' upon treatment with  $\text{SO}_2$  gas turns into green. The compound 'X' is :

### JEE Main 2026 (Online) 5th April Evening Shift

Options:

A. KI – starch

B.

$\text{KMnO}_4$

C.

$\text{Pb}(\text{CH}_3\text{COO})_2$

D.

$\text{K}_2\text{Cr}_2\text{O}_7$

**Answer: D**

### Solution:

The correct answer is **Option D:  $\text{K}_2\text{Cr}_2\text{O}_7$** .

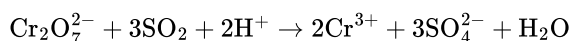
A paper dipped in dilute  $\text{H}_2\text{SO}_4$  solution of X turns **green** when treated with  $\text{SO}_2$  gas.

This means that X must be a substance which changes to **green** on reduction by  $\text{SO}_2$ .

Now,  $\text{SO}_2$  is a **reducing agent**.

Acidified potassium dichromate, that is,  $\text{K}_2\text{Cr}_2\text{O}_7$  in dilute  $\text{H}_2\text{SO}_4$ , is **orange** in colour and gets reduced to  $\text{Cr}^{3+}$ , which is **green**.

The reaction is:



So, the orange dichromate paper turns green.

Now check other options briefly:

- **A: KI-starch**

This is used to detect oxidising agents; it gives blue colour due to iodine-starch complex, not green.

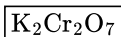
- **B:  $\text{KMnO}_4$**

Acidified  $\text{KMnO}_4$  gets decolourised by  $\text{SO}_2$ , not green.

- **C:  $\text{Pb}(\text{CH}_3\text{COO})_2$**

Lead acetate paper is used for detecting  $\text{H}_2\text{S}$ , giving black colour.

Hence, the compound  $X$  is



## Question 18

The correct statements among the following are :

- A. Basic vanadium oxide is used in the manufacture of  $\text{H}_2\text{SO}_4$ .
- B. The spin-only magnetic moment value of the transition metal halide employed in Ziegler-Natta polymerization is 2.84 BM .
- C. The p-block metal compound employed in Ziegler-Natta polymerization has the metal in +3 oxidation state.
- D. The number of electrons present in the outer most ' d ' orbital of metal halide employed in Wacker process is 8.

Choose the correct answer from the options given below :

### JEE Main 2026 (Online) 6th April Morning Shift

Options:

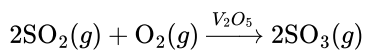
- A.  
A and B Only
- B.  
A, C and D Only
- C.  
C and D Only
- D.  
B, C and D Only

**Answer: C**

**Solution:**

**Statement A:** Basic vanadium oxide is used in the manufacture of  $\text{H}_2\text{SO}_4$ .

1. The manufacture of sulphuric acid ( $\text{H}_2\text{SO}_4$ ) by the Contact Process is a very important reaction.
2. In this process, the key step is the catalytic oxidation of  $\text{SO}_2$  to  $\text{SO}_3$ :



3. The catalyst used here is Vanadium pentoxide,  $\text{V}_2\text{O}_5$ .

4. Now, we need to check the nature of this oxide. Let's find the oxidation state of Vanadium (V) in  $\text{V}_2\text{O}_5$ .

Let the oxidation state of V be  $x$ .

$$2(x) + 5(-2) = 0$$

$$2x - 10 = 0$$

$$x = +5$$

5. As a general rule from the d-block elements chapter, metal oxides in higher oxidation states are acidic or amphoteric in nature. Since Vanadium is in its highest oxidation state (+5),  $\text{V}_2\text{O}_5$  is amphoteric, meaning it reacts with both acids and bases. It is not a basic oxide.

So, statement A is **incorrect**.

**Statement B: The spin-only magnetic moment value of the transition metal halide employed in Ziegler-Natta polymerization is 2.84 BM.**

1. Ziegler-Natta catalysts are used for the polymerization of alkenes, like converting ethene to polyethene.

2. A typical Ziegler-Natta catalyst is a mixture of titanium tetrachloride ( $\text{TiCl}_4$ ) and triethylaluminium ( $\text{Al}(\text{C}_2\text{H}_5)_3$ ).

3. The transition metal halide here is  $\text{TiCl}_4$ . Let's find the magnetic moment for the metal ion in this compound.

4. In  $\text{TiCl}_4$ , the oxidation state of Titanium (Ti) is +4.

5. The atomic number of Ti is 22. Its electronic configuration is  $[\text{Ar}]3d^24s^2$ .

6. The electronic configuration of the  $\text{Ti}^{4+}$  ion is  $[\text{Ar}]3d^0$ .

7. The number of unpaired electrons ( $n$ ) is 0.

8. The spin-only magnetic moment ( $\mu$ ) is calculated using the formula:

$$\mu = \sqrt{n(n+2)} \text{ B.M.}$$

For  $\text{Ti}^{4+}$ , since  $n = 0$ ,

$$\mu = \sqrt{0(0+2)} = 0 \text{ B.M.}$$

9. The value given in the statement is 2.84 B.M. A value of  $\mu \approx 2.84$  B.M. corresponds to 2 unpaired electrons ( $n = 2$ ), because  $\sqrt{2(2+2)} = \sqrt{8} \approx 2.83$  B.M.

10. Since the magnetic moment of Ti in  $\text{TiCl}_4$  is 0 B.M., this statement does not hold for the common example of a Ziegler-Natta catalyst.

So, statement B is **incorrect**.

**Statement C: The p-block metal compound employed in Ziegler-Natta polymerization has the metal in +3 oxidation state.**

1. As we just discussed, the Ziegler-Natta catalyst contains a compound like triethylaluminium,  $\text{Al}(\text{C}_2\text{H}_5)_3$ .

2. Aluminium (Al) belongs to Group 13, which is in the p-block of the periodic table.

3. Let's find the oxidation state of Al in  $\text{Al}(\text{C}_2\text{H}_5)_3$ . The ethyl group ( $\text{C}_2\text{H}_5$ ) is an alkyl group and carries a charge of -1 when bonded to a metal.

Let the oxidation state of Al be  $x$ .

$$x + 3(-1) = 0$$

$$x = +3$$

4. So, the p-block metal (Al) is indeed in the +3 oxidation state.

Therefore, statement C is **correct**.

**Statement D: The number of electrons present in the outer most 'd' orbital of metal halide employed in Wacker process is 8.**

1. The Wacker process is used for the oxidation of ethene to ethanal (acetaldehyde) using a catalyst.
2. The catalyst used is Palladium(II) chloride,  $PdCl_2$ , in an aqueous medium.
3. The metal halide here is  $PdCl_2$ .
4. In  $PdCl_2$ , the oxidation state of Palladium (Pd) is +2.
5. Palladium (Pd) has atomic number 46. Its electronic configuration is  $[Kr]4d^{10}5s^0$ .
6. To get the configuration of the  $Pd^{2+}$  ion, we remove two electrons. The configuration becomes  $[Kr]4d^8$ .
7. The outermost 'd' orbital for Palladium is the 4d orbital. The number of electrons in this 4d orbital is 8.

Therefore, statement D is **correct**.

**Conclusion:**

We found that statements C and D are correct, while A and B are incorrect.

Looking at the options, we need to choose the one that says "C and D Only".

The correct answer is **Option C**.

## Question 19

**The correct order of first ( $\Delta_i H_1$ ) and second ( $\Delta_i H_2$ ) ionisation enthalpy values of Cr and Mn are :**

A.  $\Delta_i H_1$  : Cr > Mn

B.  $\Delta_i H_2$  : Cr > Mn

C.  $\Delta_i H_1$  : Mn > Cr

D.  $\Delta_i H_2$  : Mn > Cr

**Choose the correct answer from the options given below :**

### JEE Main 2026 (Online) 6th April Evening Shift

**Options:**

A.

A and B only

B.

B and C only

C.

A and D only

D.

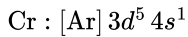
C and D only

**Answer: B**

## Solution:

- **Cr (Z = 24):**

Ground state configuration:



(due to extra stability of half-filled  $3d^5$  and  $4s^1$  configuration)

- **Mn (Z = 25):**

Ground state configuration:



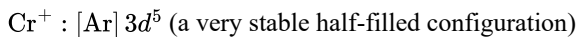
## First ionisation enthalpy, $\Delta_i H_1$

In the first ionisation process:

- **Cr:**



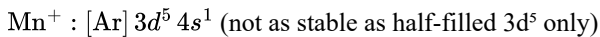
After removal of one 4s electron:



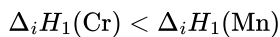
- **Mn:**



After removal of one 4s electron:

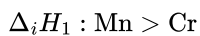


Because the resulting  $\text{Cr}^+$  ion is particularly stable, **less energy is required to form  $\text{Cr}^+$** , i.e.



Thus,

**Correct order for first ionisation enthalpy:**



(Corresponds to option C.)

## Second ionisation enthalpy, $\Delta_i H_2$

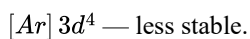
Now, let us remove the second electron.

- **Cr:**

From the already stable  $\text{Cr}^+$  ( $[\text{Ar}] 3d^5$ ):



After this,  $\text{Cr}^{2+}$  becomes:



Therefore, it requires **a large amount of energy** to remove this second electron.

- **Mn:**

From  $\text{Mn}^+$  ( $[\text{Ar}] 3d^5 4s^1$ ):



After this,  $\text{Mn}^{2+}$  becomes:

$[\text{Ar}] 3d^5$  — a stable half-filled configuration.

Thus, **less energy** is needed for the second ionisation.

Hence,

$$\Delta_i H_2(\text{Cr}) > \Delta_i H_2(\text{Mn})$$

which corresponds to option **B**.

### Combine results

- $\Delta_i H_1$ :  $\text{Mn} > \text{Cr}$  → option **C**
- $\Delta_i H_2$ :  $\text{Cr} > \text{Mn}$  → option **B**

Therefore, the correct combination is:

|              |
|--------------|
| B and C only |
|--------------|

**Final Answer: Option B**

-----

## Question20

**Given below are two statements for catalytic properties of transition metals.**

**Statement I : First row transition metals which act as catalyst utilise their 3d electrons only for formation of bonds between reactant molecules and atoms on the surface of catalyst.**

**Statement II : There is increase in the concentration of reactants on the surface of catalyst which strengthens the bonds in reacting molecules.**

**In the light of the above statements, choose the correct answer from the options given below :**

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**Options:**

A.

Both Statement I and Statement II are correct

B.

Both Statement I and Statement II are incorrect

C.

Statement I is correct but Statement II is incorrect

D.

Statement I is incorrect but Statement II is correct

**Answer: B**

### Solution:

The first-row transition metals use both their 3d and 4s electrons to form bonds with reactant molecules on the surface of the catalyst.

∴ Statement I is incorrect.

When reactants are adsorbed on the catalytic surface, their concentration on the surface increases, but this adsorption actually weakens the bonds within the reactant molecules. This helps in forming an activated complex and reduces the activation energy of the reaction.

∴ Statement II is incorrect.

-----

## Question21

**Given below are two statements :**

**Statement I :** The number of pairs among  $[\text{Ti}^{4+}, \text{V}^{2+}]$ ,  $[\text{V}^{2+}, \text{Mn}^{2+}]$ ,  $[\text{Mn}^{2+}, \text{Fe}^{3+}]$  and  $[\text{V}^{2+}, \text{Cr}^{2+}]$  in which both ions are coloured is 3 .

**Statement II :** The number of pairs among  $[\text{La}^{3+}, \text{Yb}^{2+}]$ ,  $[\text{Lu}^{3+}, \text{Ce}^{4+}]$  and  $[\text{Ac}^{3+}, \text{Lr}^{3+}]$  ions in which both are diamagnetic is 3 .

**In the light of the above statements, choose the correct from the options given below :**

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**Options:**

A.

Both Statement I and Statement II are correct

B.

Both Statement I and Statement II are incorrect

C.

Statement I is correct but Statement II is incorrect

D.

Statement I is incorrect but Statement II is correct

**Answer: A**

### Solution:

Coloured ions have unpaired electron.

(i)  $Ti^{+4} = 4s^0 3d^0$   
 $n = 0$

(ii)  $V^{+2} = 4s^0 3d^3$



$n = 3$

(iii)  $Mn^{+2} = 4s^0 3d^5$



$n = 5$

(iv)  $Fe^{+3} = 4s^0 3d^5$



$n = 5$

(v)  $Cr^{+2} = 4s^0 3d^4$



$n = 4$

The absence of unpaired electron makes the element diamagnetic.

$La^{+3} = [Xe] 5d^0 6s^0$

$4b^{+2} = [Xe] 4f^{14} 6s^0$

$Lu^{+3} = [Xe] 4f^{14} 5d^0 6s^0$

$Ce^{+4} = [Xe] 4f^0 5d^0 6s^0$

$Ac^{+3} = [Rn] 6d^0 7s^0$

$L_n^{+3} = [Rn] 5f^{14} 6d^0 7s^0$

Ans. (A)

## Question 22

**Niobium (Nb) and ruthenium (Ru) have " x " and " y " number of electrons in their respective 4 d orbitals. The value of  $x + y$  is \_\_\_\_\_.**

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**Answer: 11**

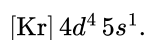
## Solution:

We need to determine the number of electrons each element has in its **4d orbitals** and then add them together.

---

### 1. Niobium (Nb, $Z = 41$ )

The ground-state electronic configuration of niobium is:

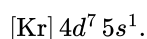


Hence, **Niobium has 4 electrons in its 4d orbitals.**

---

### 2. Ruthenium (Ru, $Z = 44$ )

The ground-state electronic configuration of ruthenium is:



Hence, **Ruthenium has 7 electrons in its 4d orbitals.**

---

### 3. Sum of 4d Electrons

$$x = 4 \quad (\text{for Nb}), \quad y = 7 \quad (\text{for Ru}).$$

Therefore,

$$x + y = 4 + 7 = 11.$$

---

**Answer: 11**

-----

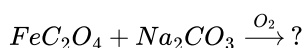
## Question23

The molar mass of the water insoluble product formed from the fusion of chromite ore ( $\text{FeCr}_2\text{O}_4$ ) with  $\text{Na}_2\text{CO}_3$  in presence of  $\text{O}_2$  is \_\_\_\_\_  $\text{g mol}^{-1}$ .

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**Answer: 160**

## Solution:



The chemical reaction can be written as



The insoluble product is  $\text{Fe}_2\text{O}_3$ .

Molar mass of  $\text{Fe}_2\text{O}_3$

Fe (Atomic mass)  $\rightarrow 55.845$  amu

Molar mass =  $55.845 \text{ g mol}^{-1}$

O (Atomic mass)  $\rightarrow 15.999$  amu

Molar mass =  $15.999 \text{ g mol}^{-1}$

Molar mass of  $\text{Fe}_2\text{O}_3 = 55.845 \text{ g mol}^{-1} \times 2 + 15.999 \text{ g mol}^{-1} \times 3$

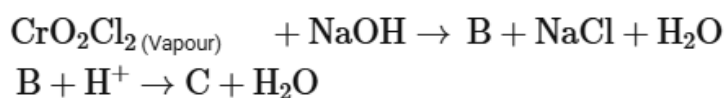
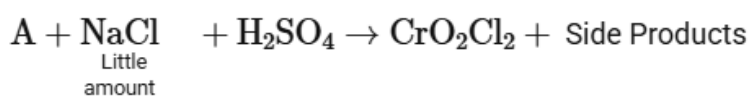
$= (111.69 + 47.997) \text{ g mol}^{-1}$

$= 159.687 \text{ g mol}^{-1} \approx 160 \text{ g mol}^{-1}$

---

## Question 24

Consider the following reactions

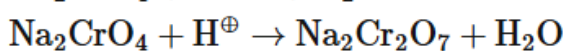
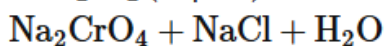
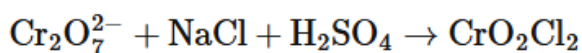


The number of terminal 'O' present in the compound 'C' is \_\_\_\_\_

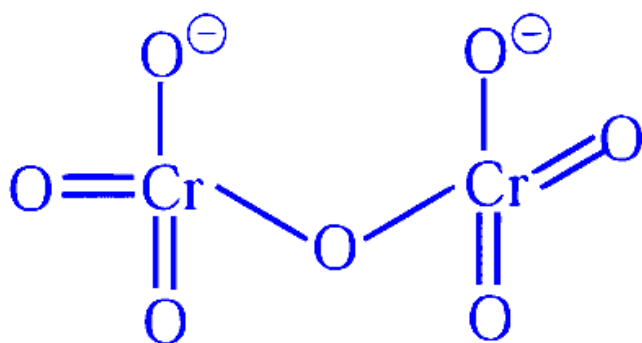
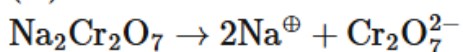
**JEE Main 2025 (Online) 3rd April Morning Shift**

**Answer: 6**

**Solution:**



(C)



No of terminal "O" = 6

---

## Question25

Among, Sc, Mn, Co and Cu , identify the element with highest enthalpy of atomisation. The spin only magnetic moment value of that element in its +2 oxidation state is \_\_\_\_\_BM (in nearest integer).

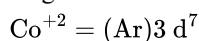
JEE Main 2025 (Online) 3rd April Evening Shift

Answer: 4

Solution:

|                                   | Sc  | Mn  | Co  | Cu  |
|-----------------------------------|-----|-----|-----|-----|
| Enthalpy of Atomisation (kJ/mole) | 326 | 281 | 425 | 339 |

Highest Co



$$n = 3$$

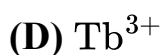
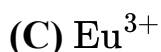
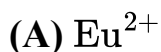
$$\mu = \sqrt{15} = 3.87$$

$$\text{Nearest integer} = 4$$

---

## Question26

Lanthanoid ions with  $4f^7$  configuration are :



Choose the correct answer from the options given below :

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**Options:**

- A. (B) and (E) only
- B. (B) and (C) only
- C. (A) and (D) only
- D. (A) and (B) only

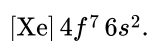
**Answer: D**

**Solution:**

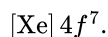
Let's analyze the electron configurations of the given lanthanide ions.

**Europium (Eu)**

The neutral atom of Eu (atomic number 63) typically has the configuration:

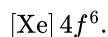


For  $\text{Eu}^{2+}$ , two electrons are removed, usually from the  $6s$  orbital, resulting in:



Thus,  $\text{Eu}^{2+}$  has a  $4f^7$  configuration.

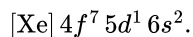
For  $\text{Eu}^{3+}$ , three electrons are removed (the two  $6s$  electrons and one  $4f$  electron), giving:



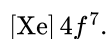
Therefore,  $\text{Eu}^{3+}$  does not have a  $4f^7$  configuration.

**Gadolinium (Gd)**

The neutral atom of Gd (atomic number 64) is usually represented as:



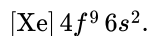
For  $\text{Gd}^{3+}$ , three electrons are removed (typically the  $5d^1$  and the two  $6s$  electrons), resulting in:



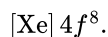
So,  $\text{Gd}^{3+}$  has a  $4f^7$  configuration.

**Terbium (Tb)**

The neutral atom of Tb (atomic number 65) typically has the configuration:



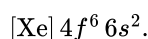
For  $\text{Tb}^{3+}$ , removal of three electrons gives:



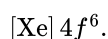
Hence,  $\text{Tb}^{3+}$  does not have a  $4f^7$  configuration.

**Samarium (Sm)**

The neutral atom of Sm (atomic number 62) generally has:



For  $\text{Sm}^{2+}$ , after removal of two electrons (likely from the  $6s$  orbital), the configuration is:



Therefore,  $\text{Sm}^{2+}$  does not have a  $4f^7$  configuration.

Thus, the ions with a  $4f^7$  configuration are:

(A)  $\text{Eu}^{2+}$

(B)  $\text{Gd}^{3+}$

The correct answer is:

**Option D – (A) and (B) only.**

-----

## Question27

**The correct set of ions (aqueous solution) with same colour from the following is:**

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**Options:**

A.  $\text{V}^{2+}$ ,  $\text{Cr}^{3+}$ ,  $\text{Mn}^{3+}$

B.  $\text{Ti}^{4+}$ ,  $\text{V}^{4+}$ ,  $\text{Mn}^{2+}$

C.  $\text{Sc}^{3+}$ ,  $\text{Ti}^{3+}$ ,  $\text{Cr}^{2+}$

D.  $\text{Zn}^{2+}$ ,  $\text{V}^{3+}$ ,  $\text{Fe}^{3+}$

**Answer: A**

**Solution:**

(1)  $\text{V}^{2+}$  (Violet),  $\text{Cr}^{3+}$  (Violet),  $\text{Mn}^{3+}$  (Violet)

(2)  $\text{Zn}^{2+}$  (Colourless),  $\text{V}^{3+}$  (Green),  $\text{Fe}^{3+}$  (Yellow)

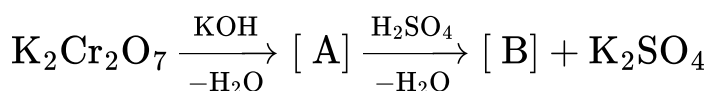
(3)  $\text{Ti}^{4+}$  (Colourless),  $\text{V}^{4+}$  (Blue),  $\text{Mn}^{2+}$  (Pink)

(4)  $\text{Sc}^{3+}$  (Colourless),  $\text{Ti}^{3+}$  (Purple),  $\text{Cr}^{2+}$  (Blue)

-----

## Question28

**Consider the following reactions**



**The products  $[\text{A}]$  and  $[\text{B}]$ , respectively are :**

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**Options:**

A.  $\text{K}_2\text{Cr}(\text{OH})_6$  and  $\text{Cr}_2\text{O}_3$

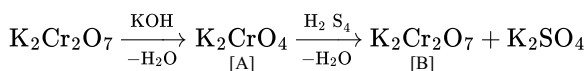
B.  $\text{K}_2\text{CrO}_4$  and  $\text{Cr}_2\text{O}_3$

C.  $\text{K}_2\text{CrO}_4$  and  $\text{K}_2\text{Cr}_2\text{O}_7$

D.  $\text{K}_2\text{CrO}_4$  and  $\text{CrO}$

**Answer: C**

**Solution:**



## Question29

**Match List - I with List - II**

|     | List - I        |       | List - II     |
|-----|-----------------|-------|---------------|
| (A) | Bronze          | (I)   | Cu, Ni        |
| (B) | Brass           | (II)  | Fe, Cr, Ni, C |
| (C) | UK silver coin  | (III) | Cu, Zn        |
| (D) | Stainless steel | (IV)  | Cu, Sn        |

**Choose the correct answer from the options given below :**

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**Options:**

A. (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

B. (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

C. (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

D. (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

**Answer: B**

**Solution:**

To match the items from List I to List II, we need to understand the composition of each material:

**Bronze** is an alloy composed of copper (Cu) and tin (Sn).

**Brass** is an alloy made of copper (Cu) and zinc (Zn).

The **UK silver coin** primarily consists of copper (Cu) and nickel (Ni).

**Stainless steel** is an alloy that includes iron (Fe), chromium (Cr), nickel (Ni), and carbon (C).

Based on these definitions, the correct matches are:

(A) Bronze corresponds to (IV) Cu, Sn

(B) Brass corresponds to (III) Cu, Zn

(C) UK silver coin corresponds to (I) Cu, Ni

(D) Stainless steel corresponds to (II) Fe, Cr, Ni, C

These pairings illustrate the composition of each material and ensure accurate alignment between List I and List II.

-----

## Question30

**Which of the following ions is the strongest oxidizing agent? (Atomic Number of Ce = 58, Eu = 63, Tb = 65, Lu = 71)**

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**Options:**

A.  $\text{Lu}^{3+}$

B.  $\text{Eu}^{2+}$

C.  $\text{Tb}^{4+}$

D.  $\text{Ce}^{3+}$

**Answer: C**

**Solution:**

$\text{Tb}^{4+}$  is strongest oxidising agent

-----

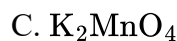
## Question31

**Preparation of potassium permanganate from  $\text{MnO}_2$  involves two step process in which the 1st step is a reaction with KOH and  $\text{KNO}_3$  to produce**

**JEE Main 2025 (Online) 24th January Morning Shift**

**Options:**

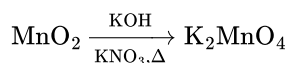
A.  $\text{K}_3\text{MnO}_4$



**Answer: C**

**Solution:**

The preparation of potassium permanganate ( $\text{KMnO}_4$ ) from manganese dioxide ( $\text{MnO}_2$ ) involves a two-step process. In the first step,  $\text{MnO}_2$  reacts with potassium hydroxide ( $\text{KOH}$ ) and potassium nitrate ( $\text{KNO}_3$ ) under heating. This reaction produces potassium manganate ( $\text{K}_2\text{MnO}_4$ ) as an intermediate product. The reaction can be depicted as follows:



## Question 32

**Match List - I with List - II.**

|     | List - I<br>(Transition metal ion) |       | List - II<br>(Spin only magnetic moment (B.M.)) |
|-----|------------------------------------|-------|-------------------------------------------------|
| (A) | $\text{Ti}^{3+}$                   | (I)   | 3.87                                            |
| (B) | $\text{V}^{2+}$                    | (II)  | 0.00                                            |
| (C) | $\text{Ni}^{2+}$                   | (III) | 1.73                                            |
| (D) | $\text{Sc}^{3+}$                   | (IV)  | 2.84                                            |

**Choose the correct answer from the options given below :**

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**Options:**

A. (A) – (III), (B) – (I), (C) – (IV), (D) – (II)

B. (A) – (II), (B) – (IV), (C) – (I), (D) – (III)

C. (A) – (III), (B) – (I), (C) – (II), (D) – (IV)

D. (A) – (IV), (B) – (II), (C) – (III), (D) – (I)

**Answer: A**

**Solution:**

$$\text{Sc}^{+3} = 3 \text{ d}^0 \quad \therefore \mu_{\text{spin}} = 0$$

$$\text{V}^{+2} = 3 \text{ d}^3 \quad \therefore \mu_{\text{spin}} = 3.87 \text{ B.M.}$$

$$\text{Ni}^{+2} = 3 \text{ d}^8 \quad \therefore \mu_{\text{spin}} = 2.84 \text{ B.M.}$$

$$\text{Ti}^{+3} = 3 d^1 \quad \therefore \mu_{\text{spin}} = 1.73 \text{ B.M}$$

---

## Question33

The metal ion whose electronic configuration is not affected by the nature of the ligand and which gives a violet colour in non-luminous flame under hot condition in borax bead test is

**JEE Main 2025 (Online) 28th January Morning Shift**

**Options:**

A.  $\text{Cr}^{3+}$

B.  $\text{Ti}^{3+}$

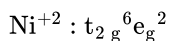
C.  $\text{Ni}^{2+}$

D.  $\text{Mn}^{2+}$

**Answer: C**

**Solution:**

$\text{Ni}^{+2}$  gives violet coloured bead in non-luminous flame under hot conditions.  $\text{Ni}^{+2}$  has  $d^8$  configuration which does not depend on nature of ligand present in octahedral complex.



## Question34

The amphoteric oxide among  $\text{V}_2\text{O}_3$ ,  $\text{V}_2\text{O}_4$  and  $\text{V}_2\text{O}_5$ , upon reaction with alkali leads to formation of an oxide anion. The oxidation state of V in the oxide anion is :

**JEE Main 2025 (Online) 28th January Evening Shift**

**Options:**

A.

+ 4

B.

+ 5

C.

+ 3

D.

+ 7

**Answer: B**

**Solution:**



In  $\text{VO}_4^{3-}$  ion, vanadium is in + 5 oxidation state

---

## Question 35

The correct option with order of melting points of the pairs (Mn, Fe), (Tc, Ru) and (Re, Os) is :

**JEE Main 2025 (Online) 29th January Morning Shift**

**Options:**

A.

$\text{Fe} < \text{Mn}$ ,  $\text{Ru} < \text{Tc}$  and  $\text{Re} < \text{Os}$

B.

$\text{Fe} < \text{Mn}$ ,  $\text{Ru} < \text{Tc}$  and  $\text{Os} < \text{Re}$

C.

$\text{Mn} < \text{Fe}$ ,  $\text{Tc} < \text{Ru}$  and  $\text{Re} < \text{Os}$

D.

$\text{Mn} < \text{Fe}$ ,  $\text{Tc} < \text{Ru}$  and  $\text{Os} < \text{Re}$

**Answer: D**

**Solution:**

Melting point of the pairs  $(M_n, F_e)$ ,  $(T_c, R_u)$ , and  $(R_e, O_s)$  :

$M_n, T_c, R_e$  are same group elements.

Also,  $F_e, R_u, O_s$ , are same group elements.

$M_n, F_e$

Fe has a higher melting point than  $M_n$ ,  $M_n < F_e$

$$\begin{bmatrix} M_n & F_e \\ T_c & R_u \\ R_e & O_s \end{bmatrix}$$

"Melting point property is directly proportional to the strength of metallic bond, which is proportional to the delocalized electrons. The more electrons are free to move around, the stronger will be the metallic bond."

For Mn and Fe,  $(\text{Mn} - d^5s^2, \text{Fe} - d^6s^2)$ .

$M_n$  has a weaker metallic bond due to its electronic configuration (half filled configuration). The half filled configuration (stable) results in less delocalization of electrons and weaker attraction between atoms, requiring less energy to break apart and melt. But, Fe has localized electrons and hence stronger metallic bonds, leading to higher melting point.

$T_c, Ru$

Tc has lower melting point than Ru,  $T_c < Ru$ . This is because of the electronic configuration of its  $d$ -orbitals.

$Tc - d^5 s^2, Ru - d^7 s^1$

Tc has half filled  $d$ -orbitals, which makes its electronic configuration stable. This configuration causes the nucleus to hold the electrons tightly, which weakens the metallic bond and it results in lower melting point.

Re, Os

Re has a higher melting point than Os,  $Os < Re$  ( $Re - d^5 s^2, Os - ds_s^6$ )

The higher melting point of Re is due to the strength of atomic bonding.

Re forms stronger metallic bonds than Os. Due to the larger size of Re, its valence electrons are not tightly held by the nucleus and hence, the valence electrons cause stronger metallic bond. As a result, melting point increases.

Osmium forms weaker metallic bonds than Re due to its crystal arrangement and electronic structure.

Comparing Re, and Os,

$R_e$  is slightly smaller in atomic radius which allows atoms to pack more efficiently than is and has stronger bonds and higher melting point.

So,

$M_n < Fe$

$T_c < Ru$

$O_s < Re$

Correct answer:

Option 4)  $Mn < Fe, TC < Ru$  and  $Os < Re$ .

## Question 36

The metal ions that have the calculated spin-only magnetic moment value of 4.9 B.M. are :

A.  $Cr^{2+}$

B.  $Fe^{2+}$

C.  $Fe^{3+}$

D.  $Co^{2+}$

E.  $Mn^{3+}$

Choose the correct answer from the options given below:

**Options:**

- A. A, D and E Only
- B. A, B and E Only
- C. A, C and E Only
- D. B and E Only

**Answer: B**

**Solution:**

To determine which metal ions have a calculated spin-only magnetic moment of 4.9 Bohr Magnetons (B.M.), we can use the formula:

$$\text{Magnetic Moment (M.M)} = \sqrt{n(n+2)} \text{ B.M.}$$

where  $n$  is the number of unpaired electrons.

Given that the magnetic moment is 4.9 B.M., we can set up the equation:

$$4.9 = \sqrt{n(n+2)}$$

Solving this equation, we find that  $n = 4$ .

Let's analyze each metal ion:

- (A)  $_{24}\text{Cr}^{2+}$ : Electronic configuration:  $[\text{Ar}]3d^4$ . This ion has 4 unpaired electrons.
- (B)  $_{26}\text{Fe}^{2+}$ : Electronic configuration:  $[\text{Ar}]3d^6$ . This ion also has 4 unpaired electrons.
- (C)  $_{26}\text{Fe}^{3+}$ : Electronic configuration:  $[\text{Ar}]3d^5$ . This ion has 5 unpaired electrons, which does not match the required  $n$  value of 4.
- (D)  $_{27}\text{Co}^{2+}$ : Electronic configuration:  $[\text{Ar}]3d^7$ . This ion has 3 unpaired electrons.
- (E)  $_{25}\text{Mn}^{3+}$ : Electronic configuration:  $[\text{Ar}]3d^4$ . This ion has 4 unpaired electrons.

Based on the number of unpaired electrons, the metal ions with a spin-only magnetic moment of approximately 4.9 B.M. are  $\text{Cr}^{2+}$ ,  $\text{Fe}^{2+}$ , and  $\text{Mn}^{3+}$ .

-----

## Question37

**Given below are two statements:**

**Statement I :  $\text{CrO}_3$  is a stronger oxidizing agent than  $\text{MoO}_3$**

**Statement II :  $\text{Cr(VI)}$  is more stable than  $\text{Mo(VI)}$**

**In the light of the above statements, choose the correct answer from the options given below**

**JEE Main 2025 (Online) 3rd April Evening Shift**

**Options:**

- A. Statement I is true but Statement II is false
- B. Both Statement I and Statement II are true
- C. Both Statement I and Statement II are false
- D. Statement I is false but Statement II is true

**Answer: A**

### Solution:

Statement I is true while Statement II is false.

Cr(VI) is less stable than Mo(VI). Consequently,  $\text{CrO}_3$  more readily reduces to  $\text{Cr}^{3+}$  compared to  $\text{MoO}_3$ , exhibiting a stronger oxidizing nature.

## Question38

**Pair of transition metal ions having the same number of unpaired electrons is**

**JEE Main 2025 (Online) 4th April Morning Shift**

**Options:**

- A.  $\text{Ti}^{3+}$ ,  $\text{Mn}^{2+}$
- B.  $\text{Ti}^{2+}$ ,  $\text{Co}^{2+}$
- C.  $\text{Fe}^{3+}$ ,  $\text{Cr}^{2+}$
- D.  $\text{V}^{2+}$ ,  $\text{Co}^{2+}$

**Answer: D**

### Solution:

|     |                  |               | Configuration          | No. of unpaired $e^-$ |
|-----|------------------|---------------|------------------------|-----------------------|
| (1) | $\text{V}^{3+}$  | $\Rightarrow$ | $[\text{Ar}]3d^3 4s^0$ | 3                     |
|     | $\text{Co}^{2+}$ | $\Rightarrow$ | $[\text{Ar}]3d^7 4s^0$ | 3                     |
| (2) | $\text{Ti}^{2+}$ | $\Rightarrow$ | $[\text{Ar}]3d^2 4s^0$ | 2                     |
|     | $\text{Co}^{2+}$ | $\Rightarrow$ | $[\text{Ar}]3d^7 4s^0$ | 3                     |
| (3) | $\text{Fe}^{3+}$ | $\Rightarrow$ | $[\text{Ar}]3d^5 4s^0$ | 5                     |
|     | $\text{Cr}^{2+}$ | $\Rightarrow$ | $[\text{Ar}]3d^4 4s^0$ | 4                     |
| (4) | $\text{Ti}^{3+}$ | $\Rightarrow$ | $[\text{Ar}]3d^1 4s^0$ | 1                     |

|  |                  |   |                                      |                                |
|--|------------------|---|--------------------------------------|--------------------------------|
|  |                  |   | Configuration                        | No. of unpaired e <sup>-</sup> |
|  | Mn <sup>2+</sup> | ⇒ | [Ar] 3d <sup>5</sup> 4s <sup>0</sup> | 5                              |

So V<sup>2+</sup> & Co<sup>2+</sup> same number of unpaired electron.

## Question 39

The first transition series metal 'M' has the highest enthalpy of atomisation in its series. One of its aquated ion (M<sup>n+</sup>) exists in green colour. The nature of the oxide formed by the above M<sup>n+</sup> ion is:

### JEE Main 2025 (Online) 7th April Morning Shift

Options:

- A. basic
- B. neutral
- C. amphoteric
- D. acidic

**Answer: A**

**Solution:**

In the 3d transition series, Vanadium (V) has the highest enthalpy of atomization. The ion V<sup>3+</sup> is known to exhibit a green color in its aqueous form. The oxide formed by the V<sup>3+</sup> ion is V<sub>2</sub>O<sub>3</sub>, which is classified as a basic oxide.

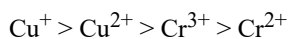
## Question 40

The correct decreasing order of spin only magnetic moment values (BM) of Cu<sup>+</sup>, Cu<sup>2+</sup>, Cr<sup>2+</sup> and Cr<sup>3+</sup> ions is :

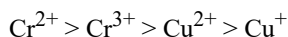
### JEE Main 2025 (Online) 8th April Evening Shift

Options:

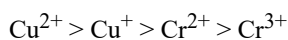
- A.
- Cr<sup>3+</sup> > Cr<sup>2+</sup> > Cu<sup>+</sup> > Cu<sup>2+</sup>
- B.



C.



D.



**Answer: C**

**Solution:**

$\text{Cu}^+ : [\text{Ar}]3d^{10}$ , Spin only magnetic moment = 0 B.M.

$\text{Cu}^{+2} : [\text{Ar}]3d^9$ , Spin only magnetic moment =  $\sqrt{3}$  B.M.

$\text{Cr}^{+2} : [\text{Ar}]3d^4$ , Spin only magnetic moment =  $\sqrt{24}$  B.M.

$\text{Cr}^{+3} : [\text{Ar}]3d^3$ , Spin only magnetic moment =  $\sqrt{15}$  B.M.

Order of  $\mu$  :  $\text{Cr}^{+2} > \text{Cr}^{+3} > \text{Cu}^{+2} > \text{Cu}^+$

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## Question41

Given below are two statements :

**Statement (I) : The 4f and 5f - series of elements are placed separately in the Periodic table to preserve the principle of classification.**

**Statement (II) : s-block elements can be found in pure form in nature. In the light of the above statements, choose the most appropriate answer from the options given below :**

**[27-Jan-2024 Shift 1]**

**Options:**

A. Statement I is false but Statement II is true

B. Both Statement I and Statement II are true

C. Statement I is true but Statement II is false

D. Both Statement I and Statement II are false

**Answer: C**

**Solution:**

s-block elements are highly reactive and found in combined state.

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## Question42

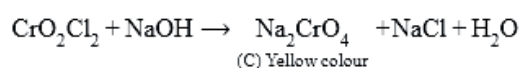
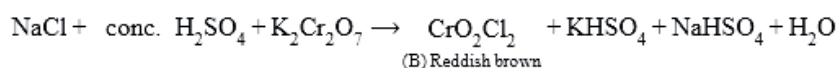
**NaCl reacts with conc.  $\text{H}_2\text{SO}_4$  and  $\text{K}_2\text{Cr}_2\text{O}_7$  to give reddish fumes (B), which react with NaOH to give yellow solution (C). (B) and (C) respectively are;**  
**[27-Jan-2024 Shift 1]**

**Options:**

- A.  $\text{CrO}_2\text{Cl}_2$ ,  $\text{Na}_2\text{CrO}_4$
- B.  $\text{Na}_2\text{CrO}_4$ ,  $\text{CrO}_2\text{Cl}_2$
- C.  $\text{CrO}_2\text{Cl}_2$ ,  $\text{KHSO}_4$
- D.  $\text{CrO}_2\text{Cl}_2$ ,  $\text{Na}_2\text{Cr}_2\text{O}_7$

**Answer: A**

**Solution:**



---

## Question43

**Given below are two statements:**

**Statement (I) : In the Lanthanoids, the formation of  $\text{Ce}^{+4}$  is favoured by its noble gas configuration.**

**Statement (II) :  $\text{Ce}^{+4}$  is a strong oxidant reverting to the common +3 state.**

**In the light of the above statements, choose the most appropriate answer from the options given below:**

**[27-Jan-2024 Shift 2]**

**Options:**

- A. Statement I is false but Statement II is true
- B. Both Statement I and Statement II are true
- C. Statement I is true but Statement II is false
- D. Both Statement I and Statement II are false

**Answer: B**

**Solution:**

Statement (1) is true,  $\text{Ce}^{+4}$  has noble gas electronic configuration.

Statement (2) is also true due to high reduction potential for  $\text{Ce}^{4+}/\text{Ce}^{3+}$  (+1.74V), and stability of  $\text{Ce}^{3+}$ ,  $\text{Ce}^{4+}$  acts as strong

oxidizing agent.

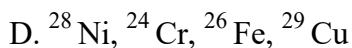
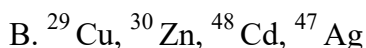
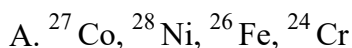
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## Question44

Choose the correct option having all the elements with  $d^{10}$  electronic configuration from the following:

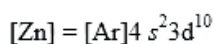
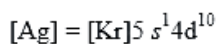
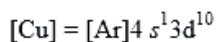
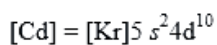
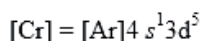
[27-Jan-2024 Shift 2]

Options:



Answer: B

Solution:

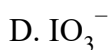
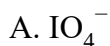


## Question45

In alkaline medium.  $\text{MnO}_4^-$  oxidises  $\text{I}^-$  to

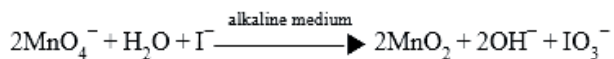
[29-Jan-2024 Shift 1]

Options:



Answer: D

Solution:



## Question46

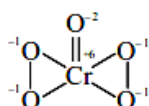
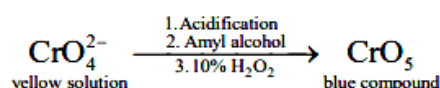
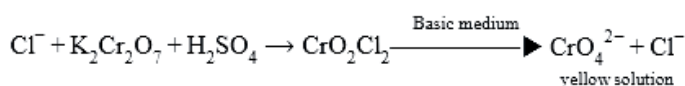
In chromyl chloride test for confirmation of  $\text{Cl}^-$  ion, a yellow solution is obtained. Acidification of the solution and addition of amyl alcohol and 10%  $\text{H}_2\text{O}_2$  turns organic layer blue indicating formation of chromium pentoxide. The oxidation state of chromium in that is  
[29-Jan-2024 Shift 1]

Options:

- A. +6
- B. +5
- C. +10
- D. +3

Answer: A

Solution:



## Question47

Which of the following statements are correct about Zn, Cd and Hg ?

- A. They exhibit high enthalpy of atomization as the d-subshell is full.
- B. Zn and Cd do not show variable oxidation state while Hg shows +I and + II.
- C. Compounds of Zn, Cd and Hg are paramagnetic in nature.
- D. Zn, Cd and Hg are called soft metals.

Choose the most appropriate from the options given below:

[29-Jan-2024 Shift 2]

Options:

- A. B, D only

B. B, C only

C. A, D only

D. C, D only

**Answer: A**

**Solution:**

(A) Zn, Cd, Hg exhibit lowest enthalpy of atomization in respective transition series.

(C) Compounds of Zn, Cd and Hg are diamagnetic in nature.

---

## Question48

Which of the following acts as a strong reducing agent? (Atomic number :  
Ce = 58, Eu = 63, Gd = 64, Lu = 71 )

[29-Jan-2024 Shift 2]

**Options:**

A.  $\text{Lu}^{3+}$

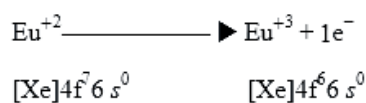
B.  $\text{Gd}^{3+}$

C.  $\text{Eu}^{2+}$

D.  $\text{Ce}^{4+}$

**Answer: C**

**Solution:**



## Question49

### Match List-I with List-II

| List-I               | List-II                 |
|----------------------|-------------------------|
| Species              | Electronic distribution |
| (A) $\text{Cr}^{-2}$ | (I) $3d^8$              |
| (B) $\text{Mn}^+$    | (II) $3d^3 4s^1$        |
| (C) $\text{Ni}^{-2}$ | (III) $3d^4$            |
| (D) $\text{V}^-$     | (IV) $3d^5 4s^1$        |

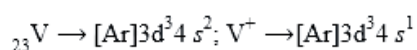
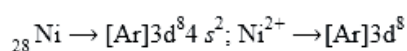
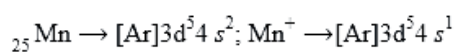
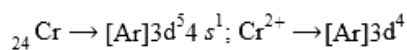
Choose the correct answer from the options given below:  
[30-Jan-2024 Shift 1]

Options:

- A. (A)-I, (B)-II, (C)-III, (D)-IV
- B. (A)-III, (B) - IV, (C) - I, (D)-II
- C. (A)-IV, (B)-III, (C)-I, (D)-II
- D. (A)-II, (B)-I, (C)-IV, (D)-III

Answer: B

Solution:



---

## Question50

Diamagnetic Lanthanoid ions are:  
[30-Jan-2024 Shift 1]

Options:

- A.  $\text{Nd}^{3+}$  and  $\text{Eu}^{3+}$
- B.  $\text{La}^{3+}$  and  $\text{Ce}^{4+}$
- C.  $\text{Nd}^{3+}$  and  $\text{Ce}^{4+}$
- D.  $\text{Lu}^{3+}$  and  $\text{Eu}^{3+}$

**Answer: B**

**Solution:**

Ce :  $[\text{Xe}]4f^1 5d^1 6s^2$ ;  $\text{Ce}^{4+}$  diamagnetic

La :  $[\text{Xe}]4f^0 5d^1 6s^2$ ;  $\text{La}^{3+}$  diamagnetic

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## Question51

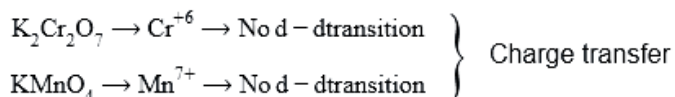
The orange colour of  $\text{K}_2\text{Cr}_2\text{O}_7$  and purple colour of  $\text{KMnO}_4$  is due to  
[30-Jan-2024 Shift 2]

**Options:**

- A. Charge transfer transition in both.
- B.  $d \rightarrow d$  transition in  $\text{KMnO}_4$  and charge transfer transitions in  $\text{K}_2\text{Cr}_2\text{O}_7$ .
- C.  $d \rightarrow d$  transition in  $\text{K}_2\text{Cr}_2\text{O}_7$  and charge transfer transitions in  $\text{KMnO}_4$ .
- D.  $d \rightarrow d$  transition in both.

**Answer: A**

**Solution:**



## Question52

Alkaline oxidative fusion of  $\text{MnO}_2$  gives " A " which on electrolytic oxidation in alkaline solution produces B. A and B respectively are:  
[30-Jan-2024 Shift 2]

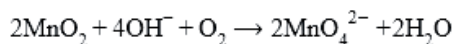
**Options:**

- A.  $\text{Mn}_2\text{O}_7$  and  $\text{MnO}_4^-$
- B.  $\text{MnO}_4^{2-}$  and  $\text{MnO}_4^-$
- C.  $\text{Mn}_2\text{O}_3$  and  $\text{MnO}_4^{2-}$
- D.  $\text{MnO}_4^{2-}$  and  $\text{Mn}_2\text{O}_7$

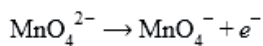
**Answer: B**

**Solution:**

Alkaline oxidative fusion of  $\text{MnO}_2$  :

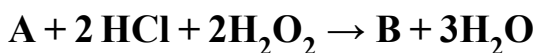
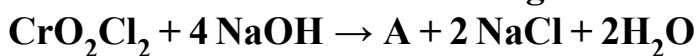


Electrolytic oxidation of  $\text{MnO}_4^{2-}$  in alkaline medium.



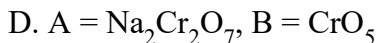
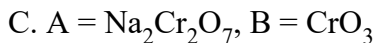
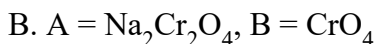
## Question53

**A and B formed in the following reactions are:**



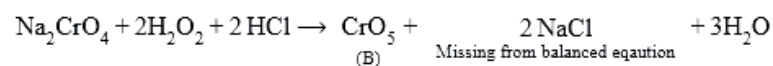
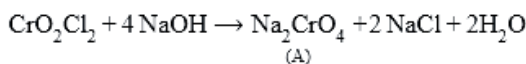
**[30-Jan-2024 Shift 2]**

**Options:**



**Answer: A**

**Solution:**



## Question54

**Identify correct statements from below:**

**A. The chromate ion is square planar.**

**B. Dichromates are generally prepared from chromates.**

**C. The green manganate ion is diamagnetic.**

**D. Dark green coloured  $\text{K}_2\text{MnO}_4$  disproportionate in a neutral or acidic medium to give permanganate.**

**E. With increasing oxidation number of transition metal, ionic character of the oxides decreases.**

**Choose the correct answer from the options given below:**

**[31-Jan-2024 Shift 1]**

**Options:**

A. B, C, D only

B. A, D, E only

C. A, B, C only

D. B, D, E only

**Answer: D**

**Solution:**

A.  $\text{CrO}_4^{2-}$  is tetrahedral

B.  $2\text{Na}_2\text{CrO}_4 + 2\text{H}^+ \rightarrow \text{Na}_2\text{Cr}_2\text{O}_7 + 2\text{Na}^+ + \text{H}_2\text{O}$

C. As per NCERT, green manganate is paramagnetic with 1 unpaired electron.

D. Statement is correct

E. Statement is correct

---

## **Question55**

**The 'Spin only' Magnetic moment for  $[\text{Ni}(\text{NH}_3)_6]^{2+}$  is  $\text{____} \times 10^{-1}$  BM.**

**( given = Atomic number of Ni : 28 )**

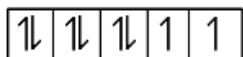
**[31-Jan-2024 Shift 1]**

**Answer: 28**

**Solution:**

$\text{NH}_3$  act as WFL with  $\text{Ni}^{2+}$

$$\text{Ni}^{2+} = 3d^8$$



No. of unpaired electron = 2

$$\mu = \sqrt{n(n+2)} = \sqrt{8} = 2.82 \text{ BM}$$

$$= 28.2 \times 10^{-1} \text{ BM}$$

$$x = 28$$

---

## Question 56

Choose the correct statements from the following

A.  $\text{Mn}_2\text{O}_7$  is an oil at room temperature

B.  $\text{V}_2\text{O}_4$  reacts with acid to give  $\text{VO}_2^{2+}$

C.  $\text{CrO}$  is a basic oxide

D.  $\text{V}_2\text{O}_5$  does not react with acid

Choose the correct answer from the options given below :

[31-Jan-2024 Shift 2]

Options:

A. A, B and D only

B. A and C only

C. A, B and C only

D. B and C only

**Answer: B**

**Solution:**

(A)  $\text{Mn}_2\text{O}_7$  is green oil at room temperature.

(B)  $\text{V}_2\text{O}_4$  dissolve in acids to give  $\text{VO}^{2+}$  salts.

(C)  $\text{CrO}$  is basic oxide

(D)  $\text{V}_2\text{O}_5$  is amphoteric it reacts with acid as well as base.

---

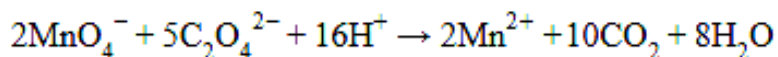
## Question 57

Number of moles of  $\text{H}^+$  ions required by 1 mole of  $\text{MnO}_4^-$  to oxidise oxalate ion to  $\text{CO}_2$  is \_\_\_\_\_

[31-Jan-2024 Shift 2]

Answer: 8

Solution:



∴ Number of moles of  $\text{H}^+$  ions required by 1 mole of  $\text{MnO}_4^-$  to oxidise oxalate ion to  $\text{CO}_2$  is 8

-----

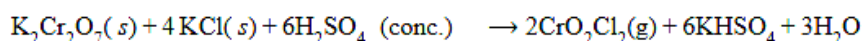
## Question58

In the reaction of potassium dichromate, potassium chloride and sulfuric acid (conc.), the oxidation state of the chromium in the product is (+)\_\_\_\_

[31-Jan-2024 Shift 2]

Answer: 6

Solution:



This reaction is called chromyl chloride test.

Here oxidation state of Cr is +6 .

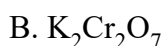
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## Question59

Which of the following compounds show colour due to d-d transition?

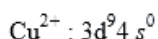
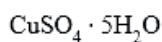
[1-Feb-2024 Shift 2]

Options:



**Answer: A**

**Solution:**



unpaired electron present so it show colour due to d-d transition.

---

## Question60

**The transition metal having highest 3<sup>rd</sup> ionisation enthalpy is :  
[1-Feb-2024 Shift 2]**

**Options:**

A. Cr

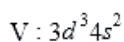
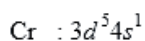
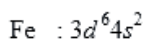
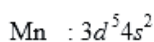
B. Mn

C. V

D. Fe

**Answer: B**

**Solution:**



So Mn has highest 3rd IE among all the given elements due to  $d^5$  configuration.

---

## Question61

**Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).**

**Assertion (A) : In aqueous solutions  $\text{Cr}^{2+}$  is reducing while  $\text{Mn}^{3+}$  is oxidising in nature.**

**Reason (R) : Extra stability to half filled electronic configuration is observed than incompletely filled electronic configuration.**

**In the light of the above statement, choose the most appropriate answer from the**

**options given below:**  
**[1-Feb-2024 Shift 2]**

**Options:**

- A. Both (A) and (R) are true and (R) is the correct explanation of (A)
- B. Both (A) and (R) are true but (R) is not the correct explanation of (A)
- C. (A) is false but (R) is true
- D. (A) is true but (R) is false

**Answer: A**

**Solution:**

$\text{Cr}^{2+}$  is reducing as its configuration changes from  $d^4$  to  $d^3$  due to formation of  $\text{Cr}^{3+}$ , which has half filled  $t_{2g}$  level, on other hand, the change  $\text{Mn}^{3+}$  to  $\text{Mn}^{2+}$  results in half filled  $d^5$  configuration which has extra stability.

-----

## Question62

**The magnetic moment of a transition metal compound has been calculated to be 3.87 B.M. The metal ion is**  
**[24-Jan-2023 Shift 1]**

**Options:**

- A.  $\text{Cr}^{2+}$
- B.  $\text{Mn}^{2+}$
- C.  $\text{V}^{2+}$
- D.  $\text{Ti}^{2+}$

**Answer: C**

**Solution:**

$\text{Cr}^{+2} : [\text{Ar}], 3d^4, 4s^0, n=4, \mu = \sqrt{4(4+2)} = \sqrt{24}$   
 $= 4.89 \text{ BM}$   
 $\text{Mn}^{+2} : [\text{Ar}], 3d^5, 4s^0, n=5, \mu = \sqrt{5(5+2)} = \sqrt{35}$   
 $= 5.91 \text{ BM}$   
 $\text{V}^{+2} : [\text{Ar}], 3d^3, 4s^0, n=3, \mu = \sqrt{3(3+2)} = \sqrt{15}$   
 $= 3.87 \text{ BM}$   
 $\text{Ti}^{+2} : [\text{Ar}], 3d^2, 4s^0, n=2, \mu = \sqrt{2(2+2)} = \sqrt{8}$   
 $= 2.82 \text{ BM}$

-----

## Question63

An ammoniacal metal salt solution gives a brilliant red precipitate on addition of dimethylglyoxime. The metal ion is :

[24-Jan-2023 Shift 1]

Options:

A.  $\text{Cu}^{2+}$

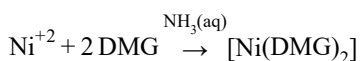
B.  $\text{Co}^{2+}$

C.  $\text{Fe}^{2+}$

D.  $\text{Ni}^{2+}$

Answer: D

Solution:



## Question64

Which one amongst the following are good oxidizing agents?

A.  $\text{Sm}^{2+}$

B.  $\text{Ce}^{2+}$

C.  $\text{Ce}^{4+}$

D.  $\text{Tb}^{4+}$

Choose the most appropriate answer from the options given below:

[24-Jan-2023 Shift 2]

Options:

A. Conly

B. D only

C. A and B only

D. C and D only

Answer: D

Solution:

$\text{Ce}^{+4}$  and  $\text{Tb}^{+4}$  act as oxidising agent.

-----

## Question65

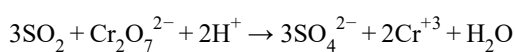
**K<sub>2</sub>Cr<sub>2</sub>O<sub>7</sub> paper acidified with dilute H<sub>2</sub>SO<sub>4</sub> turns green when exposed to**  
**[24-Jan-2023 Shift 2]**

**Options:**

- A. Carbon dioxide
- B. Sulphur trioxide
- C. Hydrogen sulphide
- D. Sulphur dioxide

**Answer: D**

**Solution:**



## Question66

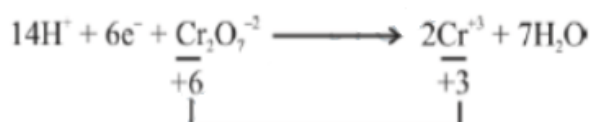
**Potassium dichromate acts as a strong oxidizing agent in acidic solution. During this process, the oxidation state changes from**  
**[25-Jan-2023 Shift 2]**

**Options:**

- A. +3 to +1
- B. +6 to +3
- C. +2 to +1
- D. +6 to +2

**Answer: B**

**Solution:**



## Question67

**During the borax bead test with CuSO<sub>4</sub>, a blue green colour of the bead was observed in oxidising flame due to the formation of**

**[29-Jan-2023 Shift 1]**

**Options:**

A.  $\text{Cu}_3\text{B}_2$

B. Cu

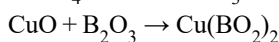
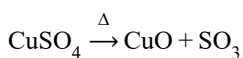
C.  $\text{Cu}(\text{BO}_2)_2$

D. CuO

**Answer: C**

**Solution:**

Blue green colour is due to formation of  $\text{Cu}(\text{BO}_2)_2$



## Question68

**The set of correct statements is:**

**(i) Manganese exhibits +7 oxidation state in its oxide.**

**(ii) Ruthenium and Osmium exhibit +8 oxidation in their oxides.**

**(iii) Sc shows +4 oxidation state which is oxidizing in nature.**

**(iv) Cr shows oxidising nature in +6 oxidation state.**

**[29-Jan-2023 Shift 2]**

**Options:**

A. (ii) and (iii)

B. (i), (ii) and (iv)

C. (i) and (iii)

D. (ii), (iii) and (iv)

**Answer: B**

**Solution:**

(i), (ii) and (iv) correct.

Manganese exhibits +7 oxidation state in its oxide. ( $\text{Mn}_2\text{O}_7$ )

Ru & Os from  $\text{RuO}_4$  &  $\text{OsO}_4$  oxide in +8 oxidation state

Cr in +6 oxidation act is oxidizing.

Sc does not show +4 oxidation state.

---

## Question69

A solution of  $\text{CrO}_5$  in amyl alcohol has a....colour

[29-Jan-2023 Shift 2]

Options:

- A. Green
- B. Orange-Red
- C. Yellow
- D. Blue

Answer: D

Solution:

A solution of  $\text{CrO}_5$  in amyl alcohol has a blue colour. So, option (4) is correct.

-----

## Question70

During the qualitative analysis of  $\text{SO}_3^{2-}$  using dilute  $\text{H}_2\text{SO}_4$ ,  $\text{SO}_2$  gas is evolved which turns  $\text{K}_2\text{Cr}_2\text{O}_7$  solution (acidified with dilute  $\text{H}_2\text{SO}_4$ ):

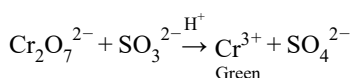
[30-Jan-2023 Shift 1]

Options:

- A. Black
- B. Red
- C. Green
- D. Blue

Answer: C

Solution:

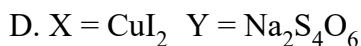
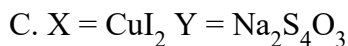
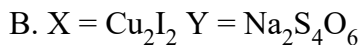
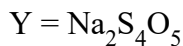
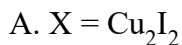


## Question71

When  $\text{Cu}^{2+}$  ion is treated with KI, a white precipitate, X appears in solution. The solution is titrated with sodium thiosulphate, the compound Y is formed. X and Y

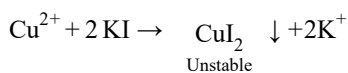
respectively are  
[31-Jan-2023 Shift 1]

Options:

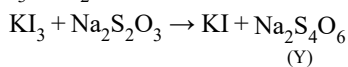
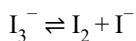
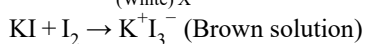
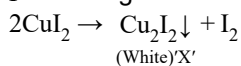


Answer: B

Solution:



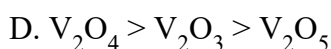
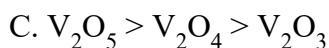
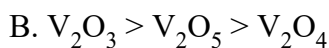
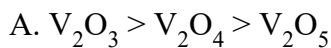
$\text{I}^-$  is strong R.A it reduces  $\text{Cu}^{2+}$  to  $\text{Cu}^+$



## Question 72

The correct order of basicity of oxides of vanadium is  
[31-Jan-2023 Shift 1]

Options:



Answer: A

Solution:

With increase in % of oxygen acidic nature of oxide of an element increase and basic nature decreases

## Question73

Highest oxidation state of Mn is exhibited in  $\text{Mn}_2\text{O}_7$ . The correct statements about  $\text{Mn}_2\text{O}_7$  are

- (A) Mn is tetrahedrally surrounded by oxygen atoms
- (B) Mn is octahedrally surrounded by oxygen atoms
- (C) Contains Mn-O-Mn bridge
- (D) Contains Mn-Mn bond.

Choose the correct answer from the options given below

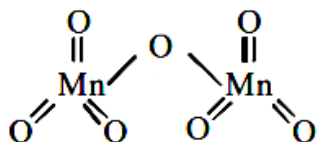
[1-Feb-2023 Shift 1]

Options:

- A. A and C only
- B. A and D only
- C. B and D only
- D. B and C only

Answer: A

Solution:



## Question74

A solution of  $\text{FeCl}_3$  when treated with  $\text{K}_4[\text{Fe}(\text{CN})_6]$  gives a prussian blue precipitate due to the formation of

[1-Feb-2023 Shift 1]

Options:

- A.  $\text{K}[\text{Fe}_2(\text{CN})_6]$
- B.  $\text{Fe}[\text{Fe}(\text{CN})_6]$
- C.  $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$
- D.  $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$

Answer: D

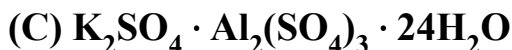
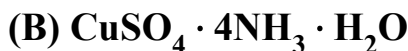
Solution:

Formation of Prussian blue complex takes place.

---

## Question 75

Which of the following are the example of double salt?



Choose the correct answer.

[1-Feb-2023 Shift 1]

Options:

A. A and C only

B. A and B only

C. A, B and D only

D. B and D only

Answer: A

Solution:

Double salt contain's two or more types of salts.  $\text{CuSO}_4 \cdot 4\text{NH}_3 \cdot \text{H}_2\text{O}$  and  $\text{Fe}(\text{CN})_2 \cdot 4\text{KCN}$  are complex compounds.

---

## Question 76

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) :  $\text{Cu}^{2+}$  in water is more stable than  $\text{Cu}^+$ .

Reason (R) : Enthalpy of hydration for  $\text{Cu}^{2+}$  is much less than that of  $\text{Cu}^+$ .

In the light of the above statements, choose the correct answer from the options given below :

[1-Feb-2023 Shift 2]

Options:

A. Both (A) and (R) are correct and (R) is the correct explanation of (A).

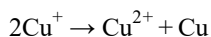
B. (A) is correct but (R) is not correct.

C. (1) is not correct but (R) is correct.

D. Both (A) and (R) are correct but (R) is not the correct explanation of (A).

**Answer: A**

**Solution:**



The stability of  $\text{Cu}^{2+}(\text{aq})$  rather than  $\text{Cu}^+(\text{aq})$ , is due to the much more negative  $\Delta_{\text{hyd}}H$  of  $\text{Cu}^{2+}(\text{aq})$  than  $\text{Cu}^+(\text{aq})$ , which more than compensates for the second ionisation enthalpy of Cu.

-----

## Question77

**Which element is not present in Nessler's reagent ?  
[1-Feb-2023 Shift 2]**

**Options:**

- A. Mercury
- B. Potassium
- C. Iodine
- D. Oxygen

**Answer: D**

**Solution:**

(Nessler's Reagent  $\rightarrow \text{K}_2[\text{HgI}_4]$ )

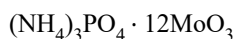
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## Question78

**In ammonium-phosphomolybdate, the oxidation state of Mo is + \_\_\_\_  
[6-Apr-2023 shift 1]**

**Answer: 6**

**Solution:**



Let X = oxidation state of Mo in  $\text{MoO}_3$

$$X + (-2) \times 3 = 0$$

$$X = +6$$

Ans: 6

-----

## Question79

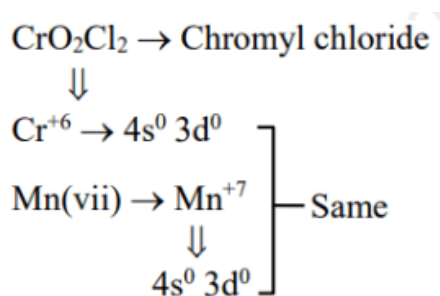
In chromyl chloride, the number of d-electrons present on chromium is same as in  
(Given at no. of Ti : 22, V: 23, Cr : 24, Mn : 25, Fe : 26)  
[8-Apr-2023 shift 1]

Options:

- A. Fe (III)
- B. V (IV)
- C. Ti (III)
- D. Mn (VII)

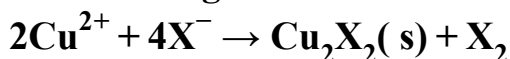
Answer: D

Solution:



## Question80

Which halogen is known to cause the reaction given below:



[8-Apr-2023 shift 1]

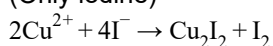
Options:

- A. All halogens
- B. Only chlorine
- C. Only Bromine
- D. Only Iodine

Answer: D

Solution:

(Only iodine)



---

## Question81

Given below are two statements :

**Statement I :** Aqueous solution of  $K_2Cr_2O_7$  is preferred as a primary standard in volumetric analysis over  $Na_2Cr_2O_7$  aqueous solution.

**Statement II :**  $K_2Cr_2O_7$  has a higher solubility in water than  $Na_2Cr_2O_7$

In the light of the above statements, choose the correct answer from the options given below:

[10-Apr-2023 shift 1]

**Options:**

- A. Statement I is false but Statement I is true
- B. Statement I is true but Statement II is false
- C. Both Statement I and Statement II are true
- D. Both Statement I and Statement II are false

**Answer: B**

**Solution:**

- (1)  $K_2Cr_2O_7$  is used as primary standard. The concentration  $Na_2Cr_2O_7$  changes in aq. solution.
- (2) It is less soluble than  $Na_2Cr_2O_7$

---

## Question82

Prolonged heating is avoided during the preparation of ferrous ammonium sulphate to

[10-Apr-2023 shift 1]

**Options:**

- A. prevent hydrolysis
- B. prevent reduction
- C. prevent breaking
- D. prevent oxidation

**Answer: C**

**Solution:**

It may oxidise ferrous ion to ferric ions.

---

## Question83

Which of the following statements are correct?

- (A) The  $M^{3+}/M^{2+}$  reduction potential for iron is greater than manganese  
(B) The higher oxidation states of first row d-block elements get stabilized by oxide ion.  
(C) Aqueous solution of  $Cr^{2+}$  can liberate hydrogen from dilute acid.  
(D) Magnetic moment of  $V^{2+}$  is observed between 4.4-5.2 BM.

Choose the correct answer from the options given below:

[10-Apr-2023 shift 1]

Options:

- A. (C), (D) only  
B. (B), (C) only  
C. (A), (B), (D) only  
D. (A), (B) only

Answer: B

Solution:

(A) The  $M^{3+}/M^{2+}$  reduction potential for manganese is greater than iron

(B)  $E_{Fe^{3+}/Fe^{2+}}^0 = +0.77$

$E_{Mn^{3+}/Mn^{2+}}^0 = +1.57$

(C)  $E_{Cr^{3+}/Cr^{2+}}^0 = -0.26$

$\therefore Cr^{2+} + H^+ \rightarrow Cr^{3+} + \frac{1}{2}H_2$

(D)  $V^{2+} = 3$  unpaired electron

Magnetic Moment = 3.87 B.M

---

## Question84

When a solution of mixture having two inorganic salts was treated with freshly prepared ferrous sulphate in acidic medium, a dark brown ring was formed whereas on treatment with neutral  $FeCl_3$ , it gave deep red colour which disappeared on boiling and a brown red ppt was formed. The mixture contains

[11-Apr-2023 shift 1]

Options:

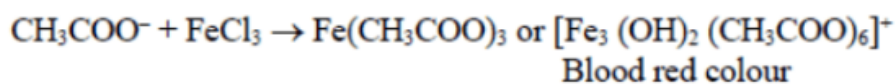
A.  $C_2O_4^{2-}$  &  $NO_3^-$

B.  $SO_3^{2-}$  &  $C_2O_4^{2-}$



**Answer: C**

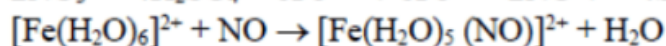
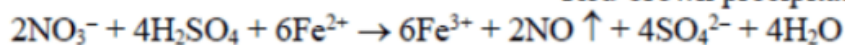
**Solution:**



Blood red colour



Red-brown precipitate



Brown

## Question85

The pair of lanthanides in which both elements have high third - ionization energy is:

[13-Apr-2023 shift 1]

**Options:**

A. Dy, Gd

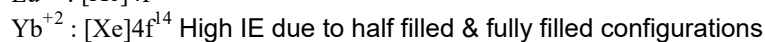
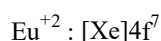
B. Eu, Gd

C. Lu, Yb

D. Eu, Yb

**Answer: D**

**Solution:**



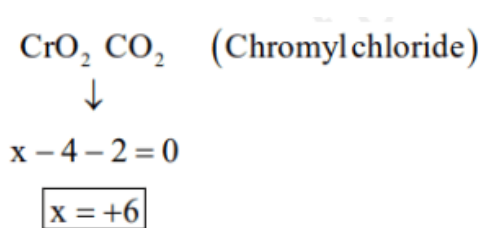
## Question86

In Chromyl chloride, the oxidation state of chromium is (+) \_\_\_\_\_.

[15-Apr-2023 shift 1]

**Answer: 6**

**Solution:**



## Question87

The difference in oxidation state of chromium in chromate and dichromate salts is

[24-Jun-2022-Shift-1]

**Answer: 0**

**Solution:**

Chromate ion  $\rightarrow \text{CrO}_4^{2-}$ , oxidation state of Cr = +6

Dichromate ion  $\rightarrow \text{CrO}_7^{2-}$ , oxidation state of Cr = +6

$\therefore$  Difference in oxidation state = zero

---

## Question88

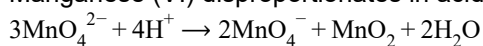
Manganese (VI) has ability to disproportionate in acidic solution. The difference in oxidation states of two ions it forms in acidic solution is \_\_\_\_\_

[24-Jun-2022-Shift-2]

**Answer: 3**

**Solution:**

Manganese (VI) disproportionates in acidic medium as



The difference in oxidation states of Mn in the products formed =  $7 - 4 = 3$

---

## Question89

**Cerium (IV) has a noble gas configuration. Which of the following is correct statement about it?**

**[25-Jun-2022-Shift-1]**

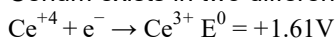
**Options:**

- A. It will not prefer to undergo redox reactions.
- B. It will prefer to gain electron and act as an oxidizing agent.
- C. It will prefer to give away an electron and behave as reducing agent.
- D. It acts as both, oxidizing and reducing agent.

**Answer: B**

**Solution:**

Cerium exists in two different oxidation state +3, +4



It shows  $\text{Ce}^{+4}$  acts as a strong oxidising agent & accepts electron.

---

## Question90

**Among the following, which is the strongest oxidizing agent?**

**[25-Jun-2022-Shift-1]**

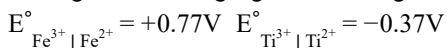
**Options:**

- A.  $\text{Mn}^{3+}$
- B.  $\text{Fe}^{3+}$
- C.  $\text{Ti}^{3+}$
- D.  $\text{Cr}^{3+}$

**Answer: A**

**Solution:**

Strongest oxidising agent have highest reduction potential value



$\text{Mn}^{+3}$  is the best oxidising agent among the given series.

---

## Question91

The metal ion (in gaseous state) with lowest spin-only magnetic moment value is  
[25-Jun-2022-Shift-2]

Options:



Answer: B

Solution:

|           | Valence shell configuration | Unpaired electrons |
|-----------|-----------------------------|--------------------|
| $V^{2+}$  | $3d^34s^0$                  | $n = 3$            |
| $Ni^{2+}$ | $3d^84s^0$                  | $n = 2$            |
| $Cr^{2+}$ | $3d^44s^0$                  | $n = 4$            |
| $Fe^{2+}$ | $3d^64s^0$                  | $n = 4$            |

---

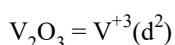
## Question92

The spin-only magnetic moment value of the most basic oxide of vanadium among  $V_2O_3$ ,  $V_2O_4$  and  $V_2O_5$  is \_\_\_\_\_ B.M. (Nearest integer)  
[26-Jun-2022-Shift-1]

Answer: 3

Solution:

The most basic oxide among  $V_2O_3$ ,  $V_2O_4$  and  $V_2O_5$  is  $V_2O_3$



$$\begin{aligned}\text{Magnetic moment} &= \sqrt{2(2+2)} = \sqrt{8} \\ &= 2.83 \approx 3\end{aligned}$$

---

## Question93

The most common oxidation state of Lanthanoid elements is +3. Which of the following is likely to deviate easily from +3 oxidation state?

[26-Jun-2022-Shift-2]

Options:

A. Ce (At. No. 58)

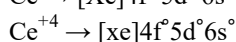
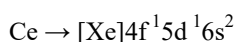
B. La (At. No. 57)

C. Lu (At. No. 71)

D. Gd (At. No. 64)

Answer: A

Solution:



Cerium in +4 oxidation state acquires inert gas configuration.

-----

## Question94

The number of statements correct from the following for Copper (at. no. 29) is/are \_\_\_\_\_

(A) Cu(II) complexes are always paramagnetic.

(B) Cu(I) complexes are generally colourless

(C) Cu(I) is easily oxidized

(D) In Fehling solution, the active reagent has Cu(I)

[27-Jun-2022-Shift-1]

Answer: 3

Solution:

(A) Cu (II) complexes are always paramagnetic as they have one unpaired electron due to  $d^9$  configuration of Cu(II)

(B) Cu(I) complexes are generally colourless due to  $d^{10}$  configuration.

(C) Cu(I) is easily oxidised to  $\text{Cu}^{+2}$  in aqueous solution  $2\text{Cu}^+ \rightarrow \text{Cu}^{+2} + \text{Cu}$

$\text{Cu}^{+1}$  disproportionates to  $\text{Cu}^{+2}$  and Cu

(  $E_{\text{cell}}^\circ > 0$  for this cell reaction in aqueous solution)

In Fehling's solution, active reagent has Cu(II) which is reduced to Cu(I) on reaction with aldehydes.

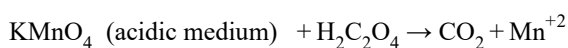
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## Question95

Acidified potassium permanganate solution oxidises oxalic acid. The spin-only magnetic moment of the manganese product formed from the above reaction is \_\_\_\_\_ B.M. (Nearest integer)  
[27-Jun-2022-Shift-1]

**Answer: 6**

**Solution:**



$\text{Mn}^{+2}$  has 5 unpaired electrons

$$\therefore \text{Spin only magnetic moment} = \sqrt{5(5+2)}$$

$$= \sqrt{5 \times 7}$$

$$= \sqrt{35}$$

$$\approx 5.92 \text{ B.M.}$$

$$\approx 6 \text{ B.M.}$$

---

## Question96

The 'f' orbitals are half and completely filled, respectively in lanthanide ions  
[Given : Atomic no. Eu, 63; Sm, 62; Tm, 69; Tb, 65; Yb, 70; Dy, 66]  
[27-Jun-2022-Shift-2]

**Options:**

A.  $\text{Eu}^{2+}$  and  $\text{Tm}^{2+}$

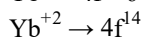
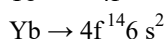
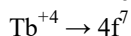
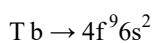
B.  $\text{Sm}^{2+}$  and  $\text{Tm}^{3+}$

C.  $\text{Tb}^{4+}$  and  $\text{Yb}^{2+}$

D.  $\text{Dy}^{3+}$  and  $\text{Yb}^{3+}$

**Answer: C**

**Solution:**



Hence, the pair  $\text{Tb}^{+4}$  and  $\text{Yb}^{+2}$  have half filled and completely filled f subshells respectively.

---

## Question97

**Dihydrogen reacts with CuO to give**  
**[28-Jun-2022-Shift-1]**

**Options:**

A.  $\text{CuH}_2$

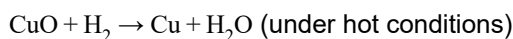
B. Cu

C.  $\text{Cu}_2\text{O}$

D.  $\text{Cu}(\text{OH})_2$

**Answer: B**

**Solution:**



## Question98

**Which one of the lanthanoids given below is the most stable in divalent form?**  
**[28-Jun-2022-Shift-1]**

**Options:**

A. Ce (Atomic Number 58)

B. Sm (Atomic number 62)

C. Eu (Atomic Number 63)

D. Yb (Atomic Number 70)

**Answer: C**

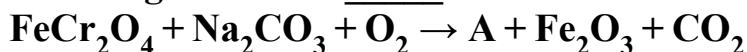
**Solution:**

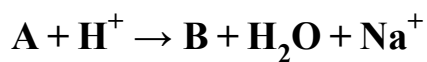
Electronic configuration of Europium (Eu) is  $[\text{Kr}]4f^7 6s^2$ . It can loose two electron to gain the half filled stable state. Hence, +2 oxidation state is most stable.

-----

## Question99

**The number of terminal oxygen atoms present in the product B obtained from the following reaction is \_\_\_\_\_**

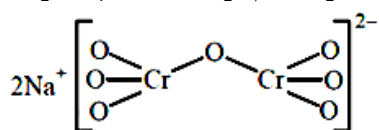
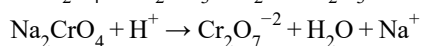
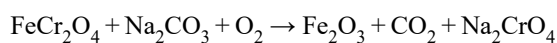




[29-Jun-2022-Shift-1]

**Answer: 6**

**Solution:**



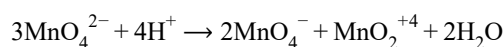
## Question100

An acidified manganate solution undergoes disproportionation reaction. The spin-only magnetic moment value of the product having manganese in higher oxidation state is \_\_\_\_\_ B.M. (Nearest integer)

[29-Jun-2022-Shift-1]

**Answer: 0**

**Solution:**



$Mn^{+7}$  = no. of unpaired electrons is ' 0 '

$\mu = 0$  B.M.

## Question101

The reaction of  $H_2O_2$  with potassium permanganate in acidic medium leads to the formation of mainly :

[25-Jul-2022-Shift-1]

**Options:**

A.  $Mn^{2+}$

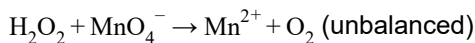
B.  $Mn^{4+}$

C.  $\text{Mn}^{3+}$

D.  $\text{Mn}^{6+}$

**Answer: A**

**Solution:**



## Question 102

Among  $\text{Co}^{3+}$ ,  $\text{Ti}^{2+}$ ,  $\text{V}^{2+}$  and  $\text{Cr}^{2+}$  ions, one if used as a reagent cannot liberate  $\text{H}_2$  from dilute mineral acid solution, its spin-only magnetic moment in gaseous state is B.M. (Nearest integer)  
[25-Jul-2022-Shift-1]

**Answer: 5**

**Solution:**

$\text{Co}^{3+}$  will not liberate  $\text{H}_2$  gas on reaction with dilute acid

$$E_{\text{Co}^{3+}/\text{Co}^{2+}}^{\circ} = +1.97$$

And  $\text{Co}^{3+}$  has electronic configuration =  $[\text{Ar}]3d^6$

$\therefore$  4 unpaired  $e^-$  are present in it

$$\therefore \text{Spin-only magnetic moment} = \sqrt{4(4+2)} = 4.92 \approx 5$$

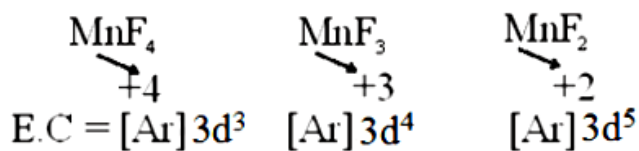
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## Question 103

The spin-only magnetic moment value of the compound with strongest oxidizing ability among  $\text{MnF}_4$ ,  $\text{MnF}_3$  and  $\text{MnF}_2$  is \_\_\_\_\_ B.M. [nearest integer]  
[26-Jul-2022-Shift-2]

**Answer: 5**

**Solution:**



Hence  $\text{MnF}_3 \Rightarrow$  strongest O.A

$$\mu = \sqrt{4(4+2)} = \sqrt{24} = 4.89 = 5$$

## Question104

Given below are two statements.

**Statement I:** Iron (III) catalyst, acidified  $\text{K}_2\text{Cr}_2\text{O}_7$  and neutral  $\text{KMnO}_4$  have the ability to oxidise  $\text{I}^-$  to  $\text{I}_2$  independently.

**Statement II:** Manganate ion is paramagnetic in nature and involves  $p\pi - p\pi$  bonding.

In the light of the above statements, choose the correct answer from the options given below.

[27-Jul-2022-Shift-1]

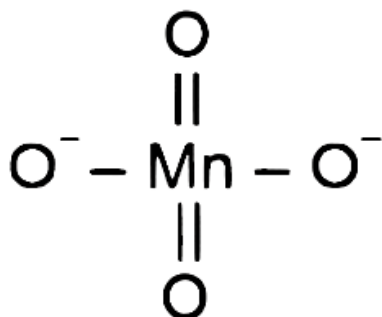
Options:

- A. Both Statement I and Statement II are true.
- B. Both Statement I and Statement II are false.
- C. Statement I is true but Statement II is false.
- D. Statement I is false but Statement II is true.

**Answer: B**

**Solution:**

Manganate ion  $\text{MnO}_4^{2-}$  has tetrahedral structure



has only  $d\pi - p\pi$  bonds.

$\text{Fe}^{3+}$  is not used as a catalyst in the conversion of  $\text{I}^-$  to  $\text{I}_2$  by  $\text{K}_2\text{Cr}_2\text{O}_7$ .  $\text{K}_2\text{Cr}_2\text{O}_7$  oxidise  $\text{I}^-$  in acidic medium easily

## Question105

The total number of Mn = O bonds in  $\text{Mn}_2\text{O}_7$  is \_\_\_\_\_.

[27-Jul-2022-Shift-1]

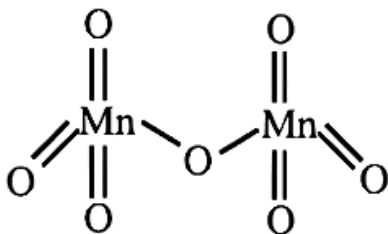
Options:

- A. 4
- B. 5
- C. 6
- D. 3

Answer: C

Solution:

Structure of  $\text{Mn}_2\text{O}_7$  is as :



$\therefore$  There are total 6 Mn = O bonds are present in  $\text{Mn}_2\text{O}_7$  compound.

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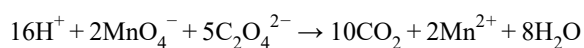
## Question106

In the titration of  $\text{KMnO}_4$  and oxalic acid in acidic medium, the change in oxidation number of carbon at the end point is \_\_\_\_\_.

[27-Jul-2022-Shift-1]

Answer: 1

Solution:



During titration of oxalic acid by  $\text{KMnO}_4$ , oxalic acid converts into  $\text{CO}_2$ .

$\therefore$  Change in oxidation state of carbon = 1

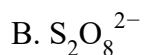
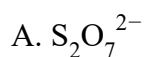
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## Question107

In neutral or alkaline solution,  $\text{MnO}_4^-$  oxidises thiosulphate to :

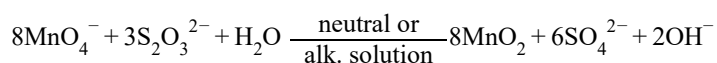
[27-Jul-2022-Shift-2]

**Options:**



**Answer: D**

**Solution:**

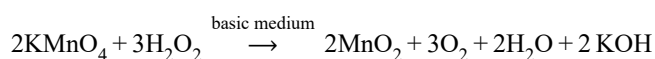


## Question108

The oxidation state of manganese in the product obtained in a reaction of potassium permanganate and hydrogen peroxide in basic medium is \_\_\_\_\_.  
[27-Jul-2022-Shift-2]

**Answer: 4**

**Solution:**



## Question109

Which of the following has least tendency to liberate  $\text{H}_2$  from mineral acids?  
[28-Jul-2022-Shift-1]

**Options:**

A. Cu

B. Mn

C. Ni

D. Zn

**Answer: A**

**Solution:**

The metal atom whose oxidation potential is less than that of hydrogen can release  $H_2$  from mineral acids.

$$E_{Zn/Zn^{+2}} = 0.76$$

$$E_{Ni/Ni^{+2}} = 0.25$$

$$E_{Mn/Mn^{+2}} = 1.18$$

$$E_{Cu^{\circ}/Cu^{+2}} = -0.34$$

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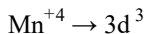
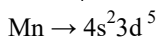
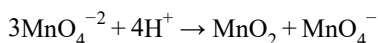
## Question110

The disproportionation of  $MnO_4^{2-}$  in acidic medium resulted in the formation of two manganese compounds A and B. If the oxidation state of Mn in B is smaller than that of A, then the spin-only magnetic moment ( $\mu$ ) value of B in BM is \_\_\_\_\_.(Nearest integer)

[28-Jul-2022-Shift-1]

**Answer: 4**

**Solution:**



$$n = 3$$

$$\mu = \sqrt{n(n+2)}$$

$$= \sqrt{3(5)}$$

$$= \sqrt{15}$$

$$= 3.87 \approx 4 \text{ B.M.}$$

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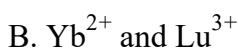
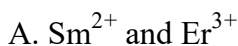
## Question111

Which of the following pair is not isoelectronic species?

(At. no. Sm, 62; Er, 68; Yb, 70; Lu, 71; Eu, 63; Tb, 65; Tm, 69)

[28-Jul-2022-Shift-2]

**Options:**



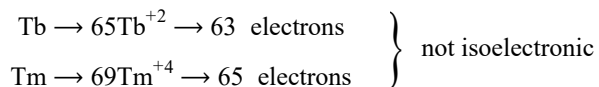
C.  $\text{Eu}^{2+}$  and  $\text{Tb}^{4+}$

D.  $\text{Tb}^{2+}$  and  $\text{Tm}^{4+}$

**Answer: 0**

**Solution:**

Species having same number of electrons are isoelectronic



---

## Question112

**The reaction of zinc with excess of aqueous alkali, evolves hydrogen gas and gives :  
[29-Jul-2022-Shift-1]**

**Options:**

A.  $\text{Zn}(\text{OH})_2$

B.  $\text{ZnO}$

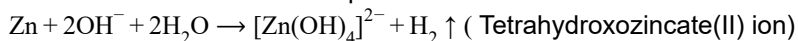
C.  $[\text{Zn}(\text{OH})_4]^{2-}$

D.  $[\text{ZnO}_2]^{2-}$

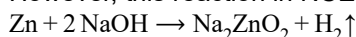
**Answer: C**

**Solution:**

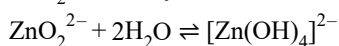
Zinc dissolves in excess of aqueous alkali.



However, this reaction in NCERT is given as



$\text{ZnO}_2^{2-}$  is anhydrous form of  $[\text{Zn}(\text{OH})_4]^{2-}$ .



So in aqueous medium best answer of this question is  $[\text{Zn}(\text{OH})_4]^{2-}$ .

---

## Question113

**In following pairs, the one in which both transition metal ions are colourless is :  
[29-Jul-2022-Shift-1]**

**Options:**

- A.  $\text{Sc}^{3+}$ ,  $\text{Zn}^{2+}$
- B.  $\text{Ti}^{4+}$ ,  $\text{Cu}^{2+}$
- C.  $\text{V}^{2+}$ ,  $\text{Ti}^{3+}$
- D.  $\text{Zn}^{2+}$ ,  $\text{Mn}^{2+}$

**Answer: A**

**Solution:**

$\text{Sc}^{+3}$  and  $\text{Zn}^{+2}$  are colourless as they contain no unpaired electron. Whereas the transition metal ions  $\text{Cu}^{+2}$ ,  $\text{Ti}^{+3}$ ,  $\text{V}^{+2}$  and  $\text{Mn}^{+2}$  are coloured as they contain unpaired electrons.

The unpaired electron from lower energy d orbital gets excited to a higher energy d orbital on absorbing light of frequency which lies in visible region. The colour complementary to light absorbed is observed.

-----

## Question114

**In neutral or faintly alkaline medium,  $\text{KMnO}_4$  being a powerful oxidant can oxidize, thiosulphate almost quantitatively, to sulphate. In this reaction overall change in oxidation state of manganese will be :  
[29-Jul-2022-Shift-1]**

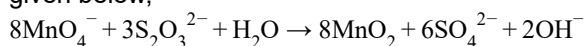
**Options:**

- A. 5
- B. 1
- C. 0
- D. 3

**Answer: D**

**Solution:**

In neutral or Faintly alkaline medium, thiosulphate is oxidised almost quantitatively to sulphate ion according to reaction given below,



Here the Mn changes from  $\text{Mn}^{+7}$  to  $\text{Mn}^{+4}$

Thus overall change in its oxidation number would be of 3 .

-----

## Question115

**Which of the following 3d-metal ion will give the lowest enthalpy of hydration ( $\Delta_{\text{hyd}} H$ ) when dissolved in water?**

**[29-Jul-2022-Shift-2]**

**Options:**

A.  $\text{Cr}^{2+}$

B.  $\text{Mn}^{2+}$

C.  $\text{Fe}^{2+}$

D.  $\text{Co}^{2+}$

**Answer: B**

**Solution:**

| Ion              | $\Delta H_{\text{Hyd}}^{\circ} (\text{kJ/mole})$ |
|------------------|--------------------------------------------------|
| $\text{Cr}^{2+}$ | -1925                                            |
| $\text{Mn}^{2+}$ | -1862                                            |
| $\text{Fe}^{2+}$ | -1998                                            |
| $\text{Co}^{2+}$ | -2079                                            |

---

## Question116

**The major components of German silver are**  
**[25 Feb 2021 Shift 2]**

**Options:**

A. Cu, Z n and Ag

B. Cu, Z n and N i

C. Ge, Cu and Ag

D. Z n, N i and Ag

**Answer: B**

**Solution:**

The major components of German silver are  
Cu : 25 – 30%, Z n : 25 – 30% and N i : 40 – 30%.

---

## Question117

In which of the following pairs, the outer most electronic configuration will be the same?

[25 Feb 2021 Shift 1]

Options:

- A.  $V^{2+}$  and  $Cr^+$
- B.  $Cr^+$  and  $Mn^{2+}$
- C.  $Ni^{2+}$  and  $Cu^+$
- D.  $Fe^{2+}$  and  $Co^+$

Answer: B

Solution:

(a)  $V^{2+} - [Ar]3d^3$ ;  $Cr^+ - [Ar]3d^5$

(b)  $Cr^+ - [Ar]3d^5$ ;  $Mn^{2+} - [Ar]3d^5$

(c)  $Ni^{2+} - [Ar]3d^8$ ;  $Cu^+ - [Ar]3d^{10}$

(d)  $Fe^{2+} - [Ar]3d^6$ ;  $Co^+ - [Ar]3d^7 4s^1$

Thus, in option (b), both ions have same outer most electronic configuration.

-----

## Question118

What is the correct order of the following elements with respect to their density ?

[24 Feb 2021 Shift 2]

Options:

- A.  $Cr < Zn < Co < Cu < Fe$
- B.  $Zn < Cu < Co < Fe < Cr$
- C.  $Zn < Cr < Fe < Co < Cu$
- D.  $Cr < Fe < Co < Cu < Zn$

Answer: C

Solution:

Generally, due to decrease in metallic radius and increase in atomic mass density increase across the period from left to right.

| Metal | Density ( $\text{g/cm}^3$ ) |
|-------|-----------------------------|
| Zn    | 7.13                        |
| Cr    | 7.19                        |
| Fe    | 7.8                         |
| Co    | 8.7                         |
| Cu    | 8.9                         |

Correct order is  $\text{Cu} > \text{Co} > \text{Fe} > \text{Cr} > \text{Zn}$ .

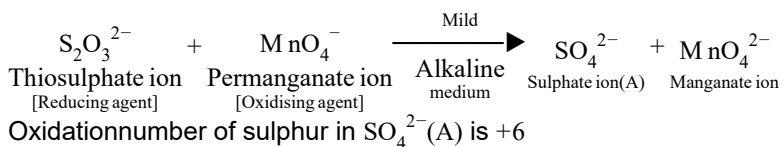
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## Question119

In mild alkaline medium, thiosulphate ion is oxidised by  $\text{MnO}_4^-$  to "A". The oxidation state of sulphur in "A" is ..... .  
[26 Feb 2021 Shift 2]

**Answer: 6**

**Solution:**



$$\left[ \begin{array}{l} \because x + 4(-2) = -2 \\ \Rightarrow x = +6 \end{array} \right]$$


---

## Question120

On treating a compound with warm dil.  $\text{H}_2\text{SO}_4$ , gas X is evolved, which turns  $\text{K}_2\text{Cr}_2\text{O}_7$  paper acidified with dil.  $\text{H}_2\text{SO}_4$  to a green compound Y. X and Y respectively are  
[26 Feb 2021 Shift 1]

**Options:**

A.  $\text{X} = \text{SO}_2$ ,  $\text{Y} = \text{Cr}_2\text{O}_3$

B.  $\text{X} = \text{SO}_3$ ,  $\text{Y} = \text{Cr}_2\text{O}_3$

D.  $X = \text{SO}_3$ ,  $Y = \text{Cr}_2(\text{SO}_4)_3$

A.  $X = \text{SO}_2$ ,  $Y = \text{Cr}_2\text{O}_3$

B.  $X = \text{SO}_3$ ,  $Y = \text{Cr}_2\text{O}_3$

C.  $X = \text{SO}_2$ ,  $Y = \text{Cr}_2(\text{SO}_4)_3$

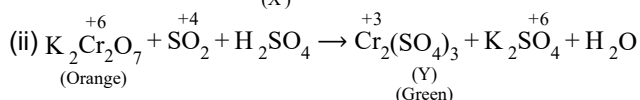
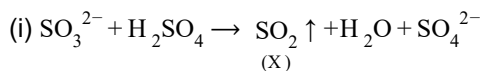
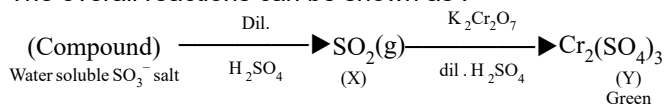
D.  $X = \text{SO}_3$ ,  $Y = \text{Cr}_2(\text{SO}_4)_3$

Cr (VI) from compound  $K_2Cr_2O_7$  on reduction changes its colour from orange to green which is the colour of Cr (III)

SO<sub>2</sub> can show both reducing and oxidising properties whereas SO<sub>3</sub> cannot show reducing property because of highest group number 16, oxidation state (+6) of sulphur in SO<sub>3</sub>.

As, X reacts with  $\text{K}_2\text{Cr}_2\text{O}_7$  in dil.  $\text{H}_2\text{SO}_4$  medium, the green coloured Cr (III) compound, Y must be Cr (III) sulphate or  $\text{Cr}_2(\text{SO}_4)_3$ .

The overall reactions can be shown as :



**Which one of the following lanthanoids does not form  $\text{M O}_2$  ? [ M is lanthanoid metal]**

**Options:**

- A. Pr  
B. Dy  
C. Nd  
D. Yb

**Solution:**

So, Yb (ytterbium) option (d) does not form  $\text{MO}_2$  type of oxide.

Note The common and predominant oxidation state of lanthanoids is +3. Consequently  $M^{4+}$  compounds are strong oxidising agents which changes to the common +3 state. Similarly, lanthanoid compounds of +2 state have a tendency to show reducing property as they get changed to +3 state easily.

**Given below are two statements:**

**Statement I  $\text{CeO}_2$  can be used for oxidation of aldehydes and ketones.**

**Statement II Aqueous solution of  $\text{EuSO}_4$  is a strong reducing agent.**

**In the light of the above statements, choose the correct answer from the options**

given below.  
[25 Feb 2021 Shift 1]

Options:

- A. Both statement I and statement II are true.
- B. Both statement I and statement II are false.
- C. Statement I is true but statement II is false.
- D. Statement I is false but statement II is true.

Answer: A

Solution:

Both statement I and statement II are true. The +3 oxidation state of lanthanide is most stable and therefore lanthanide in +4 oxidation state has strong tendency to gain electrons and converted into +3 and therefore act as strong oxidising agent, e.g.  $\text{Ce}^{4+}$ .

$\therefore \text{CeO}_2$  is used to oxidised alcohol, aldehyde and ketones. Lanthanides in +2 oxidation state has strong tendency to loss electron and converted into +3 oxidation state e.g.  $\text{Eu}^{+2}$ .  
 $\therefore \text{EuSO}_4$  acts as strong reducing agent.

-----

## Question124

The major components in "Gun Metal" are :  
[24 Feb 2021 Shift 1]

Options:

- A. Cu, Zn and Ni
- B. Cu, Sn and Zn
- C. Al, Cu, Mg and Mn
- D. Cu, Ni and Fe

Answer: B

Solution:

The major components in "Gun Metal" are Cu : 87%; Zn : 3%; Sn : 10%.

-----

## Question125

The electrode potential of  $\text{M}^{2+}/\text{M}$  of 3d-series elements shows positive value of  
[24 Feb 2021 Shift 1]

**Options:**

- A. Zn
- B. Fe
- C. Co
- D. Cu

**Answer: D**

**Solution:**

Only copper shows positive value for electrode potential of  $M^{2+}/M$  of 3d-series elements.

-----

## Question126

**The incorrect statement among the following is  
[24 Feb 2021 Shift 2]**

**Options:**

- A.  $VO_4^{3-}$  is a reducing agent.
- B.  $Cr_2O_3$  is an amphoteric oxide
- C.  $RuO_4$  is an oxidising agent.
- D. Red colour of ruby is due to the presence of  $Co^{3+}$ .

**Answer: B**

**Solution:**

Red colour of ruby is due to presence of  $Cr^{3+}$  ions in  $Al_2O_3$ .

Chromium is the trace element that causes ruby's red colour, which ranges from an orange red to a purplish red. The strength of ruby's red depends on how much chromium is present.

-----

## Question127

**The common positive oxidation states for an element with atomic number 24, are  
[17 Mar 2021 Shift 2]**

**Options:**

- A. +2 to +6
- B. +1 and +3 to +6

C. +1 and +3

D. +1 to +6

**Answer: A**

**Solution:**

Common positive oxidation states for an element with atomic number 24, are +2 to +6

Chromium (  $Z = 24$  )

Electronic configuration:  $[\text{Ar}] 4s^1 3d^5$  It has five electrons in 3d-subshell and one electron in 4s-subshell. Thus, chromium metal has six valence electrons.

Chromium can lose some or all of its valence electrons to form ions with different oxidation states. Thus, chromium shows the oxidation states of +1, +2, +3, +4, +5 and +6 respectively.

The most common oxidation states of chromium are +2, +3 and +6.

---

## Question 128

**The oxide that shows magnetic property is  
[18 Mar 2021 Shift 2]**

**Options:**

A.  $\text{SiO}_2$

B.  $\text{Mn}_3\text{O}_4$

C.  $\text{Na}_2\text{O}$

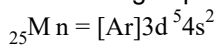
D.  $\text{MgO}$

**Answer: B**

**Solution:**

Generally, d-block elements show magnetic property due to presence of unpaired electrons in its d-block. Mn i.e. manganese shows +2 and +3 oxidation state.

Both having unpaired electrons as shown below



$\text{Mn}^{2+} (3d^5 4s^0)$  having 5 unpaired electrons in 3d orbital.

$\text{Mn}^{+3} (3d^4 4s^0)$  having 4 unpaired electrons in 3d orbital.

$\text{SiO}_2$ , p-block element compound is commonly believed to be a typical diamagnetic as no unpaired electrons are present.

$\text{MgO}$  and  $\text{Na}_2\text{O}$  also have unpaired electrons and hence are diamagnetic.

Hence,  $\text{Mn}_3\text{O}_4$  show magnetic property.

---

## Question 129

**$\text{Fe}_x$  and  $\text{Li}_y$  are known when x and y are**

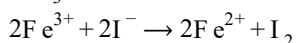
**[16 Mar 2021 Shift 2]**

**Options:**

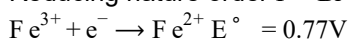
- A.  $x = \text{F, Cl, Br, I}$  and  $y = \text{F, Cl}$  and Br
- B.  $x = \text{F, Cl, Br}$  and  $y = \text{F, Cl, Br}$  and I
- C.  $x = \text{Cl, Br, I}$  and  $y = \text{F, Cl, Br}$  and I
- D.  $x = \text{F, Cl, Br, I}$  and  $y = \text{F, Cl, Br}$  and I

**Answer: A****Solution:**

$\text{FeI}_3$  does not exist because of  $\text{I}^-$ . Being a very good reducing agent,  $\text{I}^-$  reduces  $\text{Fe}^{3+}$  to  $\text{Fe}^{2+}$ .



Reducing nature order  $\text{I} > \text{Br} > \text{Cl} > \text{F}$



As standard reduction potential value of  $\text{Fe}^{3+}$  is more than  $\text{I}_2$ , so it is reduced to  $\text{Fe}^{2+}$ . Hence, except  $\text{FeI}_3$ , all other halogens can form iron halide compounds.

**Question130**

Given below are two statements.

**Statement I** The  $E^\circ$  value of  $\text{Ce}^{4+}/\text{Ce}^{3+}$  is +1.74V .

**Statement II** Ce is more stable in  $\text{Ce}^{4+}$  state than  $\text{Ce}^{3+}$  state.

In the light of the above statements, choose the most appropriate answer from the options given below.

[16 Mar 2021 Shift 1]

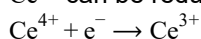
**Options:**

- A. Both statement I and statement II are correct.
- B. Statement I is incorrect but statement II is correct.
- C. Both statement I and statement II are incorrect.
- D. Statement I is correct but statement II is incorrect.

**Answer: D****Solution:**

$$E_{\text{Ce}^{4+}/\text{Ce}^{3+}}^\circ = 1.74 \text{ (data given)}$$

$\text{Ce}^{4+}$  can be reduced to  $\text{Ce}^{3+}$  and can oxidise water.



$\text{Ce}^{4+}$  is having noble gas configuration  $[\text{Xe}] 4f^0$  but it is strong oxidant and gets converted to  $\text{Ce}^{3+}$ .

Statement I is correct but statement II is incorrect.

---

## Question131

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

**Assertion (A)** Size of  $\text{Bk}^{3+}$  ion is less than  $\text{Np}^{3+}$  ion.

**Reason (R)** The above is a consequence of the lanthanoid contraction.

In the light of the above statements, choose the correct answer from the options given below

[16 Mar 2021 Shift 1]

**Options:**

- A. A is false but R is true
- B. Both A and R are true but R is not the correct explanation of A.
- C. Both A and R are true and R is the correct explanation of A.
- D. A is true but R is false.

**Answer: D**

**Solution:**

$_{97}\text{Bk}$  (Berkelium) and  $_{93}\text{Np}$  (Neptunium) are present in actinide series. Due to actinoid contraction, there is gradual decrease in size of  $\text{M}^{3+}$  ions across the period due to poor shielding of 5f-electrons. So,  $\text{Bk}^{3+}$  is smaller than that of  $\text{Np}^{3+}$  due to actinoid contraction. So, A is true and R is false.

---

## Question132

The spin only magnetic moments (in BM) for free  $\text{Ti}^{3+}$ ,  $\text{V}^{2+}$  and  $\text{Sc}^{3+}$  ions respectively are

(At.No. Sc : 21, Ti : 22, V: 23)

[25 Jul 2021 Shift 2]

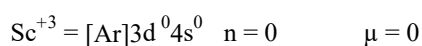
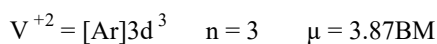
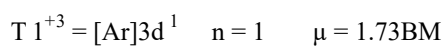
**Options:**

- A. 3.87, 1.73, 0
- B. 1.73, 3.87, 0
- C. 1.73, 0, 3.87
- D. 0, 3.87, 1.73

**Answer: B**

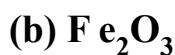
**Solution:**

$$\mu = \sqrt{n(n+2)}\text{BM}$$



## Question133

The correct order of following 3d metal oxides, according to their oxidation numbers is :



[25 Jul 2021 Shift 1]

Options:

A. (d) > (a) > (b) > (c) > (e)

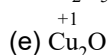
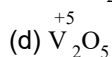
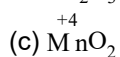
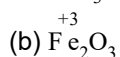
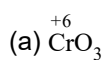
B. (a) > (c) > (d) > (b) > (e)

C. (a) > (d) > (c) > (b) > (e)

D. (c) > (a) > (d) > (e) > (b)

**Answer: C**

**Solution:**



So order of oxidation state

a > d > c > b > e

## Question134

The set having ions which are coloured and paramagnetic both is -

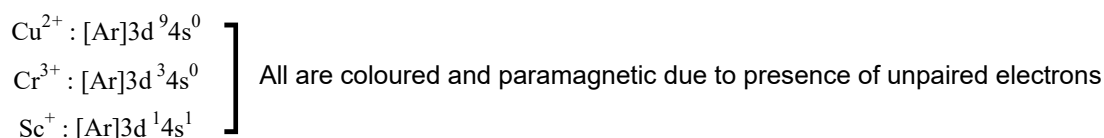
[22 Jul 2021 Shift 2]

Options:

- A.  $\text{Cu}^{2+}$ ,  $\text{Cr}^{3+}$ ,  $\text{Sc}^{+}$
- B.  $\text{Cu}^{2+}$ ,  $\text{Zn}^{2+}$ ,  $\text{Mn}^{4+}$
- C.  $\text{Sc}^{3+}$ ,  $\text{V}^{5+}$ ,  $\text{Ti}^{4+}$
- D.  $\text{Ni}^{2+}$ ,  $\text{Mn}^{7+}$ ,  $\text{Hg}^{2+}$

**Answer: A**

**Solution:**

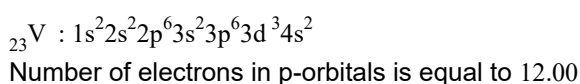


## Question135

Number of electrons that Vanadium ( $Z = 23$ ) has in p-orbitals is equal to \_\_\_\_\_  
[22 Jul 2021 Shift 2]

**Answer: 12**

**Solution:**

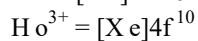
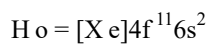


## Question136

Number of electrons present in 4f orbital of  $\text{Ho}^{3+}$  ion is \_\_\_\_\_. (Given Atomic No. of Ho = 67)  
[25 Jul 2021 Shift 2]

**Answer: 10**

**Solution:**



so number of  $e^-$  present in 4f is 10 .

---

## Question137

The nature of oxides  $\text{V}_2\text{O}_3$  and  $\text{CrO}$  is indexed as 'X' and 'Y' type respectively. The correct set of X and Y is  
[27 Aug 2021 Shift 1]

Options:

A. X = basic Y = amphoteric

B. X = amphoteric, Y = basic

C. X = acidic, Y = acidic

D. X = basic, Y = basic

Answer: D

Solution:

$\text{V}_2\text{O}_3$  is a basic oxide as on dissolving in acids, give solution of vanadium (III) complexes. In case of transition metal oxide, the metal with lower oxidation states are basic. So,  $\text{CrO}$  is also basic as oxidation state of Cr is +2.

---

## Question138

Which one of the following when dissolved in water gives coloured solution in nitrogen atmosphere?  
[26 Aug 2021 Shift 1]

Options:

A.  $\text{CuCl}_2$

B.  $\text{AgCl}$

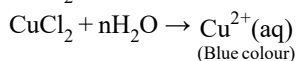
C.  $\text{ZnCl}_2$

D.  $\text{Cu}_2\text{Cl}_2$

Answer: A

Solution:

$\text{CuCl}_2$  will dissolve in water to give blue colour  $\text{Cu}^{2+}$  ions. The colour is due to the presence of unpaired  $e^-$  in  $\text{Cu}^{2+}$  ions.



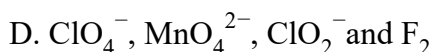
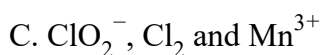
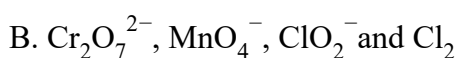
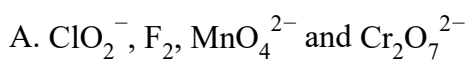
$\text{AgCl}$  and  $\text{Cu}_2\text{Cl}_2$  are insoluble in water while  $\text{ZnCl}_2$  on dissolving in water forms colourless solution due to formation of  $\text{Zn}^{2+}$  ions, which has no unpaired  $e^-$ .

---

## Question139

**In which one of the following sets all species show disproportionation reaction  
[31 Aug 2021 Shift 2]**

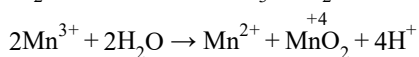
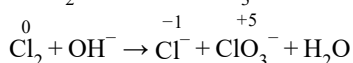
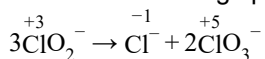
**Options:**



**Answer: B**

**Solution:**

Disproportionation reactions are redox reaction in which a compound undergoes oxidation as well as reduction. The element of reacting species is in an intermediate oxidation state and simultaneously gets oxidised and reduced.



But  $\text{MnO}_4^-$  does not get disproportionated.

Hence, option (c) is the correct answer.

---

## Question140

**Potassium permanganate on heating at 513K gives a product which is  
[27 Aug 2021 Shift 2]**

**Options:**

A. paramagnetic and colourless

B. diamagnetic and green

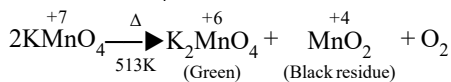
C. diamagnetic and colourless

D. paramagnetic and green

**Answer: A**

## Solution:

When  $\text{KMnO}_4$  is heated at 513K, it forms  $\text{K}_2\text{MnO}_4$  and  $\text{MnO}_2$  and  $\text{O}_2$ .



$$\text{Mn}_{25} = 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^5$$

$\text{Mn}^{+6} = 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1$ ; 1 unpaired electron, so paramagnetic.

So, option (d) is correct.

---

## Question141

The addition of dilute  $\text{NaOH}$  to  $\text{Cr}^{3+}$  salt solution will give  
[27 Aug 2021 Shift 2]

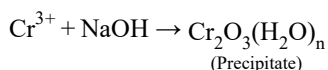
Options:

- A. a solution of  $[\text{Cr}(\text{OH})_4]^-$
- B. precipitate of  $\text{Cr}_2\text{O}_3(\text{H}_2\text{O})_n$
- C. precipitate of  $[\text{Cr}(\text{OH})_6]^{3-}$
- D. precipitate of  $\text{Cr}(\text{OH})_3$

Answer: B

## Solution:

The addition of dil.  $\text{NaOH}$  to  $\text{Cr}^{3+}$  solution will produce ppt. of  $\text{Cr}_2\text{O}_3(\text{H}_2\text{O})_n$ .



Therefore, the option (b) is correct.

---

## Question142

The  $\text{Eu}^{2+}$  ion is a strong reducing agent in spite of its ground state electronic configuration (outermost): [Atomic number of Eu = 63]  
[31 Aug 2021 Shift 2]

Options:

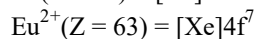
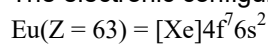
- A.  $4f^7 6s^2$
- B.  $4f^6$
- C.  $4f^7$

D.  $4f^6 6s^2$

**Answer: C**

**Solution:**

The electronic configuration of Eu and  $\text{Eu}^{2+}$  ion is as follows :



## Question143

**Which one of the following lanthanides exhibits +2 oxidation state with diamagnetic nature ?**

**(Given, Z for Nd = 60, Yb = 70, La = 57, Ce = 58)**

**[31 Aug 2021 Shift 1]**

**Options:**

A. Nd

B. Yb

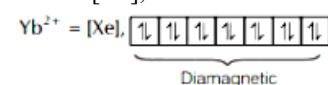
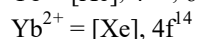
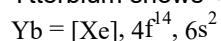
C. La

D. Ce

**Answer: B**

**Solution:**

Ytterbium shows +2 oxidation state and it is diamagnetic in nature.



Since, there is no unpaired electron, it is diamagnetic in nature.

---

## Question144

**The number of f - electrons in the ground state electronic configuration of Np(Z = 93) is ..... .**

**(Nearest integer)**

**[27 Aug 2021 Shift 1]**

**Answer: 18**

**Solution:**

The electronic configuration of Np = 93 is

$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^6 4f^{14} 5d^{10} 6s^2 6p^6 5f^4 6d^1 7s^2$

In ground state, total number of f electrons =  $14 + 4 = 18$

---

## Question 145

The number of 4f electrons in the ground state electronic configuration of  $Gd^{2+}$  is  
[Atomic number of Gd is 64.]

[26 Aug 2021 Shift 1]

**Answer: 7**

**Solution:**

The electronic configuration of Gd is as follows

$Gd = [Xe] 4f^7, 5d^1, 6s^2$

So, electronic configuration of  $Gd^{2+}$  will be

$Gd^{2+} = [Xe], 4f^7, 5d^1, 6s^0$

∴ The number of 4 f electrons in ground state configuration of  $Gd^{2+}$  is 7.

---

## Question 146

In the given chemical reaction, colours of the  $Fe^{2+}$  and  $Fe^{3+}$  ions, are respectively  
 $5Fe^{2+} + MnO_4^- + 8H^+ \longrightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$

[1 Sep 2021 Shift 2]

**Options:**

A. yellow, orange

B. yellow, green

C. green, orange

D. green, yellow

**Answer: D**

**Solution:**

The  $\text{Fe}^{2+}$  ion has green colour while  $\text{Fe}^{3+}$  is yellow in colour.

---

## Question147

**The third ionization enthalpy is minimum for:  
[Jan. 08,2020 (I)]**

**Options:**

- A. Co
- B. Fe
- C. N
- D. Mn

**Answer: B**

**Solution:**

${}_{26}\text{Fe} = [\text{Ar}]3d^6 4s^2$ . Third ionisation results into stable  $d^5$  configuration.

---

## Question148

**For the following Assertion and Reason, the correct option is:**

**Assertion:** For hydrogenation reactions, the catalytic activity increases from Group 5 to Group 11 metals with maximum activity shown by Group 7 – 9 elements.

**Reason:** The reactants are most strongly adsorbed on group 7 – 9 elements.

**[Jan. 08,2020 (II)]**

**Options:**

- A. The assertion is true, but the reason is false.
- B. Both assertion and reason are false.
- C. Both assertion and reason are true and the reason is the correct explanation for the assertion.
- D. Both assertion and reason are true but the reason is not the correct explanation for the assertion.

**Answer: A**

**Solution:**

Reactant should not be adsorbed strongly which might result into immobilisation that inhibit further adsorption on the catalyst's surface.

---

## Question149

The atomic radius of Ag is closest to:  
[Jan. 07,2020 (I)]

Options:

- A. Au
- B. Ni
- C. Cu
- D. Hg

Answer: A

Solution:

Atomic size of elements of 4d and 5d transition series are nearly same due to lanthanide contraction.

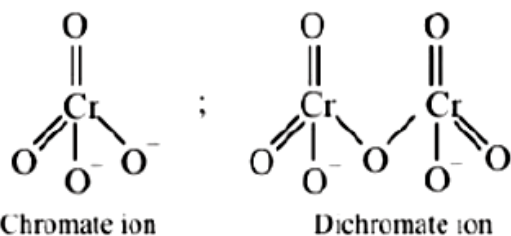
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## Question150

The sum of the total number of bonds between chromium and oxygen atoms in chromate and dichromate ions is \_\_\_\_\_.  
[NV, Jan. 09, 2020 (II)]

Answer: 12

Solution:



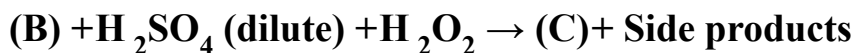
⇒ Total number of Cr and O bonds is 12 .

-----

## Question151

Consider the following reactions:

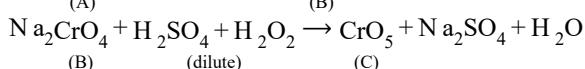
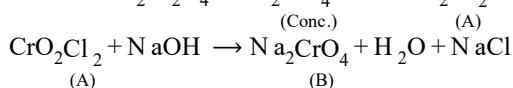
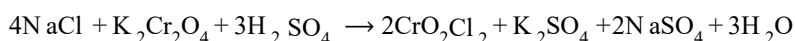




**The sum of the total number of atoms in one molecule each of (A), (B) and (C) is \_\_\_\_\_.**  
**[NV, Jan. 07, 2020 (II)]**

**Answer: 18**

**Solution:**

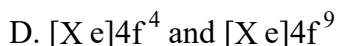
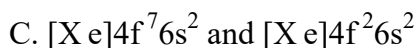
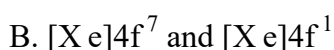
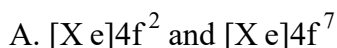


The sum of total no. of atoms in one molecules each of A,  
 B & C = 5 + 7 + 6 = 18.0

## Question152

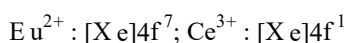
**The electronic configurations of bivalent europium and trivalent cerium are:  
 (atomic number : X e = 54, Ce = 58, E u = 63  
 [Jan. 09,2020 (I)]**

**Options:**



**Answer: B**

**Solution:**



## Question153

**The highest possible oxidation states of uranium and plutonium, respectively, are:  
 [Jan. 09,2020 (I)]**

**Options:**

- A. 6 and 7
- B. 6 and 4
- C. 7 and 6
- D. 4 and 6

**Answer: A**

**Solution:**

Maximum oxidation state shown by uranium is +6 and plutonium is 7 .

-----

## Question154

**The incorrect statement(s) among (1) – (3) is (are):**

**(1) W(VI) is more stable than Cr(VI).**

**(2) in the presence of HCl, permanganate titrations provide satisfactory results.**

**(3) some lanthanoid oxides can be used as phosphorus.**

**[Sep. 4,2020 (II)]**

**Options:**

- A. (2) and (3) only
- B. (1) and (2) only
- C. (2) only
- D. (1) only

**Answer: C**

**Solution:**

- (i) W(VI) is more stable than Cr(VI) due to smaller size of atoms and also due to lanthanide contraction.
  - (ii) Permanganate titrations in presence of HCl are unsatisfactory as HCl is oxidised to  $\text{Cl}_2$
  - (iii) Lanthanoid oxides are used as phosphors.
- 

## Question155

**The incorrect statement is:**

**[Sep. 03,2020(II)]**

**Options:**

- A. Manganate and permanganate ions are tetrahedral

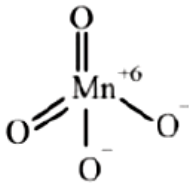
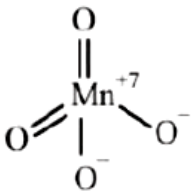
B. In manganate and permanganate ions, the  $\pi$  -bonding takes place by overlap of p -orbitals of oxygen and d – orbitals of manganese

C. Manganate and permanganate ions are paramagnetic

D. Manganate ion is green in colour and permanganate ion is purple in colour

**Answer: C**

**Solution:**

| Manganate                                                                         | Permanganate                                                                       |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| $MnO_4^{2-}$                                                                      | $MnO_4^-$                                                                          |
|  |  |
| Paramagnetic, green in colour, Tetrahedral & contains $p\pi - d\pi$ bond          | Diamagnetic, purple in colour, Tetrahedral & contains $p\pi - d\pi$ bond           |

Manganate ion is paramagnetic while permanganate ion is diamagnetic.

## Question156

The lanthanoid that does NOT show + 4 oxidation state is:  
[Sep. 06,2020(I)]

**Options:**

A. Dy

B. Ce

C. Eu

D. Tb

**Answer: C**

**Solution:**

Europium(Eu)

Atomic No. - 63

Electronic configuration -  $[Xe]4f^76s^2$

It shows only +2 and +3 oxidation state.

## Question157

**Mischmetal is an alloy consisting mainly of:**  
[Sep. 06,2020(II)]

**Options:**

- A. lanthanoid metals
- B. actinoid and transition metals
- C. lanthanoid and actinoid metals
- D. actinoid metals

**Answer: A**

**Solution:**

Mischmetal is an alloy consisting mainly of lanthanoid metals.

Alloy  $\Rightarrow$

Lan. metal  $\Rightarrow \approx 95\%$

Iron  $\Rightarrow \approx 5\%$

S, C, Ca, Al  $\Rightarrow$  traces

---

## Question158

**The correct electronic configuration and spin-only magnetic moment (BM ) of  $\text{Gd}^{3+}$  ( $Z = 64$ ), respectively, are:**  
[Sep. 05,2020(I)]

**Options:**

- A.  $[\text{Xe}] 4f^7$  and 8.9
- B.  $[\text{Xe}] 4f^7$  and 7.9
- C.  $[\text{Xe}] 5f^7$  and 8.9
- D.  $[\text{Xe}] 5f$  and 7.9

**Answer: B**

**Solution:**

Electronic configuration of 64

$\text{Gd} = [\text{Xe}] 4f^7 5d^1 6s^2$

${}_{64}\text{Gd}^{3+} = [\text{Xe}] 4f^7$

No. of unpaired electron = 7

$\mu = \sqrt{n(n+2)}\text{BM} = \sqrt{63} = 7.93\text{BM}$

---

## Question159

The pair of metal ions that can give a spin only magnetic moment of 3.9BM for the complex  $[M(H_2O)_6]Cl_2$ , is:

[Jan. 12, 2019(I)]

Options:

- A.  $V^{2+}$  and  $Co^{2+}$
- B.  $V^{2+}$  and  $Fe^{2+}$
- C.  $Co^{2+}$  and  $Fe^{2+}$
- D.  $Cr^{2+}$  and  $Mn^{2+}$

Answer: A

Solution:

Given  $\mu = 3.9BM$

$$\mu = \sqrt{n(n+2)}B \cdot M \therefore 3.9 = \sqrt{n(n+2)}; n = 3$$

So, the central metal ion has 3 unpaired electrons.

$\therefore$  Configuration is either  $d^3$  or  $d^7$

As  $H_2O$  is a weak field ligand.  $V^{2+}$  and  $Co^{2+}$  will have 3 unpaired electrons.  $V^{2+}$  has  $d^3$  configuration;  $Co^{2+}$  has  $d^7$  configuration.

---

## Question160

The element that usually does NOT show variable oxidation states is:

[Jan. 11, 2019(I)]

Options:

- A. Cu
- B. Ti
- C. Sc
- D. V

Answer: C

Solution:

Sc shows oxidation state of +3 only.

---

## Question161

**The highest value of the calculated spin only magnetic moment (in BM) among all the transition metal complexes is:**

**[Jan. 9,2019(I)]**

**Options:**

A. 5.92

B. 6.93

C. 3.87

D. 4.90

**Answer: A**

**Solution:**

Magnetic moment,  $\mu = \sqrt{n(n+2)}\text{BM}$  (where, n = no. of unpaired electrons)

As transition metal atom/ion in a complex may have unpaired electrons ranging from zero to 5 . So, maximum number of unpaired electrons that may be present in a complex is 5.

$\therefore$  Maximum value of magneticmoment among all the transition metal complexes is  $\sqrt{5(5+2)} = \sqrt{35} = 5.92\text{BM}$

-----

## Question162

**The transition element that has lowest enthalpy of atomisation is:**

**[Jan. 9,2019(II)]**

**Options:**

A. F e

B. Cu

C. V

D. Z n

**Answer: D**

**Solution:**

As zinc has no unpaired of electrons to take part in the bond, it has least enthalpy of atomisation amongst the given transition elements.

-----

## Question163

**The correct order of atomic radii is :**

**[Jan. 12, 2019 (II)]**

**Options:**

- A.  $N > Ce > Eu > Ho$
- B.  $Ho > N > Eu > Ce$
- C.  $Ce > Eu > Ho > N$
- D.  $Eu > Ce > Ho > N$

**Answer: D**

**Solution:**

Atomic radii follows the order  
 $Eu > Ce > Ho > N$   
 185pm 182pm 177pm 71pm

---

## Question 164

**The effect of lanthanoid contraction in the lanthanoid series of elements by and large means:**

**[Jan. 10, 2019 (I)]**

**Options:**

- A. increase in both atomic and ionic radii
- B. decrease in atomic radii and increase in ionic radii
- C. decrease in both atomic and ionic radii
- D. increase in atomic radii and decrease in ionic radii

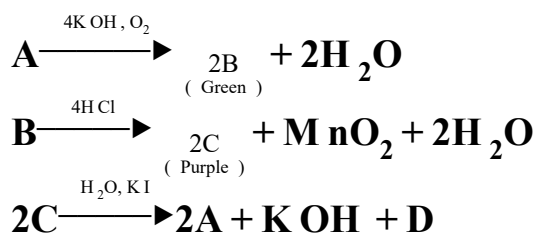
**Answer: C**

**Solution:**

Due to lanthanoid contraction, size of atoms as well as ions of lanthanoid decreases.

---

## Question 165



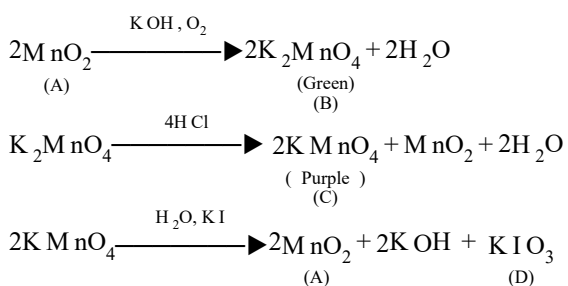
**In the above sequence of reactions, A and D, respectively, are:**  
**[Jan. 11, 2019 (II)]**

**Options:**

- A. K I and K M nO<sub>4</sub>
- B. M nO<sub>2</sub> and K I O<sub>3</sub>
- C. K I O<sub>3</sub> and M nO<sub>2</sub>
- D. K I and K<sub>2</sub>M nO<sub>4</sub>

**Answer: B**

**Solution:**



## Question166

**The pair that has similar atomic radii is :**  
**[April 12,2019 (II)]**

**Options:**

- A. M n and Re
- B. T i and H f
- C. Sc and N i
- D. M o and W

**Answer: D**

**Solution:**

Mo and W belong to group-6 and period 5 (4d series) and 6( 5d series) respectively. Due to lanthanoid contraction, radius of Mo and W are almost same i.e. 0.140nm and 0.141nm respectively.

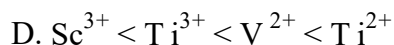
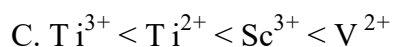
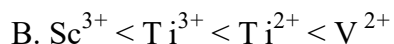
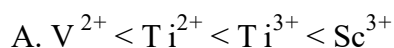
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## Question167

**Consider the hydrated ions of T i<sup>2+</sup>, V<sup>2+</sup>, T i<sup>3+</sup>, and Sc<sup>3+</sup>. The correct order of their spin-only magnetic moments is:**

**[April 12,2019 (II)]**

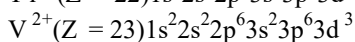
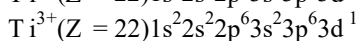
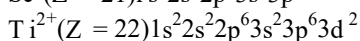
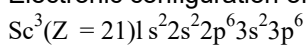
**Options:**



**Answer: B**

**Solution:**

Electronic configuration of the given transition metal ions are :



Since, magnetic moment is directly proportional to the number of unpaired electrons. The correct increasing order of magnetic moment is

$Sc^{3+} < Ti^{3+} < Ti^{2+} < V^{2+}$  because they have 0,1,2 and 3 unpaired electrons respectively.

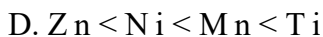
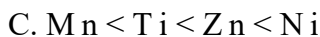
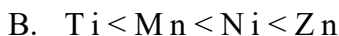
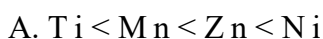
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## Question168

**The correct order of the first ionization enthalpies is:**

**[April 10,2019 (II)]**

**Options:**



**Answer: B**

**Solution:**

I.E. increases on moving left to right in a period.  $\therefore Ti < Mn < Ni < Zn$

-----

## Question169

**The INCORRECT statement is:**

**[April 10,2019 (II)]**

**Options:**

- A. the gemstone, ruby, has  $\text{Cr}^{3+}$  ions occupying the octahedral sites of beryl.
- B. the spin-only magnetic moment of  $[\text{Ni}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$  is 2.83BM
- C. the color of  $[\text{CoCl}(\text{NH}_3)_5]^{2+}$  is violet as it absorbs the yellow light.
- D. the spin-only magnetic moments of  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  and  $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$  are nearly similar.

**Answer: A**

**Solution:**

Ruby is aluminium oxide ( $\text{Al}_2\text{O}_3$ ) containing about 0.5 – 1%  $\text{Cr}^{3+}$  ions, which are randomly distributed in place of  $\text{Al}^{3+}$  ions.

-----

## Question170

**Match the catalysts (Column I) with products (Column II).**

| Column I                               | Column II               |
|----------------------------------------|-------------------------|
| Catalyst                               | Product                 |
| $\text{V}_2\text{O}_5$                 | Polyethylene            |
| $\text{TiCl}_4/\text{Al}(\text{Me})_3$ | ethanol                 |
| $\text{PdCl}_2$                        | $\text{H}_2\text{SO}_4$ |
| Iron Oxide                             | $\text{NH}_3$           |

**[April 9,2019 (I)]**

**Options:**

- A. (A) – (iii); (B) – (iv); (C) – (i); (D) – (ii)
- B. (A) – (ii); (B) – (iii); (C) – (i); (D) – (iv)
- C. (A) – (iii); (B) – (i); (C) – (ii); (D) – (iv)
- D. (A) – (iv); (B) – (iii); (C) – (ii); (D) – (i)

**Answer: C**

**Solution:**

(A)  $\text{V}_2\text{O}_5 \rightarrow$  Preparation of  $\text{H}_2\text{SO}_4$  in contact process

(B)  $\text{TiCl}_4 + \text{Al}(\text{Me})_3 \rightarrow$  Polyethylene (Ziegler-Natta catalyst)

- (C)  $\text{PdCl}_2 \rightarrow \text{Ethanol}$  (Wacker's process)  
(D) Iron oxide  $\rightarrow \text{NH}_3$  in (Haber's process)
- 

## Question 171

The statement that is **INCORRECT** about the interstitial compounds is:  
[April 8, 2019 (II)]

Options:

- A. they are chemically reactive.
- B. they are very hard.
- C. they have metallic conductivity.
- D. they have high melting points.

**Answer: A**

**Solution:**

Interstitial compounds are inert, i.e., they are chemically non-reactive.

---

## Question 172

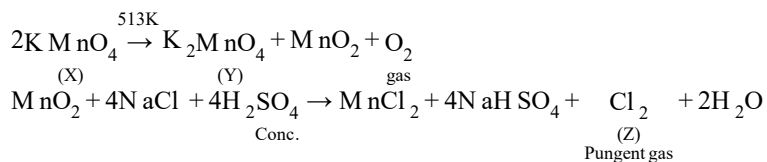
Thermal decomposition of a Mn compound (X) at 513K results in compound Y,  $\text{MnO}_2$  and a gaseous product.  $\text{MnO}_2$  reacts with  $\text{NaCl}$  and concentrated  $\text{H}_2\text{SO}_4$  to give a pungent gas Z. X, Y, and Z, respectively, are :  
[April 12, 2019 (II)]

Options:

- A.  $\text{KMnO}_4$ ,  $\text{K}_2\text{MnO}_4$  and  $\text{Cl}_2$
- B.  $\text{K}_2\text{MnO}_4$ ,  $\text{KMnO}_4$  and  $\text{SO}_2$
- C.  $\text{K}_3\text{MnO}_4$ ,  $\text{K}_2\text{MnO}_4$  and  $\text{Cl}_2$
- D.  $\text{K}_2\text{MnO}_4$ ,  $\text{KMnO}_4$  and  $\text{Cl}_2$

**Answer: A**

**Solution:**



## Question173

**The maximum number of possible oxidation states of actinoides are shown by:  
[April 9, 2019 (II)]**

**Options:**

- A. Nobelium (No) and lawrencium (Lr)
- B. Actinium (Ac) and thorium (Th)
- C. Berkelium (Bk) and californium (Cf)
- D. Neptunium (Np) and plutonium (Pu)

**Answer: D**

**Solution:**

Actinoids Oxidation state shown

Th +3, +4

Ac +3

Pu +3, +4, +5, +6, +7

Np +3, +4, +5, +6, +7

Bk +3, +4

Cm +3, +4, +5

Ir +3

∴ Maximum oxidation state is shown by Np and Pu.

## Question174

**The lanthanide ion that would show colour is:  
[April 8, 2019 (II)]**

**Options:**

- A. Gd<sup>3</sup>
- B. Sm<sup>3</sup>
- C. La<sup>3+</sup>
- D. Lu<sup>3</sup>

**Answer: B**

**Solution:**

$\text{Sm} = 4f^6 6s^2$   
 $\text{Sm}^{3+} = 4f^5 =$  Partially filled f orbital  
 $\therefore \text{Sm}^{3+}$  will be coloured  
 $\text{Lu}^{3+} = 4f^{14} =$  colourless.

---

## Question175

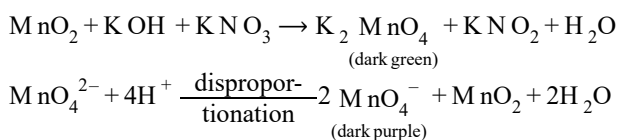
When  $\text{X O}_2$  is fused with an alkali metal hydroxide in presence of an oxidizing agent such as  $\text{K N O}_3$ , a dark green product is formed which disproportionate in acidic solution to afford a dark purple solution. X is:  
 [Online April 16, 2018]

Options:

- A. M n
- B. Cr
- C. V
- D. T i

**Answer: A**

**Solution:**



## Question176

Which of the following ions does not liberate hydrogen gas on reaction with dilute acids?  
 [Online April 9, 2017]

Options:

- A.  $\text{T i}^{2+}$
- B.  $\text{V}^{2+}$
- C.  $\text{Cr}^{2+}$
- D.  $\text{M n}^{2+}$

**Answer: D**

## Solution:

Ions  $E^0(V)$

Ti<sup>2+</sup> - 0.37

V<sup>2+</sup> - 0.26

Cr<sup>2+</sup> - 0.41

Mn<sup>2+</sup> + 1.57

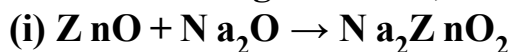
Negative value of  $E^0$  means these metals liberate hydrogen from dilute acid.

( $M^{2+} + 2H^+ - e^- \rightarrow M^{3+} + H_2$ )

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## Question 177

In the following reactions, ZnO is respectively acting as a/an:



[2017]

Options:

A. base and acid

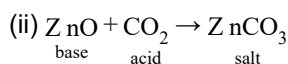
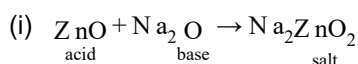
B. base and base

C. acid and acid

D. acid and base

Answer: D

## Solution:



## Question 178

The reaction of zinc with dilute and concentrated nitric acid, respectively, produces:  
[2016]

Options:

A. NO and N<sub>2</sub>O

B. NO<sub>2</sub> and N<sub>2</sub>O

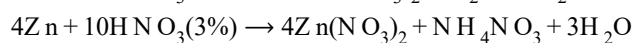
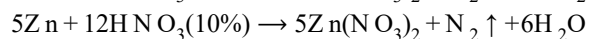
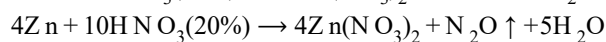
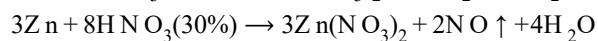
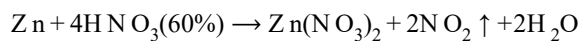
C. N<sub>2</sub>O and NO<sub>2</sub>

D.  $\text{N O}_2$  and  $\text{N O}$

**Answer: C**

**Solution:**

Reaction of  $\text{Zn}$  with different concentration of  $\text{HNO}_3$  are as follows:



Hence option (c) is correct.

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## Question179

**Which one of the following species is stable in aqueous solution?  
[Online April 9,2016]**

**Options:**



**Answer: B**

**Solution:**

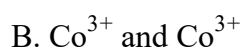
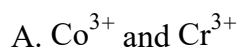
$\text{MnO}_4^{2-}$  manganese is in +6 oxidation state which is having highest stability.

-----

## Question180

**The transition metal ions responsible for colour in ruby and emerald are,  
respectively:  
[Online April 10,2016]**

**Options:**



C.  $\text{Cr}^{3+}$  and  $\text{Cr}^{3+}$

D.  $\text{Cr}^{3+}$  and  $\text{Co}^{3+}$

**Answer: C**

**Solution:**

A ruby is a crystal of alumina, aluminum oxide ( $\text{Al}_2\text{O}_3$ ) containing a trace of chromium (III) ions replacing some of the aluminum ions. In ruby, each  $\text{Al}^{3+}$  ion and  $\text{Cr}^{3+}$  ion is surrounded by six oxide ions in an octahedral arrangement. The origin of the color of emeralds is similar to that of the color of rubies. However, the bulk of an emerald crystal is composed of beryl, beryllium aluminum silicate ( $\text{Be}_3\text{Al}_2(\text{SiO}_3)_6$ ) instead of the alumina which forms rubies. The color is produced by chromium (III) ions, which replace some of the aluminum ions in the crystal. In emeralds, the  $\text{Cr}^{3+}$  is surrounded by six silicate ions, rather than the six oxide ions in ruby. Therefore, the color (green) of emeralds is different from that of ruby.

-----

## Question181

**Which of the following compounds is metallic and ferromagnetic?  
[2016]**

**Options:**

A.  $\text{VO}_2$

B.  $\text{MnO}_2$

C.  $\text{TiO}_2$

D.  $\text{CrO}_2$

**Answer: D**

**Solution:**

Out of all the four given metallic oxides,  $\text{CrO}_2$  is attracted by magnetic field very strongly. The effect persists even when the magnetic field is removed. Thus  $\text{CrO}_2$  is metallic and ferromagnetic in nature.

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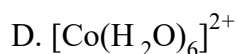
## Question182

**When concentrated  $\text{HCl}$  is added to an aqueous solution of  $\text{CoCl}_2$ , its colour changes from reddish pink to deep blue. Which complex ion gives blue colour in this reaction?**

**[Online April 11, 2015]**

**Options:**

A.  $[\text{CoCl}_4]^{2-}$



**Answer: A**

**Solution:**

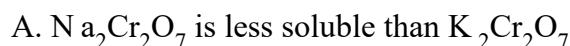
Aqueous solution of  $\text{CoCl}_2$  contains  $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$  which is pinkish in colour so option (d) is incorrect. Reduction potential of  $\text{Co}^{3+} \rightarrow \text{Co}^{2+}$  is high so option(b) is incorrect.  $\text{Co}^{2+}$  does not oxidises easily to  $\text{Co}^{3+}$ . It is general case that symmetrical substituted octahedral complexes are less deeper in colour than tetrahedral complexes. So  $[\text{CoCl}_4]^{2-}$  is deep blue in colour.

---

## Question183

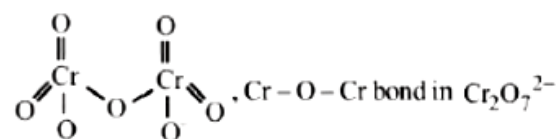
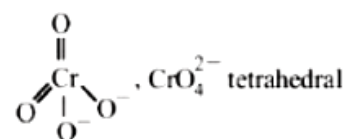
**Which of the following statements is false?**  
**[Online April 11, 2015]**

**Options:**



**Answer: A**

**Solution:**



$\text{Na}_2\text{Cr}_2\text{O}_7$  is used as a primary standard in volumetry.

Hydration energy of  $\text{Na}^{\oplus}$  is greater than  $\text{K}^{\oplus}$ . Because of smaller size of  $\text{Na}^{\oplus}$ ,  $\text{Na}_2\text{Cr}_2\text{O}_7$  is more soluble than  $\text{K}_2\text{Cr}_2\text{O}_7$

---

## Question184

**Chloro compound of Vanadium has only spin magnetic moment of 1.73BM . This Vanadium chloride has the formula:**

**[Online April 9,2014]**

**Options:**



**Answer: B**

**Solution:**

$$\text{Magnetic moment} = \sqrt{n(n+2)}$$

where  $n$  = number of unpaired electrons  $\sqrt{n(n+2)} = 1.73$

$$\therefore n = 1$$

Electronic configuration of V is  $[\text{Ar}]4s^23d^3$ .

For only one unpaired electron it has to be  $\text{V}^{4+}$ .

Hence, the formula of vanadium chloride is  $\text{VCl}_4$ .

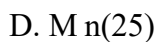
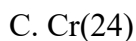
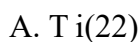
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## Question185

**Which one of the following exhibits the large number of oxidation states?**

**[Online April 12,2014]**

**Options:**



**Answer: D**

**Solution:**

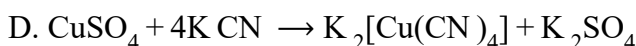
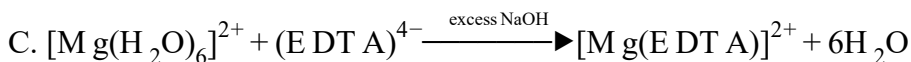
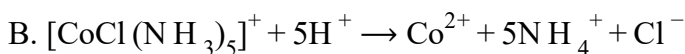
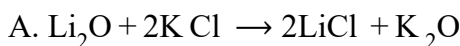
Manganese exhibits the large number of oxidation states. The most common oxidation states of Mn are +2,+3,+4,+6 and +7

---

## Question186

**The equation which is balanced and represents the correct product(s) is:  
[2014]**

**Options:**



**Answer: B**

**Solution:**

The equation in option (b) is correct since both charges as well as atoms are balanced. For the rest,  
(a) Given reaction is unfavourable in the forward direction ( $\text{K}_2\text{O}$  is unstable, while  $\text{Li}_2\text{O}$  is stable)

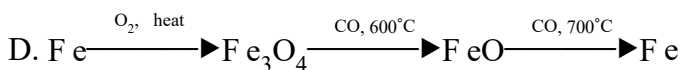
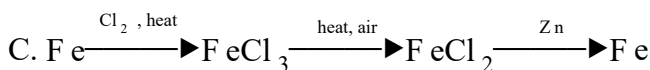
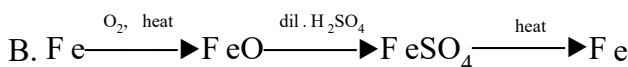
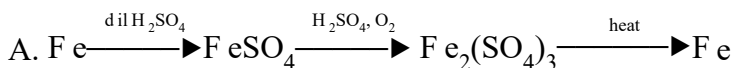
(c) Given reaction is not balanced w.r.t. charge.

(d) Given reaction will give  $\text{K}_3[\text{Cu}(\text{CN})_4]$  as product instead of  $\text{K}_2[\text{Cu}(\text{CN})_4]$

## Question187

**Which series of reactions correctly represents chemical reactions related to iron and its compound?  
[2014]**

**Options:**



**Answer: D**

**Solution:**

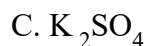
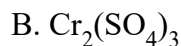
In equation (a)  $\text{Fe}_2(\text{SO}_4)_3$ , and in equation (b)  $\text{FeSO}_4$  on decomposition will form oxide instead of Fe. In equation (c)  $\text{FeCl}_3$  cannot be reduced when heated in air. Hence equation (d) is correct.

## Question188

Which of the following is not formed when  $\text{H}_2\text{S}$  reacts with acidic  $\text{K}_2\text{Cr}_2\text{O}_7$  solution?

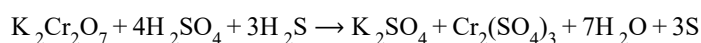
[Online April 9, 2014]

Options:



Answer: A

Solution:

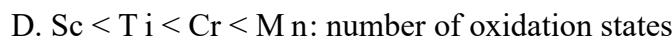
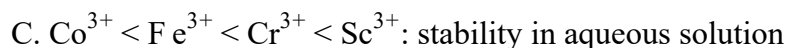
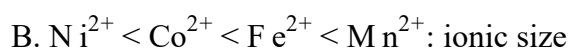
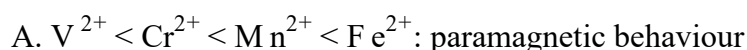


## Question 189

Which of the following arrangements does not represent the correct order of the property stated against it?

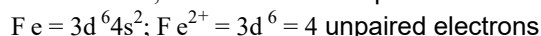
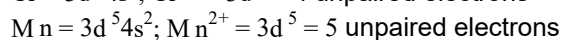
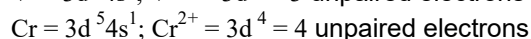
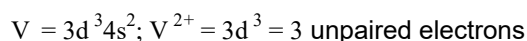
[2013]

Options:

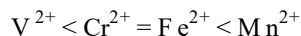


Answer: A

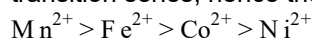
Solution:



Hence the correct order of paramagnetic behaviour



(b) For the same oxidation state, the ionic radii generally decreases as the atomic number increases in a particular transition series, hence the order is



(c) Larger size, least hydrated more stable in aqueous solution. As we move across the period ( $\text{Sc}^{3+} \rightarrow \text{Cr}^{3+} \rightarrow$

$\text{Fe}^{3+} \rightarrow \text{Co}^{3+}$ ), the ionic size usually decreases.  $\text{Sc}^{3+}$  with the large size as least hydrated and hence more stable.

(d) Sc – (+2), (+3)  
Ti – (+2), (+3), (+4)  
Cr – (+2), (+3), (+4), (+5), (+6)  
Mn – (+2), (+3), (+4), (+5), (+6), (+7)  
i.e. Sc < Ti < Cr < Mn

---

## Question190

Four successive members of the first row transition elements are listed below with atomic numbers. Which one of them is expected to have the highest  $E_{M^{3+}/M^{2+}}$  value?  
[2013]

Options:

- A. Cr(Z = 24)
- B. Mn(Z = 25)
- C. Fe(Z = 26)
- D. Co(Z = 27)

Answer: D

Solution:

$E^{\circ}Cr^{3+}/Cr^{2+} = -0.41V$   
 $E^{\circ}Fe^{3+}/Fe^{2+} = +0.77V$   
 $E^{\circ}Mn^{3+}/Mn^{2+} = +1.57V$ ,  $E^{\circ}Co^{3+}/Co^{2+} = +1.97V$

---

## Question191

The element with which of the following outer electron configuration may exhibit the largest number of oxidation states in its compounds:  
[Online April 9,2013]

Options:

- A.  $3d^5 4s^2$
- B.  $3d^8 4s^2$
- C.  $3d^7 4s^2$
- D.  $3d^6 4s^2$

Answer: A

Solution:

The element with outer electron configuration  $3d^5 4s^2$  is Mn which exhibits oxidation states from +2 to +7 .

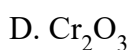
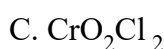
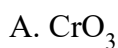
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## Question192

**Potassium dichromate when heated with concentrated sulphuric acid and a soluble chloride, gives brown-red vapours of :**

**[Online April 9, 2013]**

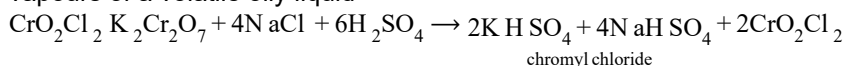
**Options:**



**Answer: C**

**Solution:**

Solid potassium dichromate when heated with concentrated sulphuric acid and a soluble chloride gives orange red vapours of a volatile oily liquid

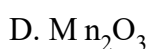
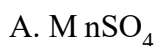


## Question193

**When a small amount of  $\text{KMnO}_4$  is added to concentrated  $\text{H}_2\text{SO}_4$ , a green oily compound is obtained which is highly explosive in nature. Compound may be:**

**[Online April 23, 2013]**

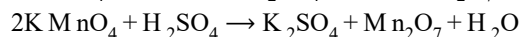
**Options:**



**Answer: B**

**Solution:**

$\text{KMnO}_4$  reacts with  $\text{H}_2\text{SO}_4$  to form  $\text{Mn}_2\text{O}_7$  which is highly explosive substance.



## Question194

**Identify incorrect statement:**

**[Online April 23, 2013]**

**Options:**

- A.  $\text{Cu}_2\text{O}$  is colourless.
- B. Copper (I) compounds are colourless except when colour results from charge transfer.
- C. Copper (I) compounds are diamagnetic.
- D.  $\text{Cu}_2\text{S}$  is black.

**Answer: A**

**Solution:**

$\text{Cu}_2\text{O}$  is yellow in colour.

---

## Question195

**Iron exhibits +2 and +3 oxidation states. Which of the following statements about iron is incorrect?**

**[2012]**

**Options:**

- A. Ferrous oxide is more basic in nature than the ferric oxide.
- B. Ferrous compounds are relatively more ionic than the corresponding ferric compounds.
- C. Ferrous compounds are less volatile than the corresponding ferric compounds.
- D. Ferrous compounds are more easily hydrolysed than the corresponding ferric compounds.

**Answer: D**

**Solution:**

$\text{Fe}^{3+}$  is easily hydrolysed than  $\text{Fe}^{2+}$  due to more positive charge.

---

## Question196

**Which pair of elements with the given atomic numbers is expected to have similar properties?**

**[Online May 19,2012]**

**Options:**

A. 40,72

B. 20,36

C. 10,28

D. 11,12

**Answer: A**

**Solution:**

Zr(40), Hf (72) have similar properties as they belong to same group and have same valence shell electronic configuration.

-----

## **Question197**

**Which is not the correct statement?**

**(At. nos. Ce = 58, Lu = 71, La = 57, Y b = 70 )**

**[Online May 7,2012]**

**Options:**

A. Colour of  $\text{Yb}^{3+}$  ion is pink.

B.  $\text{La}^{3+}$  is diamagnetic.

C.  $\text{Ce}^{4+}$  has  $f^0$  configuration.

D.  $\text{Lu}^{3+}$  had  $f^{14}$  configuration.

**Answer: A**

**Solution:**

Option (a) is incorrect as  $\text{Yb}^{3+}$  is colorless.

-----

## **Question198**

**Magnetic moment of  $\text{Gd}^{3+}$  ion ( $Z = 64$ ) is**

**[Online May 12,2012]**

**Options:**

- A. 3.62BM
- B. 9.72BM
- C. 7.9BM
- D. 10.60BM

**Answer: C**

**Solution:**

$$\begin{aligned}\text{Gd (64)} &= [\text{Xe}]4f^7 5d^1 6s^2 \\ \text{Gd}^{3+} &= [\text{Xe}]4f^7 5d^0 6s^0 \\ \text{i.e. no. of unpaired electrons} &= 7 \\ \mu &= \sqrt{n(n+2)} = \sqrt{7(7+2)} \\ &= \sqrt{63} = 7.93\text{BM}\end{aligned}$$


---

## Question199

**Which of the following forms stable +4 oxidation state?**  
[Online May 26,2012]

**Options:**

- A. La(Z = 57)
- B. Eu(Z = 63)
- C. Ce(Z = 58)
- D. Gd (Z = 64)

**Answer: C**

**Solution:**

Only Ce(Z = 58) shows stable (+4) oxidation state.

---

## Question200

**The number of unpaired electrons in Gadolinium [Z = 64] is**  
[Online May 26,2012]

**Options:**

- A. 3
- B. 8
- C. 6

D. 2

**Answer: B**

**Solution:**

$\text{Gd (64)} = [\text{Xe}]4f^7 5d^1 6s^2$   
 $\therefore \text{No. of unpaired electrons} = 8$

---

## Question201

**The correct order of  $E^\circ \text{M}^{2+}/\text{M}^{2+}$  values with negative sign for the four successive elements Cr, Mn, Fe and Co is [2010]**

**Options:**

- A.  $\text{Mn} > \text{Cr} > \text{Fe} > \text{Co}$
- B.  $\text{Cr} < \text{Fe} > \text{Mn} > \text{Co}$
- C.  $\text{Fe} > \text{Mn} > \text{Cr} > \text{Co}$
- D.  $\text{Cr} > \text{Mn} > \text{Fe} > \text{Co}$

**Answer: A**

**Solution:**

Across the first transition series, the negative values for standard electrode potential decrease except for Mn due to the stable  $d^5$  configuration. Thus, correct order is  
 $\text{Mn}^{2+} > \text{Cr}^{2+} > \text{Fe}^{2+} > \text{Co}^{2+}$

---

## Question202

**Knowing that the chemistry of lanthanoids(Ln) is dominated by its +3 oxidation state, which of the following statements is incorrect? [2009]**

**Options:**

- A. The ionic size of Ln (III) decreases in general with increasing atomic number
- B. Ln (III) compounds are generally colourless.
- C. Ln (III) hydroxides are mainly basic in character.
- D. Because of the large size of the Ln(III) ions the bonding in its compounds is predominantly ionic in character.

**Answer: B**

**Solution:**

Most of the  $\text{Ln}^{3+}$  compounds except  $\text{La}^{3+}$  and  $\text{Lu}^{3+}$  are coloured due to the presence of unpaired f -electrons.

---

## Question203

**Amount of oxalic acid present in a solution can be determined by its titration with  $\text{KMnO}_4$  solution in the presence of  $\text{H}_2\text{SO}_4$ . The titration gives unsatisfactory result when carried out in the presence of  $\text{HCl}$ , because  $\text{HCl}$**   
**[2008]**

**Options:**

- A. gets oxidised by oxalic acid to chlorine
- B. furnishes  $\text{H}^+$  ions in addition to those from oxalic acid
- C. reduces permanganate to  $\text{Mn}^{2+}$
- D. oxidises oxalic acid to carbon dioxide and water

**Answer: C**

**Solution:**

The titration of oxalic acid with  $\text{KMnO}_4$  in presence of  $\text{HCl}$  gives unsatisfactory result because of the fact that  $\text{KMnO}_4$  can also oxidise  $\text{HCl}$  along with oxalic acid.  $\text{HCl}$  on oxidation gives  $\text{Cl}_2$  and  $\text{HCl}$  reduces  $\text{KMnO}_4$  to  $\text{Mn}^{2+}$ , thus the correct answer is (c).

---

## Question204

**Larger number of oxidation states are exhibited by the actinoids than those by the lanthanoids, the main reason being**  
**[2008]**

**Options:**

- A. 4f orbitals more diffused than the 5f orbitals
- B. lesser energy difference between 5f and 6d than between 4f and 5d orbitals
- C. more energy difference between 5f and 6d than between 4f and 5d orbitals
- D. more reactive nature of the actinoids than the lanthanoids

**Answer: B**

## Solution:

Note : The main reason for exhibiting larger number of oxidation states by actinoids as compared to lanthanoids is lesser energy difference between 5f and 6d orbitals as compared to that between 4f and 5d orbitals. In case of actinoids we can remove electrons from 5f as well as from 6d and due to this actinoids exhibit larger number of oxidation state than lanthanoids.

-----

## Question 205

**Identify the incorrect statement among the following:**  
**[2007]**

**Options:**

- A. 4f and 5f orbitals are equally shielded.
- B. d -Block elements show irregular and erratic chemical properties among themselves.
- C. La and Lu have partially filled d -orbitals and no other partially filled orbitals.
- D. The chemistry of various lanthanoids is very similar.

**Answer: A**

## Solution:

4f Orbital is nearer to nucleus as compared to 5f orbital therefore, shielding of 4f is more than 5f.

-----

## Question 206

**The actinoids exhibit more number of oxidation states in general than the lanthanoids. This is because**  
**[2007]**

**Options:**

- A. the 5f orbitals extend further from the nucleus than the 4f orbitals
- B. the 5f orbitals are more buried than the 4f orbitals
- C. there is a similarity between 4f and 5f orbitals in their angular part of the wave function
- D. the actinoids are more reactive than the lanthanoids.

**Answer: A**

## Solution:

Note: More the distance between nucleus and outer orbitals, lesser will be force of attraction on them. Distance between nucleus and 5f orbitals is more as compared to distance between 4f orbital and nucleus. So actinoids exhibit more

number of oxidation states in general than the lanthanoids.

---

## Question207

The "spin-only" magnetic moment [in units of Bohr magneton, ( $\mu_B$ )] of  $\text{Ni}^{2+}$  in aqueous solution would be (At. No. Ni = 28 )  
[2006]

Options:

- A. 6
- B. 1.73
- C. 2.82
- D. 4.90

Answer: C

Solution:

The number of unpaired electrons in  $\text{Ni}^{2+}(\text{aq}) = 2$ . Water is a weak ligand, hence no pairing will take place  $\therefore$  spin magnetic moment  $= \sqrt{n(n+2)} = \sqrt{2(2+2)} = \sqrt{8} = 2.82$

---

## Question208

Lanthanoid contraction is caused due to  
[2006]

Options:

- A. the same effective nuclear charge from Ce to Lu
- B. the imperfect shielding on outer electrons by 4f electrons from the nuclear charge
- C. the appreciable shielding on outer electrons by 4f electrons from the nuclear charge
- D. the appreciable shielding on outer electrons by 5d electrons from the nuclear charge

Answer: B

Solution:

The configuration of lanthanides shows that the additional electron enters the 4f subshell. The shielding of one 4f electron by another is very little or imperfect. The imperfect shielding of f electrons is due to the shape of f orbitals which is very much diffused. Thus as the atomic number increases, the nuclear charge increases by unity at each step, while no comparable increase in the mutual shielding effect of 4f occurs. This causes a contraction in the size of the 4f subshell as a result of which atomic and ionic radii decrease gradually from La to Lu.

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## Question209

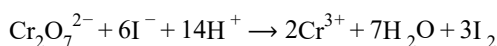
The oxidation state of chromium in the final product formed by the reaction between KI and acidified potassium dichromate solution is:  
[2005]

Options:

- A. +3
- B. +2
- C. +6
- D. +4

Answer: A

Solution:



## Question210

The value of the 'spin only' magnetic moment for one of the following configurations is 2.84BM . The correct one is  
[2005]

Options:

- A.  $d^5$  (in strong ligand field)
- B.  $d^3$  (in weak as well as in strong fields)
- C.  $d^4$  (in weak ligand fields)
- D.  $d^4$  (in strong ligand fields)

Answer: D

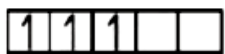
Solution:

$d^5 \rightarrow$  strong ligand field



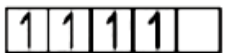
$$\mu = \sqrt{n(n+2)} = \sqrt{1(1+2)} = \sqrt{3} = 1.73 BM$$

$d^3$  – in weak as well as in strong field



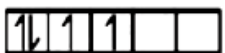
$$\mu = \sqrt{3(3+2)} = \sqrt{15} = 3.87 B.M$$

$d^4$  – in weak ligand field



$$\mu = \sqrt{4(4+2)} = \sqrt{24} = 4.89$$

$d^4$  – in strong ligand field



$$\mu = \sqrt{2(2+2)} = \sqrt{8} = 2.82$$

---

## Question211

**Which of the following factors may be regarded as the main cause of lanthanide contraction?**  
[2005]

**Options:**

- A. Greater shielding of 5d electrons by 4f electrons
- B. Poorer shielding of 5d electrons by 4f electrons
- C. Effective shielding of one of the 4f electrons by another in the subshell
- D. Poor shielding of one of the 4f electrons by another in the subshell

**Answer: D**

**Solution:**

Lanthanide contraction is associated with the intervention of the 4f orbitals which must be filled before the 5d series of elements begin. The filling of 4f before 5d orbitals results in a regular decrease in atomic radii called lanthanoid contraction which essentially compensates for the expected increase in atomic size with increasing atomic number. The factor responsible for the lanthanoid contraction is the imperfect shielding of one electron by another in the same set of orbitals.

---

## Question212

**The lanthanide contraction is responsible for the fact that**  
[2005]

**Options:**

- A. Zr and Zn have the same oxidation state

- B. Zr and Hf have about the same radius
- C. Zr and Nb have similar oxidation state
- D. Zr and Y have about the same radius

**Answer: B**

**Solution:**

Note : In vertical columns of transition elements, there is an increase in size from first member to second member as expected but from second member to third member, there is very small change in size and some times sizes are same. This is due to lanthanide contraction this is the reason for Zr and Hf to have same radius.

## Question 213

Heating mixture of  $\text{Cu}_2\text{O}$  and  $\text{Cu}_2\text{S}$  will give

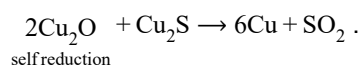
[2005]

**Options:**

- A.  $\text{Cu}_2\text{SO}_3$
- B.  $\text{CuO} + \text{CuS}$
- C.  $\text{Cu} + \text{SO}_3$
- D.  $\text{Cu} + \text{SO}_2$

**Answer: D**

**Solution:**



## Question 214

Calomel ( $\text{Hg}_2\text{Cl}_2$ ) on reaction with ammonium hydroxide gives

[2005]

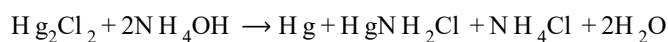
**Options:**

- A.  $\text{HgO}$
- B.  $\text{Hg}_2\text{O}$
- C.  $\text{NH}_2 - \text{Hg} - \text{Hg} - \text{Cl}$

D.  $\text{HgNH}_2\text{Cl}$

**Answer: D**

**Solution:**



## Question 215

**Of the following outer electronic configurations of atoms, the highest oxidation state is achieved by which one of them ?  
[2004]**

**Options:**

A.  $(n-1)d^3ns^2$

B.  $(n-1)d^5ns^1$

C.  $(n-1)d^8ns^2$

D.  $(n-1)d^5ns^2$

**Answer: D**

**Solution:**

$(n-1)d^5ns^2$  attains the maximum O.S. of +7

---

## Question 216

**The correct order of magnetic moments (spin only values in B.M.) among the following is  
[2004]**

**Options:**

A.  $[\text{Fe}(\text{CN})_6]^{4-} > [\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-}$

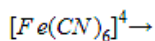
B.  $[\text{MnCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-}$

C.  $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-}$

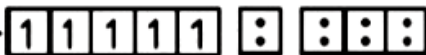
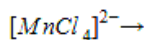
D.  $[\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$  (Atomic nos. : Mn = 25, Fe = 26, Co = 27)

**Answer: C**

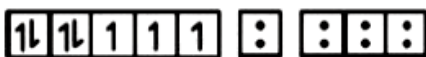
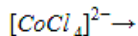
## Solution:



No of unpaired electron = 0



No of unpaired electrons = 5



No of unpaired electrons = 3

Note : The greater the number of unpaired electrons, greater the magnitude of magnetic moment. Hence the correct order will be  $[MnCl_4]^{2-} > [CoCl_4]^{2-} > [Fe(CN)_6]^{4-}$

---

## Question217

**Cerium (Z = 58) is an important member of the lanthanoids. Which of the following statements about cerium is incorrect?**

**[2004]**

**Options:**

- A. The +4 oxidation state of cerium is not known in solutions
- B. The +3 oxidation state of cerium is more stable than the +4 oxidation state
- C. The common oxidation states of cerium are +3 and +4
- D. Cerium (IV) acts as an oxidizing agent

**Answer: A**

## Solution:

The +4 oxidation state of cerium is also known in solution.

---

## Question218

**Ammonia forms the complex ion  $[Cu(NH_3)_4]^{2+}$  with copper ions in alkaline solutions but not in acidic solutions. What is the reason for it?**

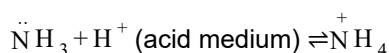
**[2003]**

**Options:**

- A. In acidic solutions, protons coordinate with ammonia molecules forming  $\text{NH}_4^+$  ions and thus  $\text{NH}_3$  molecules are not available
- B. In alkaline solutions insoluble  $\text{Cu}(\text{OH})_2$  is precipitated which is soluble in excess of any alkali
- C. Copper hydroxide is an amphoteric substance
- D. In acidic solutions hydration protects copper ions

**Answer: A**

**Solution:**



## Question219

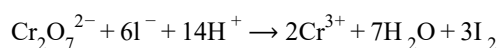
**A red solid is insoluble in water. However it becomes soluble if some K I is added to water. Heating the red solid in a test tube results in liberation of some violet coloured fumes and droplets of a metal appear on the cooler parts of the test tube. The red solid is**  
**[2003]**

**Options:**

- A.  $\text{HgI}_2$
- B.  $\text{HgO}$
- C.  $\text{Pb}_3\text{O}_4$
- D.  $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$

**Answer: A**

**Solution:**



## Question220

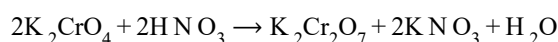
**What would happen when a solution of potassium chromate is treated with an excess of dilute nitric acid?**  
**[2003]**

**Options:**

- A.  $\text{Cr}_2\text{O}_7^{2-}$  and  $\text{H}_2\text{O}$  are formed
- B.  $\text{CrO}_4^{2-}$  is reduced to +3 state of Cr
- C.  $\text{CrO}_4^{2-}$  is oxidized to +7 state of Cr
- D.  $\text{Cr}^{3+}$  and  $\text{Cr}_2\text{O}_7^{2-}$  are formed

**Answer: A**

**Solution:**



## Question221

**Which one of the following nitrates will leave behind a metal on strong heating?  
[2003]**

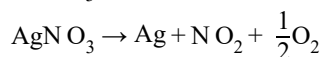
**Options:**

- A. Copper nitrate
- B. Manganese nitrate
- C. Silver nitrate
- D. Ferric nitrate

**Answer: C**

**Solution:**

$\text{AgNO}_3$  on heating till red hot decomposes as follows:



## Question222

**The radius of  $\text{La}^{3+}$  (Atomic number of La = 57 ) is  $1.06\text{\AA}$ . Which one of the following given values will be closest to the radius of  $\text{Lu}^{3+}$  (Atomic number of Lu = 71 ) ?  
[2003]**

**Options:**

- A.  $1.40\text{\AA}$
- B.  $1.06\text{\AA}$

C.  $0.85\text{\AA}$

D.  $1.60\text{\AA}$

**Answer: C**

**Solution:**

Ionic radii  $\propto \frac{1}{Z}$

Thus,  $\frac{Z_2}{Z_1} = \frac{71}{57} = \frac{1.06}{(\text{ionic radii of Lu}^{3+})}$

$\therefore$  Ionic radii of  $\text{Lu}^{3+} = 0.85\text{\AA}$

---

## Question223

**A reduction in atomic size with increase in atomic number is a characteristic of elements of [2003]**

**Options:**

A. d -block

B. f -block

C. radioactive series

D. high atomic masses

**Answer: B**

**Solution:**

f -Block elements show a regular decrease in atomic size due to lanthanide/actinide contraction.

---

## Question224

**Which of the following ions has the maximum magnetic moment? [2002]**

**Options:**

A.  $\text{Mn}^{+2}$

B.  $\text{Fe}^{+2}$

C.  $\text{Ti}^{+2}$

D.  $\text{Cr}^{+2}$

**Answer: A**

**Solution:**

$\text{Mn}^{2+}$  – 5 unpaired electrons

$\text{Fe}^{2+}$  – 4 unpaired electrons

$\text{Ti}^{2+}$  – 2 unpaired electrons

$\text{Cr}^{2+}$  – 4 unpaired electrons

Note : Magnetic moment  $\propto$  Number of unpaired electrons

---

## Question 225

**The most stable ion is  
[2002]**

**Options:**

A.  $[\text{Fe}(\text{OH})_5]^3$

B.  $[\text{Fe}(\text{Cl})_6]^3$

C.  $[\text{Fe}(\text{CN})_6]^{3-}$

D.  $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$

**Answer: C**

**Solution:**

The cyano and hydroxo complexes are far more stable than those formed by halide ion. This is due to the fact that  $\text{CN}^-$  and  $\text{OH}^-$  are strong Lewis bases (nucleophiles). Further  $[\text{Fe}(\text{OH})_5]^{3-}$  is not formed. hence most stable ion is  $[\text{Fe}(\text{CN})_6]^{3-}$

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## Question 226

**When  $\text{KMnO}_4$  acts as an oxidising agent and ultimately forms**

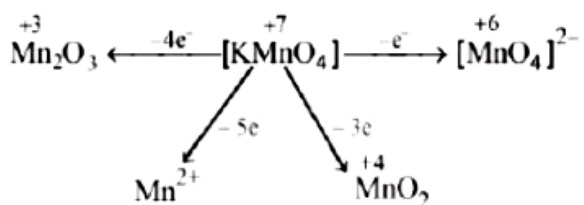
**$[\text{MnO}_4]^{-2}$ ,  $\text{MnO}_2$ ,  $\text{Mn}_2\text{O}_3$ ,  $\text{Mn}^{+2}$  then the number of electrons transferred in each case respectively is  
[2002]**

**Options:**

- A. 4,3,1,5
- B. 1,5,3,7
- C. 1,3,4,5
- D. 3,5,7,1 .

**Answer: C**

**Solution:**



## Question227

**Most common oxidation states of Ce (cerium) are [2002]**

**Options:**

- A. +2,+3
- B. +2,+4
- C. +3,+4
- D. +3,+5 .

**Answer: C**

**Solution:**

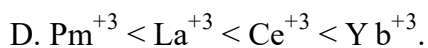
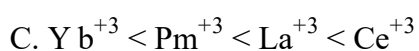
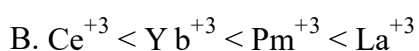
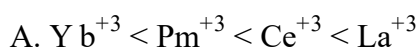
Common oxidation states of Ce(cerium) are +3 and +4

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## Question 228

Arrange  $\text{Ce}^{+3}$ ,  $\text{La}^{+3}$ ,  $\text{Pm}^{+3}$  and  $\text{Yb}^{+3}$  in increasing order of their ionic radii.  
[2002]

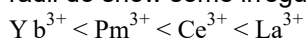
Options:



Answer: A

Solution:

In lanthanides there is a regular decrease in the atomic radii as well as ionic radii of trivalent ions as the atomic number increases from Ce to Lu. This decrease in size of atoms and ions is known as lanthanide contraction. Although the atomic radii do show some irregularities but ionic radii decrease from La to Lu. Thus the correct order is.



86.8pm 97pm 102pm 103pm

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